RESEARCH COMPANION

BFGS on Convex Nonsmooth Functions

Global and local convergence, deterministic failure,
and the complete mathematical record

A convergence theory that accommodates gradient jumps.

The mixed benchmark, homogeneous and composite families,
moving comparison geometry, and deterministic counterexamples.

Unmodified full BFGS. Armijo–weak-Wolfe.
Differentiable accepted iterates and actual matrix proofs.

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The convergence question and this companion

The question is what convergence ordinary full BFGS can guarantee when a convex objective has gradient jumps. Its practical success invites a theory, but the smooth argument cannot simply treat every secant as a bounded Hessian acting on a short step. Lewis–Overton's experiments and Challenge 7.1 place this problem in a broader almost-sure semialgebraic setting [2]. The present work gives deterministic positive and negative answers for specified convex classes. It does not claim to settle every part of that challenge.

The mixed objective $F(u, v) = u^2 + |v|$ gives a concrete measure of progress. Guo–Lewis developed extensions of Powell's smooth-convex argument and proved convergence on the Euclidean norm [10, Theorems 3.2, 4.1, 4.3 and Corollary 4.2]. Their Corollary 4.4 proves only that a subsequence for the mixed objective approaches a point on $v = 0$, leaving whole-sequence convergence to zero open. The horizontal error u^2 can remain positive on that line. Gebken's published 2026 introduction still identifies this benchmark as unresolved, while its criticality theorem assumes iterate and matrix behavior [5, Theorem 3.1 and (B1)].

The centered smooth–sublinear result here proves that every infinite admissible full-BFGS run on $u^2 + |v|$ converges to zero, for arbitrary positive-definite initialization and fixed valid Armijo–weak-Wolfe parameters. It is a consequence of a family argument in arbitrary dimension. The sublinear part is modeled exactly toward its center; adding a locally $C^{1,1}$ convex term leaves only a quadratic inward remainder. Compact initial sublevels inside the structural region give value convergence, and uniqueness gives convergence of the whole sequence. The same route covers coercive degree-one homogeneous functions, including coercive finite maxima of linear functions. The Euclidean norm is an attributed known case. The full noncompact homogeneous value alternatives are retained separately: they conclude objective-value convergence without any conclusion about the iterates, and they do not subsume Xie–Wächter's finite detection of unbounded descent in the two-piece affine model [1, Theorem IV.1].

Local convergence extends this positive picture. Finitely many polyhedral cells with smooth extensions supply a uniform inward model after a low sublevel traps the iterates near the minimizer. The algorithm need not identify a cell. Curved switching sets can defeat that radial condition; the planar theorem constructs a comparison field along positive contours instead. Its nondegenerate smooth-max corollary and explicit three-quadratic example show a convergence conclusion beyond the radial class. The fixed-normal theorem in arbitrary dimension and the flat-contour parabolic family use different hypotheses and remain independent parts of the theory.

The negative results establish a complementary limitation. A fixed unit-first implementation can generate a shrinking cycle around a nonstationary corner. The construction certifies every rejected and accepted query, and then embeds copies so that one fixed objective has failing starts arbitrarily close to its unique minimizer, all with identity initialization. One embedding has four globally convex $C^{2,1}$ branches with bounded Hessians and no uniform positive lower-curvature assumption. A different embedding preserves uniform strong convexity with $C^{1,1}$ branches, three of which lose second differentiability at the minimizer. These are two distinct exclusions of deterministic neighborhood guarantees. They neither establish positive-probability failure nor produce a finite uniformly strongly convex C^2 local counterexample.

The common analytical tool is an adjacent-degree hierarchy for actual principal determinants of full-BFGS matrices. Building on the classical trace/determinant approach [16, 15, 14], it forces the arithmetic mean of reciprocal metric-gradient values to diverge under the stated bounded-gradient and vanishing-decrease hypotheses. A comparison energy turns this matrix restriction into convergence when objective geometry preserves visibility of nonoptimality, controls the secant correction and permits transport at a summable cost. Matrix degeneration is allowed throughout.

The radial and moving-field arguments demonstrate ways this mechanism succeeds; the scalar nonnecessity results prevent it from being read as a complete characterization.

Routes through the complete theory

The article proves the centered and coercive homogeneous global results, the local finite-partition theorem and a genuinely nonradial planar application, alongside both independent local failure constructions. This companion retains the full corrected record, including conclusions with incomparable hypotheses beyond the article's selection. Shared matrix foundations are proved once in Chapter 2.

Chapter 1 is the scope and implication map. Chapters 3 and 4 develop the comparison-energy and inward-model mechanisms. For the global family questions, read the centered applications in Chapters 4 and 6, followed by the noncompact homogeneous theory in Chapter 7. For moving geometry, read the affine-critical preparation in Chapter 8 and the complete planar proof in Chapter 9. Chapters 10 and 11 retain the distinct flat-contour and fixed-normal extensions; neither is absorbed by the planar strict-contour argument. Chapter 5 tests the limits of the certificate itself. Chapters 12 and 13 keep the two local negative regimes separate. Appendix A spells out the finite rational certificate common to their seed construction. Chapter 14 and Appendix B distinguish mathematical theorems, exact formal contracts, conditional results and execution evidence.

Original source snapshots, corrections and sealed audits are preserved. Appendix B records source versions and complete theorem-level correspondence. Definitions, assumptions, questions and historical execution records are distinguished from proved conclusions.

Attribution and research assistance. The literature record distinguishes known special cases, standard tools, related results and unresolved priority. The Euclidean-norm result of Guo–Lewis and the two-piece affine escape theorem of Xie–Wächter are prior work, not new discoveries here [10, 1]. Exact access levels, versions and search limits are in `editorial/LITERATURE_LEDGER.json` and the narrative revision's primary-source comparison ledger; a metadata check does not constitute reading a proof. Nothing in a Lean audit certifies novelty.

OpenAI ChatGPT/Codex systems and, from October 2026, Anthropic Claude (in OpenCode) assisted mathematical development and correction, proof reconstruction, Lean formalization, audit software, literature search and manuscript preparation. This is substantive research assistance. The editorial reviews recorded here are automated author-side reviews, not independent human peer review. Human authors must approve the mathematics, citations, responsibility and AI-use statements, authorship metadata, and any submission or deposit. No such approval is asserted in this edition.

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CHAPTER 1

Scope, quantifiers and the result map

The scope map separates objective convergence, convergence to the solution set and convergence of the full sequence. It also separates universal admissible-run statements from the particular deterministic executions used in the counterexamples.

Corrected Lean verification edition. This inventory states the operative mathematical scope. The original source record is preserved separately; unsupported historical execution and archive assertions are not repeated as facts.

Status and reading convention

Every retained mathematical assertion is classified in the claim/proof index, and each asserted result is paired there with its formal statement and Lean proof. Named results, intermediate identities, substantive unnamed deductions and actual objective applications are included. Definitions, assumptions, explicit open questions and process evidence are separately classified. Some informal proof routes have been replaced by different formal proofs; these changes are disclosed rather than described as literal translations.

The chapters use ordinary chapter numbering and shared algebra in Chapter 2. The original chapter sources remain at `BFGS_MASTER_REPORT_LaTeX/report/chapters/`; the corrected sources, edit ledger and complete correspondence index are delivered together. No unavailable provenance archive is required to reproduce the formal proofs.

Lean checks proofs under its standard foundational axioms. The audit records transitive dependencies and excludes custom mathematical axioms and unchecked proof mechanisms. This is not a proof of Lean's consistency, a human-referee certification or a publication-priority claim. Background bibliographic pointers are separate from formal proof dependencies.

1.1 Algorithm, quantifiers, and notions of convergence

The algorithm is **ordinary full BFGS** in finite-dimensional Euclidean space:

$$g_k = \nabla F(x_k), \qquad s_k = -\alpha_k H_k g_k, \qquad y_k = g_{k+1} - g_k, \qquad H_0 \succ 0,$$

$$H_{k+1} = \left(I - \frac{s_k y_k^T}{y_k^T s_k}\right) H_k \left(I - \frac{y_k s_k^T}{y_k^T s_k}\right) + \frac{s_k s_k^T}{y_k^T s_k}.$$

An *infinite admissible run* has nonzero gradients at differentiable accepted iterates, positive lengths, and fixed Armijo–weak-Wolfe constants $0 < c_1 < c_2 < 1$:

$$F(x_{k+1}) \le F(x_k) + c_1 g_k^T s_k, \qquad g_{k+1}^T s_k \ge c_2 g_k^T s_k.$$

No damping, restart, skipped update, memory truncation, proximal operation, imposed spectral limit, or assumed iterate convergence is introduced. Positive theorems quantify over all such infinite runs unless a more specific implementation is stated.

Three conclusions must remain distinct:

$$F(x_k) \to F_*; \qquad \operatorname{dist}(x_k, X_*) \to 0; \qquad x_k \to x_*.$$

On a compact sublevel, objective convergence implies distance to the solution set tends to zero. Uniqueness then gives whole-sequence convergence. Affine-fiber nonuniqueness has a separate lifting theorem; arbitrary nonunique solution sets do not inherit that conclusion.

A zero gradient at a differentiable point is successful termination. A ray with no differentiable acceptable point is not an infinite admissible run. The positive statements must not silently be strengthened into universal all-ray existence or finite oracle complexity.

1.2 The negative results that must both be retained

1.2.1 N1. Deterministic local failure with globally $C^{2,1}$ convex branches

There is one fixed coercive $F : \mathbb{R}^2 \to \mathbb{R}$, with a unique nonsmooth minimizer at zero, of the form

$$F = \max_{0 \le i \le 3} f_i, \qquad f_i \in C^{2,1}(\mathbb{R}^2), \qquad 0 \preceq \nabla^2 f_i \preceq 64I.$$

Every branch Hessian is globally Lipschitz. With the fixed initialization $H_0 = I$, a sequence of bad starts approaches the minimizer. The **actual deterministic unit-first weak-Wolfe doubling/bisection search** produces nonstationary converged limits:

$$x_0^{(j)} \to 0, \qquad x_k^{(j)} \to p_j \neq 0, \qquad F(x_k^{(j)}) \downarrow F(p_j) > 0, \qquad 0 \notin \partial F(p_j).$$

The same objective, starts, and trial decisions work for every fixed $0 < c_1 \le 1/3$, $c_1 < c_2 < 1$. Every accepted and rejected query is differentiable. Accepted lengths repeat $1, 1/4, 1/16$, with at most five trials per iteration. The entire query trajectory can remain in an arbitrarily small neighborhood of the true minimizer. Initial objective gaps also tend to zero.

This is a failure of a deterministic neighborhood guarantee, not a claim that every close start fails. It is proved by an exactly isolated algebraic cycle, exact matrix closure, a scalar convex interpolation, all-trial containment, and scaling induction. **Proof: Chapter 12.**

1.2.2 N2. Deterministic local failure with positive lower curvature

There is also a strongly convex finite objective, with a unique nonsmooth minimizer and compact sublevels, such that

$$0 < \mu I \preceq \nabla^2 F(x) \preceq LI$$

wherever its classical Hessian exists, yet the same specified deterministic implementation admits bad starts arbitrarily close to the minimizer. A single fixed affine change of coordinates makes the common initial matrix I.

One version is a maximum of four globally $C^{1,1}$, uniformly strongly convex, gradient-Lipschitz branches. Three of the four branches **are not twice differentiable at the minimizer**. Another version has one constant SPD classical Hessian wherever it exists; it is constructed from a countably piecewise-affine nonsmooth component rather than from four smooth branches.

The source fixed $c_1 = 10^{-4}$, $c_2 = 1/2$; the same objective and bad starts serve every $0 < c_1 \le 1/3$, $c_1 < c_2 < 1$. Its bad seed is an existence object on a certified hyperbolic stable manifold, not a displayed rounded point claimed to lie exactly on that manifold. **Proof: Chapter 13.**

N1 and N2 are not merged into a fictitious stronger theorem. N1 has more branch differentiability and no uniform positive lower curvature. N2 has positive lower curvature and less differentiability at the minimizer. Neither is constructed from finitely many uniformly strongly convex C^2 branches through the optimizer, so neither proves local failure for that class.

1.2.3 N3. Simple global counterexamples retained as corollaries

The deterministic seed gives failure on a coercive finite maximum of four affine functions, with a unique minimizer. This retains global failure in the finite polyhedral and semialgebraic class. It is not superseded by an accumulation-based local example.

Adding a common positive quadratic to the three-piece seed yields

$$Q_\mu(x) = \frac{\mu}{2}\|x\|^2 + \max_{i=0,1,2} g_i^T x.$$

For every $\mu > 0$, suitable initial states produce, for every fixed $0 < c_1 \le 1/3$, $c_1 < c_2 < 1$, the same deterministic nonstationary convergence to zero. Each branch is quadratic, analytic, and has Hessian exactly μI. The true minimizer lies elsewhere. This is **global failure**, and is consistent with local convergence near the true minimizer. The stable-manifold argument is required; the unperturbed orbit cannot simply be reused. **Proofs: Section 12.3.1 and Chapter 13.**

1.3 General positive mechanisms: retain distinct hypotheses

1.3.1 P1. Objective-level transported certificates

For differentiable endpoints define $g = \nabla F(x)$, $y = \nabla F(z) - g$, $s = z - x$, $d = -g^T s$, $\Delta = F(x) - F(z)$, and $t = y^T s$. An eligible pair satisfies

$$d > 0, \qquad c_1 d \le \Delta \le d, \qquad t \ge (1 - c_2)d.$$

On each compact positive objective band, it suffices that a finite-valued PSD field W satisfy

$$g^T W(x) g \ge m > 0, \qquad y^T W(z) y \le Ct,$$

$$\Pi_s W(z) \Pi_s \preceq \Pi_s W(x) \Pi_s,$$

for every eligible pair with sufficiently small d. The latter two constants and the field may depend on the band and on the fixed line-search parameters. No continuity, uniform boundedness, or positive definiteness of W is required.

Then every admissible run on the compact sublevel has value and solution-set convergence; uniqueness gives iterate convergence. A local structural neighborhood gives a trapped low-sublevel basin. A condition on one-sided shells around possible positive limiting levels suffices as well.

The normalization with exact projected nonexpansion is equivalent to allowing $(1 + Ld)$: multiply W by $\exp((L/c_1)(F - F_*))$. **Proof: Chapter 3; normalization and geometry: Chapter 5.**

RS implies the radial specialization $W(x) = (x - x_*)(x - x_*)^T$. Its equivalent inward Taylor condition is useful for checking functions, but is not a separate strongest certificate theorem. The inclusion of RS is strict. The transported condition itself is not necessary for actual convergence.

1.3.2 P2. Moving frames with controlled cumulative transport

A frame satisfying $V_{k+1} = V_k T_k + s_k b_k^T$ has an exact BFGS comparison-energy identity. Uniform visibility, bounded secant correction, and bounded cumulative transport products and inverses exclude positive-level stagnation along runs with bounded gradients. This trajectory mechanism need not produce a single position-only certificate. **Proof: Chapter 2.**

1.3.3 P3. Imperfect transport in fixed smooth directions

If a fixed subspace S satisfies $\|P_S y_k\|^2 \le L_S y_k^T s_k$, the actual update gives $\operatorname{tr}(P_S B_k) = O(k)$ for $B_k = H_k^{-1}$. Frame residuals are then permitted:

$$V_{k+1} = V_k T_k + s_k b_k^T + R_k,$$

$$\|T_k\|^2 \le 1 + Cd_k, \qquad \operatorname{range} R_k \subset S, \qquad \|R_k\|_F^2 \le Cd_k.$$

Together with uniform visibility and bounded secant correction, along runs with bounded gradients and $\sum d_k < \infty$, the resulting $O(N^2)$ error budget is defeated by the superquadratic reciprocal theorem below. This extends the proof mechanism; **a strict inclusion between the existential function classes of P1 and P3 has not been proved**. **Proof: Chapter 2; objective applications: Chapter 11.**

1.4 Strongest concrete local and global positive families

1.4.1 P4. Radially regular and finite-partition geometry

On a compact positive band, RS is equivalent to uniformly quadratic inward first-order modeling error:

$$0 \le F(x - \theta r) - F(x) + \theta \nabla F(x)^T r \le (M_\eta/2)\theta^2, \qquad r = x - x_*.$$

The constants may deteriorate as the band approaches the minimum. This yields local convergence for a finite convex function whose neighborhood of a unique minimizer has a **finite polyhedral partition with $C^{1,1}$ extensions on each cell**. Individual cell extensions need not be convex outside their cells; the full objective is convex.

Consequences include finite convex PLQ objectives on polyhedral cells; smooth convex plus finite polyhedral terms; appropriate absolute-value, hinge, quantile, fused and generalized ℓ_1 terms; and local mixtures of Euclidean group residuals. A finite maximum of arbitrary quadratics is not automatically PLQ.

Centered local representations

$$F(x_* + z) = \psi(z) + \gamma(z)$$

are covered when ψ is convex $C^{1,1}_{\mathrm{loc}}$ and γ is finite convex degree-one homogeneous. A global representation on a compact initial sublevel gives global value convergence; uniqueness gives iterate convergence. Neither original summand must separately be minimized at the center.

Also retained are finite sums of powered nonnegative gauges, and increasing convex smooth transforms of coercive gauges. Different powers, including $1 < p < 2$, and coupled linear maps are permitted under the stated source hypotheses. General spectral regularization at a nonzero rank-deficient solution is not automatic. **Proofs: Chapters 4 and 6.**

For smooth-plus-polyhedral objectives, retain the explicit sufficient basin threshold obtained by minimizing the objective over the active regions of pieces inactive at the optimizer. For nonunique affine fibers, retain the exact linear-image lifting formula. These are not replaced by a generic uniqueness statement.

1.4.2 P5. Planar affine-critical cluster points

Let $F : \mathbb{R}^2 \to \mathbb{R}$ be finite convex, have an attained minimum and a compact sublevel above the minimum, and have strictly convex positive sublevels. Away from affine-critical points, assume either local convex C^2 smoothness or a local maximum of two convex C^2 branches whose active gradients are linearly independent.

Then every infinite admissible run has an affine-critical cluster point. Adding no false affine criticality on the local basin gives convergence. For finite smooth maxima, affine independence

and strictly positive optimal barycentric weights at a unique optimizer make the required local geometry automatic in the plane. Strictly convex low contours then suffice; uniformly strongly convex branches are a stronger convenient assumption. **Complete proof: Chapter 9.**

This does not cover every planar finite maximum: rank-degenerate ties and flat positive contours are outside the stated theorem. Nor does it prove the arbitrary-dimensional rotating-normal conjecture.

1.4.3 An explicit supersession established in this consolidation

The older theorem for

$$F(u,v) = \max\{a(u) + pv, b(u) - qv\}, \qquad p, q > 0,$$

with convex C^2 profiles, coercive $\phi = (qa + pb)/(p + q)$ having unique minimizer u_*, and $a'', b'' > 0$ off u_*, is a **corollary of P5**, not an independent strongest theorem.

Here is the implication. The profiles are strictly convex. If two different points of a positive sublevel have different horizontal coordinates, strict profile convexity puts the interior of their segment strictly below the level. If their horizontal coordinates agree, their vertical coordinates differ; the two nonzero normal slopes again make every strict convex combination satisfy both inequalities strictly. Thus every positive sublevel is strictly convex. At a tie, the determinant of the active gradients is $-qa'(u) - pb'(u) = -(p + q)\phi'(u)$. It is nonzero except at u_*, where the tied point is precisely the optimizer. Smooth points are already regular. Hence P5 applies globally, and its localization argument gives the old local result.

This is a disclosed deduction between the stated theorems. The operative implication and its formal proof are included here and in the Lean index; no unavailable independent archive is needed.

1.4.4 P6. Arbitrarily many parabolic branches, including flat contours

For

$$F(u,v) = \max_{i \le m}\{a_i u^2 + b_i v\}, \qquad a_i > 0, \quad \min_i b_i < 0 < \max_i b_i,$$

all admissible full-BFGS runs converge globally to zero. Branches with $b_i = 0$ are allowed. Such branches can create flat contour segments, so P5 does **not** subsume the whole family. Its explicit certificate also proves strict inclusion beyond eligible-pair RS. **Full proof: Chapter 10.**

1.4.5 P7. Arbitrary-dimensional fixed-normal geometry

Let $P = \{x : a^T x = b\}$ be an exceptional hyperplane through a unique minimizer, with $S = a^\perp$. On a neighborhood of a compact initial sublevel, assume strong convexity, a single-valued Lipschitz tangential subgradient projection $P_S \partial F(x) = \{A(x)\}$, and local $C^{1,1}$ regularity off P. Every admissible run converges.

The full normal gradient need not have a uniform Lipschitz constant near the hyperplane. A local version uses a low trapped sublevel. For $F(u,v) = \psi(u,v) + h(v)$, full strong convexity can be weakened to tangential strong convexity of ψ, strict positive contours, compactness, and a unique optimizer with $v_* = 0$.

A separate cluster theorem permits a closed collection of **parallel genuinely nonsmooth interfaces**. It forces one affine-critical cluster point under strong convexity and a Lipschitz tangential projection. Add no false affine criticality to conclude optimality. The single-hyperplane theorem is not a literal specialization: it allows differentiable but non-$C^{1,1}$ exceptional points.

The arbitrary-dimensional result is not a planar reduction or a block-diagonal algorithm. Its real restriction is the fixed normal direction. The explicit normal-curvature-spike examples lie outside RS. **Complete proofs and infinite-series oracle: Chapter 11.**

1.4.6 P8. Scalar convergence

In one dimension, finite convex objectives with a unique minimizer and compact initial sublevel have convergence of every infinite admissible positive-metric Armijo–weak-Wolfe run, independently of the BFGS update. This remains nonredundant because it covers functions lacking every local transported certificate. **Proof and nonnecessity example: Chapter 5.**

1.5 Noncompact homogeneous theory remains independent

For every finite convex degree-one homogeneous objective, every infinite admissible run satisfies

$$F(x_k) \to 0 \quad \text{or} \quad F(x_k) \to -\infty.$$

No iterate, step, matrix, or length bound is assumed, and the supporting gradient set may be infinite. Bounded-below objectives therefore have optimal value convergence; coercive ones have $x_k \to 0$.

For finite max-linear objectives, retain three refinements: a finite-value-limit run has zero in the affine hull of its recurrent gradients; recurrent gradient span of dimension at most two gives whole-iterate convergence on the finite-value branch; and purely flat-direction noncompactness admits an explicit metric-projection limit. General affine-fiber lifting gives the same type of result for reduced nonhomogeneous objectives.

The noncompact zero-value iterate branch in recurrent dimension at least three is **not completely classified**. A compact-level certificate theorem cannot replace these noncompact statements. **Full proofs: Chapter 7.**

The two-piece finite-escape theorem from Xie–Wächter is pre-existing background and remains stronger in its finite-termination conclusion than the new many-piece infinite-run escape statement. That paper [1] is prior work, not a new result of this report.

1.6 Shared algebra and quantitative record

The principal-minor inequality is proved once in Chapter 2:

$$e_j(H_{k+1}) \le \frac{\|y_k\|^2}{\beta_k^2 q_k} e_{j+1}(H_k) + \frac{\alpha_k}{\beta_k} e_j(H_k), \quad q_k = g_k^T H_k g_k, \quad \beta_k = \frac{y_k^T s_k}{d_k}.$$

With bounded gradients, fixed weak Wolfe and $d_k \to 0$, both the geometric and arithmetic means of q_k^{-1} diverge. Under the stronger assumption $\sum d_k < \infty$, the final recorded strengthening is

$$\boxed{\frac{1}{N^2} \sum_{k<N} q_k^{-1} \to \infty.}$$

Retain the explicit finite-horizon bound as well; an asymptotic statement does not replace its finite constants. The former $\liminf q_k = 0$ and $\Omega(N^2)$ headlines are retired as main results. They remain consequences used by older proofs.

For a fixed reference point x_* and $E_k = (x_k - x_*)^T B_k (x_k - x_*)$, the fixed-reference identity is

$$E_{k+1} = E_k - \frac{[g_k^T (x_k - x_*)]^2}{q_k} + \frac{[y_k^T (x_{k+1} - x_*)]^2}{y_k^T s_k}.$$

Along runs with bounded gradients, a cumulative correction $o(\sum q_k^{-1})$ excludes a positive gap. In particular, by the final reciprocal result, an $O(N^2)$ cumulative correction suffices. This last

sentence is an explicit consolidation consequence of the two displayed inequalities, not a claim of an additional source theorem.

Big-O versions of the following rates appeared in the source notes. Theorem 2.3 strengthens them to $o(N^{-1/2})$ for uniformly bounded radial correction along runs with bounded gradients, $o(N^{-1})$ for bounded-below degree-one homogeneous objectives, and $o(\log N/N)$ for positive-definite quadratic plus coercive gauge. These are disclosed consolidation deductions, proved in Chapter 6 while preserving the stronger explicit finite-horizon inequalities. They are not uniform rates over arbitrary certificates, sharpness statements, or line-search evaluation bounds.

1.7 Geometric findings retained independently of convergence

The condition

$$0 \in \operatorname{aff} \partial F(p) \Longrightarrow 0 \in \partial F(p)$$

means that every nonoptimal nearby point admits a direction along which all subgradients predict the same strictly negative rate. Affine independence and strictly positive optimal coefficients make this true locally for finite smooth maxima, with uniform positive-band affine separation. Pointwise separation alone need not be uniform when ranks change.

For finite convex semialgebraic objectives, the affine-critical set has finitely many connected components, and the objective is constant on each. Thus only finitely many affine-critical values occur, giving no false affine criticality below a sufficiently low threshold. This is a geometric proposition, not a blanket BFGS theorem. **Proofs: Chapter 8.**

The incoming-jump obstruction rules out all PSD certificates when the target gradient is a vanishing-cost linear combination of several eligible jumps arriving at the same target. It neither assumes regularity of the certificate nor constructs one common BFGS history. Scalar accumulating-kink examples show certificate failure despite convergence. These distinctions are retained in Chapter 5.

1.8 Verification and unresolved boundaries

The delivered verification records contain actual pinned Lean builds, exact theorem contracts, module-wide declaration inventories and transitive axiom checks. Chapter 14 identifies the process logs and reproduction commands. Earlier claims about unprovided scripts, old execution counts, precisions and timings remain unsupported historical text in the original snapshot.

The exact deterministic cycle, its smooth embedding, the actual return-map derivative and stable manifold, principal determinants, frame identities, geometric joining and infinite tail arguments have Lean proofs. Their logical roles are distinct. A rounded initial point is not asserted to belong to the stable manifold, and a finite numerical experiment is not used as a proof of an infinite trajectory.

The remaining central gaps are: arbitrary-dimensional rotating-normal smooth-max geometry; the general local theorem for finite uniformly strongly convex C^2 branches without the recorded nondegeneracy/geometric assumptions; generic or positive-measure failure; other fixed line-search implementations; arbitrary nonunique solution-set iterate convergence; noncompact high-rank homogeneous zero-value iterates; and sharp rates or universal matrix asymptotics.

The blanket question is not answered by calling all semialgebraic functions safe: global failure already occurs for affine maxima. The planar and fixed-normal positive theorems have precise additional assumptions. The latest lower-curvature counterexamples likewise do not establish local failure in a finite analytic or semialgebraic description.

1.9 Supersession policy

Earlier item	Operative treatment
Prescribed acceptable-length counterexamples	Replaced by the final deterministic-bracketing seed and all-trial embeddings.
Alternative deterministic seeds	Do not mix constants. Only the final $(1,0),(-1,-1/8),(c,d)$ seed is canonical.
Upper-regularity local failure versus lower-curvature local failure	Retain both: differentiability and lower-curvature assumptions are incomparable.
Original scalar-ridge convergence headline	Corollary of the later planar theorem; the implication and its Lean proof are operative.
All parabolic multi-branch convergence	Retain: zero-normal-slope pieces can violate the planar strict-contour hypothesis.
Separate ℓ_1 and $u^2+\lvert v\rvert$ headlines	Examples of broad radial/partition families; retain special rate and matrix-limit facts as corollaries.
RS as the broadest sufficient condition	Replaced by the transported criterion; retain its natural equivalent and verifiable subclasses.
Exact versus smooth-subspace-imperfect frame transport	Retain the distinct hypotheses; no unproved strict inclusion of function classes is asserted.
Earlier reciprocal lower bounds	Use the superquadratic conclusion under summability, but retain the weaker-assumption theorem and finite-horizon constants.
Noncompact homogeneous theorems	Retain independently of compact-level results.
Certificate counterexamples	Keep the general obstruction and strongest nonnecessity example; do not label these as bad BFGS trajectories.

The scope of the positive and negative programs is now substantially clearer, but neither gives a necessary-and-sufficient classification of nonsmooth full-BFGS convergence.

Relation to earlier work

The following comparisons are bibliographic scholarship, separate from the formal theorem assertions.

The bibliographic notes distinguish established results and proof ingredients from the particular full-BFGS statements in this chapter. They record the primary-literature comparison through 11 September 2026, rather than certify publication priority. Such historical and comparative notes are separate from the mathematical assertions paired with Lean proofs. The detailed source/version register is delivered in `bfgs_lean_stage1/literature_audit/`; internal topic labels are not independently published sources.

The positive families of Section 1.4 include established special cases: Guo–Lewis proves Euclidean-norm convergence in arbitrary finite dimension [10, Corollary 4.2]. The broader applications retain their own hypotheses and proofs. The existence of a known special case does not identify the complete family theorem with that earlier result.

CHAPTER 2

BFGS algebra and reciprocal budgets

The same matrix calculation supports every positive result in this volume. We establish it once for actual full BFGS matrices, then retain the distinct qualitative, finite-horizon and summable-budget consequences.

2.1 Notation, elementary consequences, and localization

This chapter consolidates the common algebra of the source notes. No outside BFGS convergence theorem is substituted for this algebra. All results inherit the infinite-admissible-run convention in the inventory.

Put

$$B_k = H_k^{-1}, \qquad q_k = g_k^T H_k g_k > 0, \qquad d_k = -g_k^T s_k = \alpha_k q_k,$$
$$t_k = y_k^T s_k, \qquad \Delta_k = F(x_k) - F(x_{k+1}), \qquad c = 1 - c_2 > 0.$$

Convexity, Armijo and weak Wolfe give

$$c_1 d_k \le \Delta_k \le d_k, \qquad t_k \ge c d_k > 0. \tag{2.1}$$

In particular $\sum d_k < \infty$ on a run with finite limiting value. The Hessian form is

$$B_{k+1} = B_k - \frac{g_k g_k^T}{q_k} + \frac{y_k y_k^T}{t_k}. \tag{2.2}$$

Metric Cauchy–Schwarz proves

$$M_k := B_k - \frac{g_k g_k^T}{q_k} \succeq 0, \qquad M_k s_k = 0. \tag{2.3}$$

For example $(g_k^T z)^2 \le q_k z^T B_k z$, and $B_k s_k = -\alpha_k g_k$ proves the second identity.

A finite convex function has uniformly bounded subgradients on any compact set K. Indeed, choose $\epsilon > 0$ and bound its continuous values on $K + \epsilon\overline{B}$. The supporting inequality at x evaluated at $x + \epsilon u$ bounds $g^T u$ for every unit u, hence $\|g\|$. This does not require differentiability throughout K.

Lemma 2.1 (A low sublevel is an invariant local basin)**.** *Let F be finite convex and have a unique minimizer x_*. In positive dimension, choose $r > 0$ so that $\overline{B}_r(x_*)$ lies in the specified structural neighborhood. The sphere is nonempty, and*

$$\omega_r = \min_{\|z\|=r} \{F(x_* + z) - F(x_*)\} > 0,$$

and $F(x_0) - F(x_*) < \omega_r$ places the entire initial sublevel in $B_r(x_*)$.

Proof. For $\|x - x_*\| \ge r$, convexity along the radial segment gives

$$F(x) - F(x_*) \ge \frac{\|x - x_*\|}{r}\omega_r \ge \omega_r.$$

Continuity, compactness of the sphere, and uniqueness give $\omega_r > 0$. Armijo keeps every accepted iterate in the initial sublevel. No bound on H_0 is needed, and rejected trials need not remain in the neighborhood. This argument also proves boundedness of every sublevel for a finite convex function with an attained unique minimizer. In dimension zero the space is a singleton, so the trapping and compact-sublevel conclusions hold directly; no minimum over an empty positive-radius sphere is used. □

Lemma 2.2 (Strict contours give vanishing steps on a positive limiting level)**.** *Suppose the run is contained in a compact sublevel, $F(x_k) \downarrow \ell > F_*$, and the boundary of $\{F \le \ell\}$ contains no nontrivial line segment. Then $s_k \to 0$.*

Proof. Armijo and compactness give $d_k \to 0$. If the accepted steps do not vanish, compactness supplies consecutive-pair limits $x \neq z$ with $F(x) = F(z) = \ell$. Convexity bounds values on each accepted segment above by $F(x_k)$, while the supporting inequality bounds them below by $F(x_k) - d_k$. Passing to the limit proves $F = \ell$ on $[x, z]$. A point with value below ℓ shows that this segment lies in the frontier of $\{F \le \ell\}$, contradicting the strict-contour hypothesis. This is the formal proof route; it does not require choosing a limiting gradient. □

2.2 The principal-minor inequality: complete algebra

For SPD H, let $e_j(H)$ be the elementary symmetric polynomial of degree j of its eigenvalues, with $e_{n+1} = 0$. Thus $e_1 = \mathrm{tr}$ and $e_n = \det$.

Theorem 2.1 (Principal-minor hierarchy)**.** *For $H \succ 0$ and an ordinary inverse-BFGS update with $s = -\alpha Hg$, $\alpha > 0$, $q = g^T Hg > 0$ and $\beta = -y^T Hg/q > 0$,*

$$e_j(H_+) \le \frac{\|y\|^2}{\beta^2 q} e_{j+1}(H) + \frac{\alpha}{\beta} e_j(H), \qquad 1 \le j \le n. \tag{2.4}$$

Proof. Set $v = Hg$, $\kappa = g^T Hy = -\beta q$, $P = I - vy^T/\kappa$ and $\mathcal{F} = PHP^T$. The update is

$$H_+ = \mathcal{F} + \frac{\alpha}{\beta q} vv^T.$$

Let Z have j orthonormal columns, and put

$$A = Z^T HZ,\ a = Z^T Hg,\ b = Z^T Hy,\ r = y^T Hy, \quad u = a^T A^{-1} a, \quad w = a^T A^{-1} b, \quad z = b^T A^{-1} b.$$

The compression is $Z^T \mathcal{F} Z = A - (ab^T + ba^T)/\kappa + (r/\kappa^2) aa^T$. A rank-two determinant expansion gives

$$\det(Z^T \mathcal{F} Z) = \frac{\det A}{\kappa^2}\big((\kappa - w)^2 + u(r - z)\big).$$

If Gram_H denotes the Gram matrix in the H inner product, Schur complements give

$$\det \mathrm{Gram}_H(y, Z) = (r - z) \det A,$$

$$\det \mathrm{Gram}_H(g, y, Z) = \big((q - u)(r - z) - (\kappa - w)^2\big) \det A.$$

Consequently

$$\det(Z^T \mathcal{F} Z) = \frac{q \det \mathrm{Gram}_H(y, Z) - \det \mathrm{Gram}_H(g, y, Z)}{\beta^2 q^2} \le \frac{\det \mathrm{Gram}_H(y, Z)}{\beta^2 q}. \tag{2.5}$$

All Gram determinants are nonnegative. Adding the rank-one term changes the compressed determinant by exactly

$$\frac{\alpha}{\beta q} \det A\, u \le \frac{\alpha}{\beta} \det A,$$

because $u \le q$. The identities remain valid for singular larger Gram matrices or singular $Z^T \mathcal{F} Z$; only A must be invertible, which it is.

Choose an orthonormal eigenbasis of H, with eigenvalues λ_i, and sum over coordinate j-subsets I. Then

$$\sum_{|I|=j} \det \mathrm{Gram}_H(y, E_I) = \sum_{|J|=j+1} \left(\prod_{i \in J} \lambda_i\right) \sum_{i \in J} y_i^2 \le \|y\|^2 e_{j+1}(H).$$

The rank-one increases sum to at most $(\alpha/\beta) e_j(H)$. This proves (2.4). For $j = n$, the calculation gives the exact identity $\det H_+ = (\alpha/\beta) \det H$. □

2.3 Reciprocal growth: the strongest recorded versions

Theorem 2.2 (Bounded gradients and vanishing predicted decrease)**.** *If an infinite full-BFGS run has bounded gradients, fixed weak Wolfe and $d_k \to 0$, then*

$$\left(\prod_{k<N} q_k^{-1}\right)^{1/N} \to \infty, \qquad \frac{1}{N}\sum_{k<N} q_k^{-1} \to \infty.$$

Proof. Choose $G > 0$ with $\|g_k\| \le G$ and write $K_0 = 4G^2/c^2$. Since $\beta_k = -y_k^T H_k g_k/q_k = t_k/d_k \ge c$ and $\|y_k\| \le 2G$, the hierarchy gives

$$e_j(H_{k+1}) \le q_k^{-1}\{K_0 e_{j+1}(H_k) + (d_k/c)e_j(H_k)\}.$$

Fix $\eta > 0$ and a tail $k \ge J$ on which $d_k/c \le \eta$. Define

$$V_\eta(H) = \sum_{j=1}^{n} (K_0/\eta)^{j-1} e_j(H) \ge \operatorname{tr} H.$$

Multiplying and summing yields $V_\eta(H_{k+1}) \le (2\eta/q_k)V_\eta(H_k)$, while $q_k \le G^2 V_\eta(H_k)$. Iterating through the penultimate step of a block and using the last endpoint gives

$$\prod_{k=J}^{J+m-1} q_k^{-1} \ge \frac{1}{G^2 V_\eta(H_J)(2\eta)^{m-1}}.$$

Taking mth roots and then allowing the fixed η to be arbitrarily small proves divergence of the geometric mean; a fixed prefix has no effect. Arithmetic–geometric mean proves the other assertion. □

Theorem 2.3 (Finite horizon and superquadratic growth under summability)**.** *Assume bounded gradients, fixed weak Wolfe, and $\sum_k d_k < \infty$. For a fixed tail J, let $D_J = \sum_{k\ge J} d_k > 0$. For every $\eta > 0$ and $m \ge 2$, with G, K_0 and V_η as in the proof of Theorem 2.2,*

$$\sum_{k=J}^{J+m-1} q_k^{-1} \ge m[G^2 V_\eta(H_J)]^{-1/m} \left(\eta + \frac{D_J}{c(m-1)}\right)^{-(m-1)/m}. \tag{2.6}$$

In particular,

$$\frac{1}{N^2}\sum_{k<N} q_k^{-1} \longrightarrow \infty. \tag{2.7}$$

Proof. Without any tail restriction on d_k, the weighted hierarchy gives

$$V_\eta(H_{k+1}) \le \frac{\eta + d_k/c}{q_k} V_\eta(H_k).$$

Iterate for $m-1$ steps, use the endpoint bound for q_{J+m-1}, and apply arithmetic–geometric mean:

$$\prod_{k=J}^{J+m-1} q_k \le G^2 V_\eta(H_J) \prod_{k=J}^{J+m-2} (\eta + d_k/c) \le G^2 V_\eta(H_J)\left(\eta + \frac{D_J}{c(m-1)}\right)^{m-1}.$$

Arithmetic–geometric mean for the reciprocals proves (2.6). Choose $\eta = D_J/[c(m-1)]$. With J fixed, $V_\eta(H_J)$ grows at most polynomially in m, of degree $n-1$. Its mth root tends to one. Hence

$$\liminf_{N\to\infty} N^{-2}\sum_{k<N} q_k^{-1} \ge \frac{c}{2D_J}.$$

The fixed prefix does not affect the normalization. Finally let $J \to \infty$; then $D_J \to 0$. This proves (2.7). No uniform bound on H_J as J varies is used: the tail is fixed before the horizon limit. □

These conclusions do not say $H_k \to 0$, $q_k \to 0$ pointwise, or that eigenvectors converge. The finite-horizon bound and the weaker-assumption geometric-mean theorem are retained alongside the strongest summable-budget asymptotic statement.

2.4 Fixed-reference and moving-frame energy identities

For any fixed reference point x_* define $r_k = x_k - x_*$, $a_k = g_k^T r_k$, $E_k = r_k^T B_k r_k$ and

$$R_k = \frac{(y_k^T r_{k+1})^2}{t_k}.$$

Expansion of (2.2) at $r_{k+1} = r_k - \alpha_k H_k g_k$ gives exactly

$$E_{k+1} = E_k - \frac{a_k^2}{q_k} + R_k. \tag{2.8}$$

The cancellation is $-2\alpha a + \alpha^2 q - (a - \alpha q)^2/q = -a^2/q$. If x_* is a minimizer, convexity gives $a_k \geq F(x_k) - F_*$. Therefore

$$[F(x_N) - F_*]^2 \sum_{k<N} q_k^{-1} \leq E_0 + \sum_{k<N} R_k. \tag{2.9}$$

Thus, along runs with bounded gradients, $\sum_{k<N} R_k = o(\sum_{k<N} q_k^{-1})$ excludes a positive gap. With the summability assumptions of Theorem 2.3, an $O(N^2)$ cumulative correction suffices. This last observation is an explicit consolidation deduction from the recorded energy and reciprocal results; no new regularity hypothesis is inferred.

For $V_k \in \mathbb{R}^{n\times r}$, suppose

$$V_{k+1} = V_k T_k + s_k b_k^T.$$

Set $A_k = V_k^T B_k V_k$, $a_k = V_k^T g_k$, $c_k = V_{k+1}^T y_k$. By (2.3),

$$A_{k+1} = T_k^T \left(A_k - \frac{a_k a_k^T}{q_k} \right) T_k + \frac{c_k c_k^T}{t_k}. \tag{2.10}$$

Suppose on a hypothetical nonoptimal tail $\|a_k\| \geq \epsilon > 0$, $\|c_k\|^2/t_k \leq C$ and the cumulative products $P_J = I$, $P_{k+1} = P_k T_k$ have bounded norms and bounded inverse norms. Congruence by P_{k+1}^{-1} and a trace yield

$$\operatorname{tr} \widetilde{A}_{k+1} = \operatorname{tr} \widetilde{A}_k - \frac{\|P_k^{-T} a_k\|^2}{q_k} + \frac{\|P_{k+1}^{-T} c_k\|^2}{t_k}, \quad \widetilde{A}_k = P_k^{-T} A_k P_k^{-1}.$$

Positivity gives $\sum_{J\leq k<N} q_k^{-1} = O(N)$, contradicting Theorem 2.2 under its hypotheses. This is a trajectory criterion; one still has to derive its hypotheses from the objective. Neither P_k nor T_k is a BFGS matrix.

2.5 Imperfect transport with a derived smooth-subspace budget

Theorem 2.4 (Residuals in fixed smooth directions)**.** *Assume bounded gradients, fixed weak Wolfe and $\sum d_k < \infty$. For a fixed subspace S, suppose*

$$\|P_S y_k\|^2 \leq L_S t_k.$$

A tail cannot simultaneously admit frames with

$$\|V_k^T g_k\|^2 \geq m > 0, \qquad \|V_{k+1}^T y_k\|^2 \leq C_0 t_k,$$

$$V_{k+1} = V_k T_k + s_k b_k^T + R_k,$$

$$\|T_k\|_{\mathrm{op}}^2 \leq 1 + C_T d_k, \quad \operatorname{range} R_k \subset S, \quad \|R_k\|_F^2 \leq C_R d_k.$$

No full-rank frame or invertible transport is required.

Proof. Taking the trace of (2.2) on S gives

$$\operatorname{tr}(P_S B_{k+1}) = \operatorname{tr}(P_S B_k) - \frac{\|P_S g_k\|^2}{q_k} + \frac{\|P_S y_k\|^2}{t_k} \le \operatorname{tr}(P_S B_k) + L_S.$$

Consequently $\operatorname{tr}(P_S B_k) \le C_S + L_S k$. This bound is derived, not imposed.

Let $E_k = \operatorname{tr}(V_k^T B_k V_k)$ and use M_k from (2.3). Then

$$E_{k+1} = \operatorname{tr}\Big[(V_k T_k + R_k)^T M_k (V_k T_k + R_k)\Big] + \frac{\|V_{k+1}^T y_k\|^2}{t_k}.$$

Since M_k is positive semidefinite, choose a real factor C_k with $M_k = C_k^T C_k$. The trace is the sum of squares of the entries of $C_k(V_k T_k + R_k)$. Entrywise Young's inequality with parameter d_k, followed by the Euclidean operator bound on right multiplication by T_k, bounds this by

$$(1 + d_k)\|T_k\|^2 \left(E_k - \frac{\|V_k^T g_k\|^2}{q_k}\right) + (1 + d_k^{-1}) \operatorname{tr}(R_k^T M_k R_k) + C_0.$$

The parenthesized energy is nonnegative, and

$$\operatorname{tr}(R_k^T M_k R_k) \le \|R_k\|_F^2 \operatorname{tr}(P_S B_k) \le C_R d_k (C_S + L_S k).$$

On a tail with $d_k \le 1$,

$$E_{k+1} \le (1 + C_1 d_k)(E_k - m/q_k) + C_2(k+1).$$

Normalize by $P_{k+1} = P_k(1 + C_1 d_k)$ with $P_J = 1$. Finite $\sum d_k$ gives $1 \le P_k \le \bar{P} < \infty$. Summing and using final-energy positivity gives

$$\frac{m}{\bar{P}} \sum_{k=J}^{N-1} q_k^{-1} \le E_J + C_2 \sum_{k=J}^{N-1} (k+1) = O(N^2),$$

contrary to Theorem 2.3. □

Only residuals in the fixed subspace are controlled. Euclidean smallness alone does not control a residual in a direction where B_k may grow without this estimate. No strict inclusion of all existential objective-certificate classes is claimed.

Relation to earlier work

The following comparisons are bibliographic scholarship, separate from the formal theorem assertions.

The BFGS update, secant equation, positive-definiteness preservation and determinant identity are classical. Trace/determinant analysis is developed in Nocedal [14, Section 5, equations (5.7), (5.15) and (5.21)] and the earlier Byrd–Nocedal work [15]. The determinant endpoint in this chapter belongs to that established theory. The complete adjacent-degree principal-minor inequality is distinguished from it; the literature audit found no exact match, without establishing publication priority. Gram determinants, Schur complements and Cauchy–Binet are standard tools.

Product bounds and arithmetic–geometric mean already occur in BFGS convergence proofs: see Schöpfer [11, proof of Theorem 3.7] and Jin–Jiang–Mokhtari [18, Appendices C–D]. Their smooth hypotheses and normalized quantities differ from those in this chapter. Here the actual principal-minor hierarchy supplies reciprocal bounds for $q_k = g_k^T H_k g_k$, including the finite-horizon and summable-decrease conclusions; those specific conclusions were not located in the accessed literature.

The energy cancellations in this chapter are algebraic consequences of the classical Hessian update [14, Section 5]; congruence and quadratic-form inequalities are established tools. Their use in a moving-frame criterion still requires the separately stated visibility, secant and transport hypotheses.

Summable multiplicative-error estimates and normalization have precedents in Combettes–Vũ [21, Lemmas 2.2–2.3]. The argument in this chapter additionally derives the restricted trace bound and an $O(N^2)$ cumulative allowance. Its additive errors need not be summable, so the earlier lemmas alone do not prove this residual theorem.

CHAPTER 3

Transported comparison energies

The reciprocal budget becomes an objective theorem only when geometry controls a comparison energy. Visibility, secant control and transport are imposed on objective-level pairs, not inferred from a presumed limit of the BFGS matrices.

3.2 Algorithm and objective-pair terminology

Let $F : \mathbb{R}^n \to \mathbb{R}$ be finite and convex, let $F_* = \min F$ be attained, and let

$$K = \{x : F(x) \le b\}, \qquad b > F_*,$$

be compact. Ordinary full BFGS uses differentiable accepted endpoints,

$$g_k = \nabla F(x_k), \quad s_k = -\alpha_k H_k g_k, \quad x_{k+1} = x_k + s_k, \quad y_k = g_{k+1} - g_k,$$

with $H_0 \succ 0$, $\alpha_k > 0$, and

$$H_{k+1} = (I - s_k y_k^{\mathsf{T}}/t_k) H_k (I - y_k s_k^{\mathsf{T}}/t_k) + s_k s_k^{\mathsf{T}}/t_k, \qquad t_k = y_k^{\mathsf{T}} s_k.$$

The fixed Armijo–weak-Wolfe conditions are

$$F(x_{k+1}) \le F(x_k) + c_1 g_k^{\mathsf{T}} s_k, \qquad g_{k+1}^{\mathsf{T}} s_k \ge c_2 g_k^{\mathsf{T}} s_k, \qquad 0 < c_1 < c_2 < 1.$$

An infinite admissible run has nonzero gradients and well-defined accepted steps. These are the algorithmic quantifiers in Xie–Wächter [1]. No particular line-search implementation, damping, restart, or proximal operation is added.

For differentiable endpoints x, z, write

$$g = \nabla F(x), \quad y = \nabla F(z) - g, \quad s = z - x,$$

$$d = -g^{\mathsf{T}} s, \qquad \Delta = F(x) - F(z), \qquad t = y^{\mathsf{T}} s.$$

Call the pair *eligible* if

$$d > 0, \qquad c_1 d \le \Delta \le d, \qquad t \ge (1 - c_2) d. \tag{3.2}$$

Every accepted step is eligible. The upper bound $\Delta \le d$ follows from convexity. Eligibility does not assert compatibility of an arbitrary chain with BFGS; imposing a condition on all such pairs is an objective-only sufficient assumption.

For $s \neq 0$, define the Euclidean orthogonal projector

$$\Pi_s = I - \frac{s s^{\mathsf{T}}}{s^{\mathsf{T}} s}.$$

All matrices in the next assumption are objective-pair data, not H_k.

3.3 The transported quadratic secant condition

Theorem 3.1 (One-sided transported-certificate convergence)**.** *For each $\eta > 0$ with nonempty band*

$$K_\eta = \{x \in K : F(x) \ge F_* + \eta\},$$

suppose there is a finite-valued symmetric positive-semidefinite field $W_\eta(x)$ on differentiable points of K_η, and constants

$$m_\eta > 0, \quad C_\eta < \infty, \quad L_\eta \geq 0, \quad \delta_\eta > 0,$$

with these properties:

$$g^\mathsf{T} W_\eta(x) g \geq m_\eta, \tag{3.V}$$
$$y^\mathsf{T} W_\eta(z) y \leq C_\eta t, \tag{3.S}$$
$$\Pi_s W_\eta(z) \Pi_s \preceq (1 + L_\eta d) \Pi_s W_\eta(x) \Pi_s. \tag{3.T}$$

The last two inequalities need hold only for eligible pairs $x, z \in K_\eta$ with $d \leq \delta_\eta$. Then every infinite admissible full-BFGS run starting in K satisfies

$$F(x_k) \to F_*, \qquad \operatorname{dist}(x_k, \arg\min F) \to 0.$$

If the minimizer is unique, the entire iterate sequence converges to it.

Condition (V) says that the comparison directions retain a nonzero component of the gradient on a positive band. Condition (S) controls the secant correction in those directions. Condition (T) controls their expansion only in directions perpendicular to the proposed step.

There is no uniform positive-definiteness requirement on W_η. Uniform boundedness of W_η is convenient in examples but is not used by the proof; it is enough that its values are finite. Constants and the field may depend on the positive band and the fixed line-search parameters. They may degenerate as $\eta \downarrow 0$.

3.3.1 Proof

Put

$$B_k = H_k^{-1}, \qquad q_k = g_k^\mathsf{T} H_k g_k, \qquad d_k = \alpha_k q_k.$$

The ordinary Hessian update is

$$B_{k+1} = B_k - \frac{g_k g_k^\mathsf{T}}{q_k} + \frac{y_k y_k^\mathsf{T}}{t_k}. \tag{3.3}$$

Its first two terms form a PSD matrix

$$M_k = B_k - g_k g_k^\mathsf{T} / q_k \succeq 0,$$

by metric Cauchy–Schwarz. Also

$$M_k s_k = 0, \qquad M_k = \Pi_{s_k} M_k \Pi_{s_k}. \tag{3.4}$$

This is why (T) may ignore the step direction.

Armijo implies $\sum d_k < \infty$. Gradients are bounded on K: a finite convex function has uniformly bounded subgradients on a compact set, as follows by comparing its values at x and $x + \epsilon u$ on a slightly enlarged compact set. Thus the shared foundation in Chapter 2 applies and gives

$$\frac{1}{N} \sum_{k<N} \frac{1}{q_k} \longrightarrow \infty. \tag{3.5}$$

Suppose the objective gap tends to $\ell > 0$. Use the field for $\eta = \ell/2$. On a tail $k \geq k_0$, all pairs lie in the band and $d_k \leq \delta_\eta$. Suppress η and set

$$E_k = \operatorname{tr}(B_k W(x_k)) \geq 0, \qquad a_k = 1 + L d_k.$$

Using (3.3)–(3.4), the Loewner inequality (T), and (S),

$$\begin{aligned} E_{k+1} &= \operatorname{tr}(M_k W(x_{k+1})) + y_k^{\mathsf{T}} W(x_{k+1}) y_k / t_k \\ &\le a_k \operatorname{tr}(M_k W(x_k)) + C \\ &= a_k \left(E_k - \frac{g_k^{\mathsf{T}} W(x_k) g_k}{q_k} \right) + C \\ &\le a_k (E_k - m/q_k) + C. \end{aligned} \tag{3.6}$$

Let $P_{k_0} = 1$, $P_{k+1} = P_k a_k$. Since $\sum d_k < \infty$,

$$1 \le P_k \le \exp\left(L \sum_{j \ge k_0} d_j \right) =: \overline{P} < \infty.$$

Divide (3.6) by P_{k+1} and sum:

$$\frac{m}{\overline{P}} \sum_{k=k_0}^{N-1} \frac{1}{q_k} \le E_{k_0} + C(N - k_0).$$

This contradicts (3.5). The gap therefore tends to zero. Compactness gives convergence to the solution set, and uniqueness gives whole-sequence convergence. □

The formal principal theorem uses the same derived trace recurrence with a harmless larger additive bound $C_+(k+1)$, where $C_+ = \max\{C, 0\}$. The normalized energy estimate then gives an $O(N^2)$ reciprocal-sum bound and contradicts Chapter 2's superquadratic theorem. Its additional summability requirement is already a consequence of Armijo and the finite limiting value. Thus no extra hypothesis is added. The sharper linear telescoping estimate above follows from the separately proved scalar transport budget; it is not necessary to the formal principal theorem's contradiction.

3.3.2 Local statement and weaker band requirements

If the assumptions hold only in a neighborhood U of a unique minimizer and K is a compact low sublevel, choose an initial level below

$$\inf_{x \in K \setminus U} F(x) > F_*.$$

Armijo then traps all accepted iterates in U. This produces a local basin independent of H_0. Rejected trial points need not stay in U. For multiple minimizers, one may instead use a neighborhood of the entire compact solution set or directly assume that the initial sublevel lies in the structural region.

The proof only needs a certificate in a small one-sided level neighborhood of each possible finite limit $\ell > F_*$. Requiring a field on every whole positive band is a cleaner, stronger formulation; certificates on shells $\ell \le F \le \ell + \epsilon_\ell$ also suffice.

3.3.3 Scalar form and freedom to rescale

Taking $W(x) = v(x)v(x)^{\mathsf{T}}$, a sufficient version of (T) is

$$v(z) = \lambda(x,z) v(x) + \beta(x,z) s, \qquad |\lambda(x,z)|^2 \le 1 + Ld.$$

Conditions (V) and (S) become

$$|g^{\mathsf{T}} v(x)|^2 \ge m, \qquad |y^{\mathsf{T}} v(z)|^2 \le Ct.$$

Unlike the moving-frame criterion of Section 2.4, there is no requirement that λ be invertible or bounded away from zero. The PSD form also accommodates multiple comparison directions and varying rank, without choosing an exact frame transport.

A bounded potential may be absorbed into W. If (T) instead has multiplier

$$\exp\{Ld + \Psi(x) - \Psi(z)\},$$

where Ψ is bounded on the band, replace W by $e^{\Psi} W$. This gives multiplier e^{Ld}, which is at most $1 + L'd$ for d sufficiently small. The other bounds change only by band-dependent constants.

3.4 Why this is genuinely weaker than RS

Given the radial secant condition (RS) of Chapter 4, take

$$W(x) = (x - x_*)(x - x_*)^\mathsf{T}.$$

Convexity gives

$$g^\mathsf{T} W(x) g = (g^\mathsf{T}(x - x_*))^2 \geq (F(x) - F_*)^2 \geq \eta^2.$$

With this W, (S) is the inequality of (RS) restricted to eligible pairs in K_η with $d \leq \delta_\eta$, so (RS) gives (S) with the same constant, while

$$\Pi_s(z - x_*) = \Pi_s(x - x_*)$$

makes (T) an equality with $L = 0$. Thus every objective covered by RS is covered by the new condition.

The reverse implication fails. Chapter 10 constructs fields satisfying the new condition for a multi-branch curved objective that violates even the eligible-pair radial condition on every positive neighborhood of its minimizer. This is strict inclusion of sufficient conditions, not merely a new notation for the old inequality.

Although Π_s is written as a Euclidean projector, (T) equivalently requires

$$a^\mathsf{T} W(z) a \leq (1 + Ld) a^\mathsf{T} W(x) a \qquad \text{for every } a \text{ with } a^\mathsf{T} s = 0.$$

Under an invertible affine change of variables with linear part A, gradients transform by $A^{-\mathsf{T}}$ and the certificate transforms by $W \mapsto AWA^\mathsf{T}$. The three conditions are therefore affine-covariant; no preferred coordinate system is essential.

The extra transport clause cannot simply be omitted: changing a comparison vector creates terms involving B_k, which may be unbounded. A small Euclidean change is not by itself a small change in the B_k energy. Projection and the multiplicative budget are precisely what make the new hypothesis sufficient without assumptions on B_k.

Relation to earlier work

The following comparisons are bibliographic scholarship, separate from the formal theorem assertions.

This is a sufficient condition for ordinary full BFGS. Multiplicative-error normalization is an established technique [21, Lemmas 2.2–2.3], also used in the rescaling discussion in this chapter and Chapter 5. The comparison field here may be singular and unbounded, and transport is projected away from the step. Consequently a uniformly coercive, bounded variable-metric convergence theorem does not supply this result or its objective-pair quantifiers.

CHAPTER 4

Radial modeling and finite partitions

The global mixed-benchmark question is answered through inward model accuracy: a centered sublinear term contributes no inward remainder, and a smooth term contributes a quadratic one. The same condition localizes to finite polyhedral partitions without requiring cell identification.

4.1 Setup: the existing condition

Let $F : \mathbb{R}^n \to \mathbb{R}$ be finite and convex, with a minimizer x_* and minimum value F_*. Let

$$K = \{x : F(x) \le b\}, \qquad b > F_*,$$

be compact. Write $r_x = x - x_*$. Fix bounds $R > 0$ and $G > 0$ such that

$$\|r_x\| \le R, \qquad \|g\| \le G \quad (x \in K,\ g \in \partial F(x)).$$

Such a subgradient bound exists because a finite convex function is locally Lipschitz on the whole space.

For differentiable $x, z \in K$, put

$$g = \nabla F(x), \qquad h = \nabla F(z), \qquad y = h - g, \qquad s = z - x,$$
$$T(x,z) = y^T s, \qquad N(x,z) = y^T r_z.$$

The supporting-plane error, or Bregman expression, is

$$D_F(z,x) = F(z) - F(x) - g^T(z - x) \ge 0.$$

Unlike the strictly convex smooth setting of many standard presentations [9], only differentiability of the second endpoint is needed to define this expression here. We have

$$T = D_F(z,x) + D_F(x,z), \qquad 0 \le T \le T_{\max} := 4GR.$$

The radial secant condition is the following positive-band condition:

$$\forall \eta > 0\ \exists C_\eta \ge 0 : \quad N(x,z)^2 \le C_\eta T(x,z) \quad \text{if } F_* + \eta \le F(z) \le F(x),\ x, z \in K,\ T > 0. \tag{4.RS}$$

Endpoints in this condition are differentiable. Constants may diverge as $\eta \downarrow 0$. The hypothesis is imposed only on nonempty bands.

The algorithm remains ordinary full BFGS with arbitrary $H_0 \succ 0$, differentiable accepted iterates, and positive steps satisfying fixed Armijo–weak-Wolfe parameters $0 < c_1 < c_2 < 1$, as in [1]. "Every run" below always means every infinite admissible run; a zero gradient is successful termination. General all-ray existence of differentiable acceptable points is not asserted.

4.2 Equivalent formulation: smooth only toward the minimizer

Theorem 4.2.1 (Inward radial smoothness is equivalent to (RS))**.** *On the compact sublevel K, condition (RS) is equivalent, up to constants, to the following hypothesis. For every $\eta > 0$, there exist $M_\eta \ge 0$ and $\theta_\eta \in (0, 1/2]$ such that, at every differentiable $x \in K$ with $F(x) \ge F_* + \eta$,*

$$0 \le F(x - tr_x) - F(x) + t\nabla F(x)^T r_x \le \frac{M_\eta}{2} t^2, \qquad 0 \le t \le \theta_\eta. \tag{4.IR}$$

No differentiability of $x - tr_x$ is required. The sampled point is a convex combination of x and x_*, so it remains in K.

The label "inward radial smoothness" is descriptive terminology in this chapter, not a claim of a standardized definition or of novelty for the terminology. The parameter t is a fraction of the distance to the minimizer. A factor $\|r_x\|^2$ can alternatively be placed in the upper bound: on a positive objective band, the two conventions are equivalent because $\eta/G \le \|r_x\| \le R$.

4.2.1 Proof that (IR) implies (RS)

Fix a positive band and write M, θ for its constants. Consider endpoints x, z on that band. If $N \ge 0$, evaluate the supporting plane at x at the inward test point $z - tr_z$. Convexity and (IR) at z give

$$tN \le D_F(z, x) + \frac{M}{2}t^2, \qquad 0 \le t \le \theta.$$

If $N < 0$, use the inward test point $x - tr_x$ and the supporting plane at z. Since

$$(g - h)^T r_x = T - N \ge |N|,$$

we instead obtain

$$t(T - N) \le D_F(x, z) + \frac{M}{2}t^2.$$

In either case there is a number $A \in [|N|, 2GR]$ and a supporting error $D \in \{D_F(z, x), D_F(x, z)\}$ such that

$$tA \le D + \frac{M}{2}t^2 \quad (0 \le t \le \theta).$$

When $M > 0$ and $A \le M\theta$, take $t = A/M$ to get $A^2 \le 2MD$. Otherwise take $t = \theta$; if $A > M\theta$, then $D \ge A\theta/2$, so

$$A^2 \le \frac{4GR}{\theta}D.$$

For $M = 0$, choosing $t = \theta$ gives an even stronger bound. Thus

$$N^2 \le CT, \qquad C = \max\left\{2M, \frac{4GR}{\theta}\right\}. \tag{4.2}$$

This proof also shows why controlling only the direction *toward* the minimizer is enough: the sign of N determines which endpoint supplies the inward probe. No outward point is needed.

4.2.2 Proof that (RS) implies (IR)

Fix $\eta > 0$ and choose

$$\theta_\eta = \min\left\{\frac{1}{2}, \frac{\eta}{4GR}\right\}.$$

For x on the η-band and $z = x - tr_x$, the Lipschitz bound on K gives

$$F(z) \ge F(x) - GtR \ge F_* + 3\eta/4.$$

First suppose both endpoints are differentiable. Put

$$a = (\nabla F(x) - \nabla F(z))^T r_x \ge 0.$$

Then

$$T(x, z) = ta, \qquad N(x, z) = -(1 - t)a.$$

Using (RS) on the $\eta/2$-band, if $a > 0$,

$$(1 - t)^2 a^2 \le C_{\eta/2} ta, \qquad a \le \frac{C_{\eta/2}t}{(1 - t)^2}.$$

Convexity supplies

$$0 \le D_F(z, x) \le ta \le 4C_{\eta/2}t^2.$$

If $a = 0$, the same conclusion follows from $D_F(z, x) \le ta = 0$. Thus (IR) holds with $M_\eta = 8C_{\eta/2}$.

For completeness, the inner endpoint may be nondifferentiable. For fixed $t < 1$, approximate x by points x_j in the interior of K such that both x_j and $x_j - t(x_j - x_*)$ are differentiable. This is possible because the differentiability set has full measure and the affine dilation is invertible. The positive-band inequalities persist, using the spare $\eta/4$ margin above. Convexity and differentiability at x imply $\nabla F(x_j) \to \nabla F(x)$. Passing to the limit in the function-value inequality proves (IR) without differentiability of the inner endpoint. Boundary points of K are approximated from its interior. The case $t = 0$ is trivial. □

4.2.3 Interpretation and a derivative-free version

For a unit direction d, inspect only the scalar function

$$\varphi_d(\rho) = F(x_* + \rho d), \qquad \rho > 0.$$

On compact positive objective bands, the equivalent conditions control the change of its scalar derivative uniformly over directions. Angular derivatives may jump. In a smooth region, the infinitesimal quantity being bounded is

$$r_x^T \nabla^2 F(x) r_x,$$

not the largest Hessian eigenvalue. For a nonsmooth function, checking this expression almost everywhere is *not enough*: a radial gradient jump contributes singular curvature and must also be excluded.

On slightly nested positive bands where both dilations remain in K, an equivalent function-value formulation is a uniform bound

$$F(x_* + (1+t)r) + F(x_* + (1-t)r) - 2F(x_* + r) \le M_\eta t^2. \tag{4.3}$$

Indeed, this bound directly bounds each nonnegative tangent-plane remainder at a differentiable middle point, yielding (IR). Conversely, apply (RS) to the outer/middle and middle/inner radial pairs. If a_+ and a_- are their scalar derivative differences in direction r, then

$$a_+ \le Ct, \qquad a_- \le Ct/(1-t)^2.$$

The central difference is at most $t(a_+ + a_-) \le 5Ct^2$ for $t \le 1/2$. Approximation by differentiable triples extends the function-value inequality to all centers. This is a local equivalence with constants allowed to change; an outward point need not remain in the original top boundary of K.

4.3 A second interpretation: what tangent planes predict at the optimum

Define, at a differentiable point,

$$b_F(x) = F_* - \big[F(x) + \nabla F(x)^T (x_* - x)\big] = \nabla F(x)^T r_x - (F(x) - F_*) \ge 0.$$

This is the amount by which the supporting plane at x underestimates the minimum when extrapolated to x_*. It is also the supporting error $D_F(x_*, x)$.

A direct calculation gives

$$N(x, z) = D_F(z, x) + b_F(z) - b_F(x). \tag{4.5}$$

Since $0 \le D_F(z, x) \le T \le T_{\max}$, the radial condition is equivalent on the same bounded domains to

$$|b_F(z) - b_F(x)|^2 \le C' T(x, z). \tag{4.SP}$$

For example, (RS) with C implies (SP) with $2C + 2T_{\max}$, and conversely. Thus the full gradient need not be Lipschitz; only the *height at the optimizer* of its supporting plane needs this curvature-relative regularity. The square root of T is not being asserted to be a metric.

For a degree-one homogeneous function minimized at zero, $b_F \equiv 0$. For

$$F(u, v) = u^2 + |v|,$$

we have $b_F(u, v) = u^2$: crossing $v = 0$ flips a gradient component but does not produce a jump in this scalar.

For a function homogeneous of degree p, Euler's identity at a differentiable point is $\nabla F(x)^T x = pF(x)$. Consequently the supporting-plane error at an arbitrary reference point x_* is

$$b_{F,x_*}(x) = (p-1)F(x) + F(x_*) - \nabla F(x)^T x_*.$$

No minimizer assumption is needed for this algebraic identity. At reference zero it reduces to $(p-1)F(x) + F(0)$; the constant term cannot be discarded without a normalization. We use homogeneity under nonnegative dilations, $F(tx) = t^p F(x)$ for $t \geq 0$, with $t^0 = 1$ also at $t = 0$. Thus degree zero forces F to be constant.

In the compact setting of this chapter, the original simplified formula is valid without assuming that the optimizer is the origin. More precisely, let $n \geq 1$, let $F : \mathbb{R}^n \to \mathbb{R}$ be finite, convex, nonnegative and homogeneous, and suppose $\{F \leq b\}$ is compact with $F(z) < b$ for some z. Then $p > 0$. Indeed, if $p \leq 0$, homogeneity at dilation two and midpoint convexity give

$$F(x) \leq \tfrac{1}{2}F(2x) + \tfrac{1}{2}F(0) = 2^{p-1}F(x) + \tfrac{1}{2}F(0),$$

whence $F(x) \leq F(0)$. Applying midpoint convexity to x and $-x$ then gives $F(x) = F(0)$ for every x, so the strict sublevel witness makes $\{F \leq b\} = \mathbb{R}^n$, a contradiction.

Positive degree gives $F(0) = 0$. Any optimizer x_* therefore has value zero. If $x_* \neq 0$, homogeneity places the unbounded ray $\{tx_* : t \geq 0\}$ inside the compact sublevel, another contradiction. Thus zero is the unique optimizer and

$$b_F(x) = (p-1)F(x).$$

The chapter's hypothesis $b > F_*$ supplies the strict witness by taking $z = x_*$. Outside this compact context, the positive constant function $F \equiv 1$ has degree zero and intercept zero, whereas $(p-1)F = -1$; its positive sublevels are noncompact. These identities explain the breadth of the sufficient classes of Section 4.6 without changing their compact-basin hypotheses.

An explicit four-piece realization. Let c, d be the exact algebraic seed parameters of Chapter 12, and let

$$F_D(u, v) = \max\{u, -u - v/8, cu + dv, v - 17\}$$

be its verified coercive four-piece objective. Define

$$\delta = c - 16d - 1 > 0, \qquad a = \frac{25}{4\delta} > 0, \qquad \mathcal{L}(u, v) = (u, 16v/5),$$

and set

$$F_R(z) = aF_D(\mathcal{L}z) + a - 1.$$

Thus F_R is the maximum of the four affine functions

$$au + a - 1, \qquad a(-u - 2v/5) + a - 1,$$
$$a(cu + 16dv/5) + a - 1, \qquad 16av/5 - 16a - 1.$$

This definition is an explicit reconstruction for this edition, not an identification of an unspecified earlier objective. Chapter 12's own optimizer remains $(-1, 16)$.

The positivity of δ follows from Chapter 12's certified parameters. Since $\mathcal{L}$ is invertible and $a > 0$, F_R is finite and convex with compact sublevels. Chapter 12's unique optimizer $(-1, 16)$ is carried to $(-1, 5)$, and the minimum value remains -1. At this optimizer the first three branches have values

$$-1, \quad -1, \quad -1 - a\delta = -29/4,$$

so their supporting errors are exactly $0, 0, 25/4$.

This objective has an actual infinite unmodified full-BFGS failure run. For any fixed parameters

$$0 < c_1 \le 1/3, \qquad c_1 < c_2 < 1,$$

let x_k, g_k, H_k, α_k be the verified run of Chapter 12. Define

$$\begin{aligned} \widehat{x}_k &= \mathcal{L}^{-1} x_k, & \widehat{g}_k &= a\mathcal{L}^T g_k, \\ \widehat{H}_k &= a^{-1}\mathcal{L}^{-1} H_k \mathcal{L}^{-T}, & \widehat{\alpha}_k &= \alpha_k. \end{aligned}$$

The gradients are the actual derivatives of F_R at the accepted points, and $\widehat{H}_0 \succ 0$. Direct substitution in the inverse-BFGS update proves that these are its exact next matrices and steps. Both the predicted and actual objective decreases and the secant curvature are multiplied by a, so Armijo and weak Wolfe hold with the same constants. This proves a transformed full-BFGS run, rather than prescribing an unrelated sequence of endpoint pairs.

The iterates satisfy $\widehat{x}_k \to 0$, while their strictly decreasing values tend to $a - 1 > -1$; zero is nonstationary. In every three phases the actual supporting errors repeat $0, 0, 25/4$. The actual positive secant curvatures are a times those of the run of Chapter 12 and tend to zero. On the fixed band

$$K = \{F_R \le F_R(\widehat{x}_0)\}, \qquad F_R \ge -1 + a,$$

all these accepted points remain above the band threshold. Applying (SP) to successive points with indices $3m + 1$ and $3m + 2$ would give $(25/4)^2 \le C' T_k$ for every cycle, a contradiction as $T_k \to 0$. Thus (SP), and by the compact-domain equivalence also (RS), fail on this actual initial sublevel. The construction specifies both the objective and the infinite full-BFGS trajectory.

4.5 A broader verifiable family: polyhedrally piecewise smooth convex functions

Theorem 4.5.1 (Finite polyhedral partitions)**.** *Let F be finite and convex, with a unique minimizer x_*. Suppose a neighborhood of x_* is covered by finitely many closed full-dimensional polyhedra P_i, with compatible function values on overlaps, and*

$$F(x) = f_i(x) \quad (x \in P_i)$$

there. Assume each f_i has a $C^{1,1}$ extension to an open neighborhood of the relevant part of its cell. Then every sufficiently low-starting infinite admissible full-BFGS run converges to x_*, for arbitrary SPD initialization.

Proof. Discard all cells not containing x_* by shrinking to a ball whose radius is below their positive distances from x_*. There are only finitely many cells. If a remaining cell contains x, it also contains the segment from x to x_* by convexity. At a differentiable point of F, its gradient agrees with the smooth gradient of each adjacent full-dimensional cell: approach from the interior of that cell and use differentiability and convexity of F.

Let L bound the Lipschitz constants of the finitely many smooth extensions on a smaller ball of radius R. For x in one such cell, the usual Taylor estimate along the inward segment gives

$$0 \le F(x - tr_x) - F(x) + t\nabla F(x)^T r_x \le \frac{L}{2} t^2 \|r_x\|^2 \le \frac{LR^2}{2} t^2.$$

The lower bound follows from convexity of F; individual smooth extensions need not be convex outside their cells. Thus uniform (IR), hence (RS), holds locally. A low compact sublevel is trapped there, and the convergence theorem of Chapter 3 (Theorem 3.1, through Section 3.4 and its local form in Section 3.3.2) applies. No cell or active-manifold identification is assumed of the iterates. □

In particular this covers every finite convex *piecewise linear-quadratic* objective, and a convex smooth loss plus a finite convex piecewise linear-quadratic regularizer. This is broader than merely having affine pieces plus one common smooth summand: different cells can have different quadratic Hessians, and the gradient jump can vary along a cell boundary (for example $u^2 + v^2 + (2 + |u|)|v|$ near the origin). Convexity of the full objective remains essential. A finite maximum of arbitrary convex quadratics need not have a polyhedral partition and is not automatically included.

4.6 Common families through the one-dimensional profile

All conclusions in this section require the indicated minimizer and compact-basin hypotheses. Local statements are not global arbitrary-initialization statements.

4.6.1 Norms, gauges, and mixed powers

For a finite sublinear γ centered at zero,

$$\gamma((1-t)z) = (1-t)\gamma(z), \qquad 0 \le t < 1.$$

There is exactly zero radial Taylor error, regardless of angular nonsmoothness. This includes polyhedral norms, Euclidean norms, the spectral norm and nuclear norm when centered at zero, and asymmetric finite gauges.

For nonnegative γ and $p \ge 1$,

$$\gamma((1-t)z)^p = (1-t)^p \gamma(z)^p.$$

On $0 \le t \le 1/2$, the second derivative in t is bounded uniformly on any bounded set of z. This includes $1 < p < 2$, without demanding a Lipschitz full gradient near zero. Therefore finite sums of differently powered, linearly composed gauges,

$$F(x_* + z) = \psi(z) + \sum_j \gamma_j(A_j z)^{p_j}, \qquad p_j \ge 1,$$

are covered when ψ is convex $C^{1,1}$ locally and the full objective has the required minimizer and compact sublevel. The individual terms need not be coercive.

4.6.2 Polyhedral losses and penalties at nonzero solutions

Locally around any optimizer, a nonzero scalar affine residual has fixed sign and is smooth; a zero residual becomes a centered absolute value or hinge. Hence smooth losses plus ℓ_1, fused/generalized ℓ_1, finite hinge and quantile losses are covered locally under uniqueness. The minimizer is not required to be zero. Finite polyhedral regularizers, and their piecewise linear-quadratic extensions from Theorem 4.5.1, are also covered. Extended-valued constraints remain outside the finite-function formulation.

4.6.3 Affine Euclidean group residuals

For terms $\|A_j x - b_j\|_2$, a nonzero optimal residual is locally smooth, while a zero optimal residual becomes $\|A_j z\|_2$, which is homogeneous. Thus sums of such terms, including overlapping Euclidean group penalties and sums of distances, added to a suitable convex smooth objective, satisfy the local condition when the minimizer is unique. This does not imply that every spectral penalty is smooth near every nonzero residual.

4.6.4 Finite maxima: a simple branch test

If locally $F = \max_i f_i$ with finitely many convex C^2 branches, it suffices that

$$(\nabla f_i(w) - \nabla f_j(w))^T (w - x_*) = 0 \quad (i, j \in I(w))$$

at every active tie near x_*. Along a fixed ray, active branches then agree in the scalar first derivative. That derivative is a continuous selection of finitely many Lipschitz branch derivatives, and has their common Lipschitz bound on compact intervals. This proves (IR) directly.

The necessity for (RS) is also clear on genuine active ties: if the active radial derivatives differ by $J > 0$, the one-dimensional central second difference at the tie is $J|t| + O(t^2)$, not $O(t^2)$. Thus radial compatibility describes the exact no-jump part of the criterion for this finite smooth-branch class. It is not a necessary condition for BFGS convergence itself.

4.7 Important boundaries: curved ridges and low-rank spectral penalties

4.7.1 The parabolic ridge

For

$$F(u, v) = \max\{u^2 + v, 2u^2 - v\}, \qquad x_* = 0,$$

the switching curve is $v = u^2/2$. At $w = (u, u^2/2)$, $u \neq 0$, the active gradient difference is $(-2u, 2)$ and its radial inner product is $-u^2 \neq 0$. A ray crosses a genuine one-dimensional kink there. For the descending pair

$$x_\epsilon = (u, u^2/2 - \epsilon), \qquad z_\epsilon = (u, u^2/2 + \epsilon/2),$$

the ratio is exactly

$$\frac{(y^T z_\epsilon)^2}{y^T(z_\epsilon - x_\epsilon)} = \frac{(u^2 - \epsilon)^2}{3\epsilon} \to \infty.$$

This fails (RS) on a positive objective band in every neighborhood of zero. It is *not* a nonconvergent BFGS trajectory and does not establish failure of BFGS.

4.7.2 A common spectral regularizer not automatically covered

Work in symmetric 2×2 matrices with the Frobenius inner product. Let

$$F(X) = \frac{1}{2}\|X - C\|_F^2 + \|X\|_*, \qquad C = \operatorname{diag}(2, 0), \quad X_* = \operatorname{diag}(1, 0).$$

This is strongly convex and uniquely minimized at X_*. Indeed $X_* - C = \operatorname{diag}(-1, 0)$ is canceled by a nuclear-norm subgradient $\operatorname{diag}(1, 0)$ there.

Take orthonormal vectors

$$v = (\cos\theta, \sin\theta), \qquad u = (-\sin\theta, \cos\theta),$$

and let $W = vv^T$, $U = uu^T$, $a = \sin^2\theta > 0$. For

$$X_\epsilon = W - \epsilon U, \qquad Z_\epsilon = W + \epsilon U,$$

both nuclear norms are differentiable, since the matrices have nonzero eigenvalues. The pair is descending: $F(Z_\epsilon) - F(X_\epsilon) = -4a\epsilon < 0$. Its gradient difference is $y = 2(1 + \epsilon)U$, so

$$T = 4\epsilon(1 + \epsilon), \qquad N = 2(1 + \epsilon)(\epsilon - a), \qquad \frac{N^2}{T} = \frac{(1 + \epsilon)(\epsilon - a)^2}{\epsilon} \to \infty.$$

Also $F(W) - F(X_*) = 2a > 0$. Choosing small θ puts such points arbitrarily close to X_*. Thus this familiar smooth-plus-nuclear-norm problem need not satisfy (RS) on any neighborhood of a nonzero rank-deficient minimizer. The nuclear norm centered at zero *is* covered. No failure of BFGS on this spectral problem is being claimed.

Relation to earlier work

The following comparisons are bibliographic scholarship, separate from the formal theorem assertions.

The Bregman expression uses established terminology [9, Definition 1] and ordinary supporting-plane calculus. Only its supporting endpoint must be differentiable; no new divergence construction is asserted.

Full-BFGS convergence on the Euclidean norm in arbitrary dimension was established by Guo–Lewis [10, Corollary 4.2]; Lewis–Overton [2, Section 5.1] analyzes the scalar absolute-value model with its stated line-search conventions. The Euclidean-norm theorem is a known case recovered here. The wider gauge and mixed-power conclusions use the radial argument under the stated minimizer and compact-basin hypotheses.

The optimizer in this example is a specialization of singular-value thresholding [12, Theorem 2.1]. That standard calculation is separate from the rotated-pair obstruction to the radial condition established in this chapter; the obstruction makes no claim of BFGS failure.

CHAPTER 5

What comparison certificates express

The certificate is a sufficient analytic condition. Its normalization and pointwise geometry help interpret it, while scalar examples separate failure of the condition from failure of BFGS.

5.1 The precise condition under study

Let $F : \mathbb{R}^n \to \mathbb{R}$ be finite and convex, with attained minimum F_*. Fix a compact sublevel $K = \{F \le b\}$, a positive band

$$K_\eta = \{x \in K : F(x) \ge F_* + \eta\}, \qquad \eta > 0,$$

and fixed $0 < c_1 < c_2 < 1$. Endpoints below are differentiable. Write

$$g = \nabla F(x), \quad g_z = \nabla F(z), \quad y = g_z - g, \quad s = z - x,$$

$$d = -g^T s, \quad \Delta = F(x) - F(z), \quad t = y^T s.$$

A pair is *eligible* when

$$d > 0, \qquad c_1 d \le \Delta \le d, \qquad t \ge (1 - c_2)d. \tag{5.E}$$

These are the necessary endpoint inequalities of the Armijo–weak-Wolfe formulation in Xie–Wächter [1]. In particular $t > 0$ and $s \neq 0$.

On each positive band, the condition asks for a finite-valued PSD field $W(x)$ and constants $m > 0$, $C < \infty$, $L \ge 0$, $\delta > 0$ such that

$$g_x^T W(x) g_x \ge m, \tag{5.V}$$

$$y^T W(z) y \le Ct, \tag{5.S}$$

$$\Pi_s W(z) \Pi_s \preceq (1 + Ld)\Pi_s W(x)\Pi_s, \qquad \Pi_s = I - ss^T/(s^T s). \tag{5.T}$$

The last two apply to all eligible pairs in the band with $d \le \delta$. The field may be singular, discontinuous, and unbounded as a function on the band; only its value at each point is finite. All constants may depend on the band and the fixed line-search parameters. No claim below silently imposes continuity or local boundedness on W.

The one-sided transported-certificate convergence theorem of Chapter 3 says that this condition implies objective convergence for every admissible infinite full-BFGS run in K, and iterate convergence under uniqueness. A local theorem requires the condition only near the minimizer, then uses a sufficiently low trapped sublevel. Failure on a large sublevel and failure in every neighborhood of a minimizer are different statements.

5.2 The expansion allowance can be removed completely

Proposition 5.2.1 (Exact nonexpansion normalization)**.** *The existential condition (V)–(T) is equivalent to one in which $L = 0$:*

$$\Pi_s \widehat{W}(z) \Pi_s \preceq \Pi_s \widehat{W}(x) \Pi_s. \tag{5.T0}$$

The other constants may change, but no domain or eligible-pair hypothesis is added.

Proof. Given a field satisfying (T), put $\kappa = L/c_1$ and

$$\widehat{W}(x) = e^{\kappa(F(x)-F_*)}W(x).$$

On the compact value band this multiplier is bounded above and away from zero, preserving finite values, (V), and (S), with changed constants. Since $\Delta \geq c_1 d$,

$$e^{-\kappa\Delta}(1 + Ld) \leq e^{-Ld}(1 + Ld) \leq 1.$$

Multiplying (T) by $e^{\kappa(F(z)-F_*)}$ gives (T0). The converse is the case $L = 0$. □

This is an equivalence, not a further relaxation. The accumulated-expansion budget was useful in the proof of the convergence theorem of Chapter 3, but is not essential to the existential definition: an objective-value rescaling absorbs it.

5.2.1 An ellipsoid interpretation

Associate with $W(x)$ the possibly degenerate centered ellipsoid

$$\mathcal{E}_x = W(x)^{1/2}\{v : \|v\| \leq 1\}.$$

Its support function is $h_{\mathcal{E}_x}(a) = \sqrt{a^T W(x) a}$. Thus the normalized condition means:

(i) $\mathcal{E}_x$ contains a direction with descent magnitude at least $\sqrt{m}$;
(ii) every incoming gradient change has magnitude at most $\sqrt{Ct}$ when tested on $\mathcal{E}_z$;
(iii) $\Pi_s\mathcal{E}_z \subseteq \Pi_s\mathcal{E}_x$ for every eligible small-decrease pair.

Equivalently, after quotienting out the executed step direction, the comparison ellipsoid can only shrink. The ellipsoids are analytical comparison directions, not BFGS steps or approximations of H_k.

A necessary rank consequence is

$$\operatorname{range} W(z) \subseteq \operatorname{range} W(x) + \operatorname{span}\{s\}. \tag{5.2}$$

Indeed, Loewner ordering implies inclusion of the ranges of the projected PSD matrices. The range of $\Pi_s W \Pi_s$ is $\Pi_s \operatorname{range} W$, because it equals the range of $(\Pi_s W^{1/2})(\Pi_s W^{1/2})^T$. Lifting the projected inclusion proves (5.2). For a rank-one field, a new comparison direction must therefore lie in the plane spanned by the old comparison direction and the step.

5.3 Removing the matrix from pointwise feasibility

Fix a band and a predicted-decrease threshold δ. For each differentiable endpoint z, let $\mathcal{I}(z)$ be all differentiable predecessors x for which (x, z) is eligible, belongs to the band, and has $d \leq \delta$. Define

$$\mathcal{D}(z) = \left\{ v : \left| [\nabla F(z) - \nabla F(x)]^T v \right| \leq \sqrt{t(x,z)} \text{ for every } x \in \mathcal{I}(z) \right\},$$

$$\nu(z) = \sup_{v \in \mathcal{D}(z)} |\nabla F(z)^T v| \in [0, \infty]. \tag{5.4}$$

An empty predecessor set gives $\mathcal{D}(z) = \mathbb{R}^n$. The set $\mathcal{D}(z)$ is the polar of the closed absolutely convex hull of the normalized incoming jumps $y/\sqrt{t}$.

Proposition 5.3.1 (Exact characterization of (V) and (S), without transport)**.** *There is a finite-valued PSD field satisfying (V) and (S) with uniform positive m and finite C if and only if*

$$\inf_z \nu(z) > 0.$$

This statement concerns only (V) and (S). It does not assert that the pointwise selections satisfy (T).

Proof. If $C > 0$ and W satisfies (V) and (S), set $h = \nabla F(z)$ and

$$v = \frac{W(z)h}{\sqrt{C}\sqrt{h^T W(z)h}}.$$

PSD Cauchy–Schwarz gives $|y^T v| \le \sqrt{t}$ for every incoming pair and $|h^T v| \ge \sqrt{m/C}$. If $C = 0$, the same argument without $\sqrt{C}$ gives a nonzero visible direction annihilating every incoming jump, which can be scaled arbitrarily.

Conversely, if $\inf \nu \ge \nu_0 > 0$, choose, for each z, a finite vector $v_z \in \mathcal{D}(z)$ with $|\nabla F(z)^T v_z| \ge \nu_0/2$. Then $W(z) = v_z v_z^T$ satisfies (S) with $C = 1$ and (V) with $m = \nu_0^2/4$. No continuity is needed for these pointwise selections. □

Thus higher rank adds no pointwise power to (V) and (S); its potential advantage is compatibility of comparison geometries across endpoints through (T). The scalar $\nu(z)$ has a direct interpretation: how much descent remains visible after all low-curvature incoming jumps have been suppressed?

5.3.1 A nonexistence test independent of the certificate

Proposition 5.3.2 (Incoming-jump obstruction)**.** *Suppose z_ϵ and finitely many predecessors $x_{j,\epsilon}$ give eligible pairs in a fixed positive band with $d_{j,\epsilon} \to 0$. Write $y_{j,\epsilon} = \nabla F(z_\epsilon) - \nabla F(x_{j,\epsilon})$ and $t_{j,\epsilon} = y_{j,\epsilon}^T(z_\epsilon - x_{j,\epsilon})$. If*

$$\nabla F(z_\epsilon) = \sum_{j=1}^{r} \lambda_{j,\epsilon} y_{j,\epsilon}, \qquad \sum_{j=1}^{r} |\lambda_{j,\epsilon}| \sqrt{t_{j,\epsilon}} \longrightarrow 0,$$

then no field satisfies (V) and (S), and hence no transported certificate exists on that band.

Proof. For any PSD field satisfying (S), the triangle inequality for the seminorm induced by $W(z_\epsilon)$ gives

$$\sqrt{\nabla F(z_\epsilon)^T W(z_\epsilon) \nabla F(z_\epsilon)} \le \sqrt{C} \sum_j |\lambda_{j,\epsilon}| \sqrt{t_{j,\epsilon}} \to 0,$$

contradicting (V). Neither transport nor boundedness or continuity of W was used. □

In particular, the obstruction holds if n incoming jumps converge to a basis of $\mathbb{R}^n$, their curvature products tend to zero, and endpoint gradients remain bounded. The coefficients expressing the gradient in that basis then remain bounded.

The predecessors need not belong to one BFGS history. This is not a loophole: the condition quantifies over all such pairs. Moreover, every eligible pair is individually realizable as a first BFGS step. If $d = -g^T s > 0$, then

$$H = \frac{ss^T}{d} + I - \frac{gg^T}{g^T g} \succ 0, \qquad Hg = -s.$$

Hence $\alpha = 1$ realizes it. Positive definiteness follows because a vector in both nullspaces would be parallel to g and orthogonal to s, and therefore zero.

5.4 Connection with tangent and normal geometry

At a nonsmooth point p, let $g_1, \dots, g_r$ be limiting active gradients. Define

$$N = \operatorname{span}\{g_i - g_j\}, \qquad U = N^\perp.$$

The projected gradients $P_U g_i$ agree. They are the common part of the first-order models that is not affected by gradient jumps. In suitable partly smooth settings N is the normal space and U

the tangent space; this interpretation and its relation to the affine span of the subdifferential are standard in partial smoothness [8].

For a certificate with a finite continuous extension to p, consider actual eligible incoming pairs on the fixed positive band, with both endpoints tending to p, $d \to 0$, $t \to 0$, and limiting gradient jump j. The secant inequality and continuity give

$$j^T W(p) j = 0, \qquad W(p) j = 0,$$

where the second conclusion uses positive semidefiniteness. Let N_{acc} be the span of all such accessible limiting jumps, and set $U_{\mathrm{acc}} = N_{\mathrm{acc}}^{\perp}$. Then

$$\operatorname{range} W(p) \subseteq U_{\mathrm{acc}}.$$

For a limiting gradient g at which visibility persists, $g^T W(p) g > 0$ implies $P_{U_{\mathrm{acc}}} g \neq 0$.

This conclusion concerns the accessible span. It does not identify N_{acc} with the earlier space N generated by all active-gradient differences. A nonzero common projection onto $U = N^{\perp}$ follows if the accessible jumps span N, or if $W(p)$ is otherwise known to annihilate all active differences. Under that additional hypothesis, the common projection is nonzero by visibility. Independently of accessibility, the following algebraic identity is exact:

$$P_U g_i = 0 \quad \Longleftrightarrow \quad 0 \in \operatorname{aff}\{g_1, \dots, g_r\}.$$

The affine-hull condition is weaker than $0 \in \operatorname{conv}\{g_1, \dots, g_r\}$ and need not imply stationarity for the convex objective.

The distinction is realized by the finite strongly convex objective

$$f(x) = \tfrac{1}{2}(x+2)^2 + |x|, \qquad (c_1, c_2) = (1/8, 1/4).$$

On every compact positive band, $W \equiv 1$ is a transported certificate. Indeed, every eligible pair with $d \leq 1/4$ stays on one smooth side of zero: a right-to-left crossing contradicts weak Wolfe, and a left-to-right crossing contradicts Armijo and the small-decrease bound. Hence $f'(z) - f'(x) = z - x$, proving (S) with $C = 1$; transport is equality, and the positive-band gradient bound proves visibility. Weak Wolfe gives $d \leq 4t/3$, so $t \to 0$ eventually makes $d \leq 1/4$; as both endpoints tend to the same point, every accessible limiting jump is therefore zero. Nevertheless the actual limiting active gradients at zero are 1 and 3, so $N = \mathbb{R}$ and $U = \{0\}$. The full-active tangential conclusion would be false here. The later no-certificate example uses the different constants $(1/4, 1/2)$ and is consistent with this distinction.

This limiting argument uses the stated finite-extension and accessibility hypotheses only in this interpretation. It is not a proof for arbitrary discontinuous or unbounded certificate fields. Proposition 5.3.2 supplies an obstruction without either regularity hypothesis. Eligibility matters: arbitrary active-gradient jumps need not be accessible under a fixed weak-Wolfe condition.

For $u^2 + |v|$ at $(u, 0)$ with $u \neq 0$, the jump space is vertical and a nonzero horizontal derivative remains. At a generic two-dimensional vertex with independent accessible jumps in both coordinates, there may be no remaining comparison direction. The common-target obstruction above makes that interpretation rigorous without continuity of a certificate.

5.6 In one dimension the condition can be fully eliminated

Proposition 5.6.1 (Exact scalar characterization)**.** *On a compact positive objective band in $\mathbb{R}$, the transported certificate exists if and only if, for some small-decrease threshold, eligible pairs satisfy*

$$\sup_{(x,z)} \frac{|F'(z) - F'(x)|}{|z - x|\,|F'(z)|^2} < \infty. \tag{5.6}$$

Equivalently, the derivative is uniformly Lipschitz on these eligible pairs:

$$|F'(z) - F'(x)| \le A_\eta |z - x|.$$

This is not a Lipschitz assertion on all pairs or across the minimizer.

Proof. In one dimension $\Pi_s = 0$, so transport imposes no restriction. From (V), $W(z) \ge m/|F'(z)|^2$. From (S), using $t = |y||s|$, $W(z)|y| \le C|s|$. These imply (5.6). Conversely, choose $W(z) = 1/|F'(z)|^2$; visibility equals one and (5.6) gives (S).

On a compact positive band, convexity gives $|F'(z)| \ge \eta/R$, where R bounds distance to a fixed minimizer, and local boundedness gives $|F'(z)| \le G$. These bounds make (5.6) equivalent to the displayed Lipschitz inequality. □

This reveals a genuine restriction: an eligible arbitrarily small crossing of a nonoptimal derivative jump rules out every certificate.

5.6.1 Convergence itself does not need this scalar restriction

Proposition 5.6.2 (Scalar convergence, independently of BFGS updates)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be finite and convex with unique minimizer and compact initial sublevel. Every infinite differentiable-iterate Armijo–weak-Wolfe run using any positive scalar metric converges to that minimizer.*

Proof. Suppose the monotone objective gap tends to $L > 0$. Convexity gives $|g_k| \ge L/R$ on the compact sublevel. In one dimension a positive-metric descent step obeys $d_k = |g_k||s_k|$. Armijo makes $\sum d_k < \infty$, hence $\sum |s_k| < \infty$. Thus the iterates converge to a nonminimizer. On a tail they lie on one side of the unique minimizer, so all gradients have the same sign. Weak Wolfe then gives $|g_{k+1}| \le c_2|g_k|$, contradicting the positive lower bound. Objective convergence and compactness with uniqueness give iterate convergence. □

Already $f(x) = \frac{1}{2}(x+2)^2 + |x|$ gives a finite strongly convex example with global convergence but, for $(c_1, c_2) = (1/4, 1/2)$, no global certificate on any sublevel containing zero in its interior. For $x = \epsilon$, $z = -\epsilon/2$ with $\epsilon > 0$,

$$g_x = 3 + \epsilon, \quad g_z = 1 - \epsilon/2, \quad y = -2 - 3\epsilon/2,$$

$$d = \frac{9}{2}\epsilon + \frac{3}{2}\epsilon^2, \quad \Delta = \frac{7}{2}\epsilon + \frac{3}{8}\epsilon^2, \quad t = 3\epsilon + \frac{9}{4}\epsilon^2.$$

These pairs are strictly eligible for $(1/4, 1/2)$ for small ϵ, and their scalar quotient in (5.6) diverges.

5.7 Local nonexistence despite global convergence

The difference between a local certificate and actual convergence remains even after shrinking every neighborhood of the optimizer.

Define $a_j = 4^{-j}$ and

$$h(x) = \frac{1}{2}x^2 + 3\sum_{j=1}^{\infty} a_j (x - a_j)_+. \tag{5.7}$$

The series converges uniformly on compact sets: its absolute value is bounded by a constant times $\sum a_j$. Every summand is convex, and $h(x) \ge x^2/2$ with $h(0) = 0$. Thus h is finite, 1-strongly convex, coercive, and uniquely minimized at zero.

At a knot $a = a_j$, the one-sided derivatives are $2a$ and $5a$. These formulas come from exact geometric sums, not a truncation. For example, when $a/4 < x < a$,

$$h(x) = \frac{1}{2}x^2 + ax - \frac{1}{5}a^2,$$

while just above a,

$$h(x) = \frac{1}{2}x^2 + 4ax - \frac{16}{5}a^2.$$

Also $h(a) = 13a^2/10$.

Fix any knot, then let $x = a + \epsilon$, $z = a - \epsilon$, with $\epsilon \downarrow 0$. Exactly,

$$g_x = 5a + \epsilon, \quad g_z = 2a - \epsilon, \quad y = -3a - 2\epsilon,$$

$$d = 10a\epsilon + 2\epsilon^2, \qquad \Delta = 7a\epsilon, \qquad t = 6a\epsilon + 4\epsilon^2.$$

The pairs are strictly eligible for $(c_1, c_2) = (1/4, 1/2)$ for small ϵ. Yet

$$\frac{|y|}{|s|\, g_z^2} = \frac{3a + 2\epsilon}{2\epsilon(2a - \epsilon)^2} \longrightarrow \infty.$$

No certificate exists on a positive band whose interior contains this knot ($b > h(a) > \eta$). Every neighborhood of zero contains such a knot. Fixing a knot first, then taking $\epsilon \to 0$, is essential: deterioration of band constants as $\eta \downarrow 0$ cannot repair the divergence on the fixed knot's band.

Nevertheless Proposition 5.6.2 proves convergence of every admissible run on h. The minimizer is differentiable for this scalar example, but the objective is nonsmooth in every neighborhood of it. Thus the transported condition is not necessary even for local convergence. This conclusion uses the explicitly defined infinite-series objective above; no unspecified archived auxiliary is needed.

5.9 Why numerical nonexistence is intrinsically delicate

Proposition 5.9.1 (Every finite sample is feasible if constants are free)**.** *Take any finite set of differentiable nonstationary endpoints and any finite set of eligible pairs between them. There are finite constants $m > 0$, $C < \infty$ and a constant field $W(x) = I$ satisfying (V), (S), and (T0) on that finite sample.*

Proof. For a nonempty endpoint sample choose m as the minimum of the positive numbers $\|\nabla F(x)\|^2$; for an empty sample set $m = 1$. Let $\mathcal{P}$ denote the finite pair sample and choose

$$C = \sum_{(x,z)\in\mathcal{P}} \frac{\|\nabla F(z) - \nabla F(x)\|^2}{t(x, z)}.$$

All denominators are positive, every summand is nonnegative, and the sum is zero for an empty pair sample. Thus C is finite and bounds each sampled quotient. The constant identity field satisfies visibility and (S), and transport is equality. □

Consequently a finite-sample feasibility problem with unrestricted constants cannot prove nonexistence. One needs a known bound on the constants, extra restrictions on the field, or an analytic limiting family that forces the required constants to infinity. The examples above supply the latter. Conversely, passing many finite tests is not an existence theorem for a uniform field on the continuum.

Relation to earlier work

The following comparisons are bibliographic scholarship, separate from the formal theorem assertions.

Absorbing multiplicative errors by rescaling is an established technique; compare [21], Lemmas 2.2–2.3. The elementary objective-value rescaling in this chapter applies to this specific, potentially singular and unbounded comparison field.

The ellipsoid interpretation and the reduction in this chapter use standard support-function duality, polar sets and PSD Cauchy–Schwarz. They are elementary deductions for the stated certificate, with no assertion that the pointwise choices satisfy transport.

The proof uses the standard triangle inequality for a PSD seminorm. The conclusion concerns this certificate's all-eligible-pairs requirement; it is neither a BFGS failure theorem nor a necessity condition for convergence.

VU and partial-smoothness theory supply the common projected-subgradient geometry [26, 8]. The argument in this chapter additionally tracks jumps accessible under the fixed line search. The span of all active-gradient differences cannot replace that accessible span without a separate justification.

Lewis–Overton analyzes convergence on the scalar absolute value for its specified bracketing/secant algorithm (Section 5.1) [2]. Their Section 4.2 also discusses accumulating corners in line-search analysis. These precedents do not assert the exact all-field certificate obstruction or the particular infinite-series example in this chapter.

CHAPTER 6

Concrete objectives, lifting and rates

These applications turn the common analytic mechanism into verifiable objective classes. The affine-fiber selection formula and quantitative conclusions retain their own initialization and budget hypotheses.

6.1 Centered smooth-plus-sublinear objectives

This chapter collects nonredundant structural consequences of the general theory. The finite-polyhedral-partition and powered-gauge radial arguments themselves are reproduced in Chapter 4. The discussion here retains the global, nonunique, explicit-basin and quantitative clauses that are not supplied by a bare local uniqueness theorem.

Theorem 6.1 (Centered smooth-plus-sublinear convergence)**.** *Let $F(x) = \psi(x) + \gamma(x - x_*)$, where ψ is convex and locally $C^{1,1}$ on an open convex neighborhood of a compact initial sublevel, and γ is finite convex and positively homogeneous of degree one. Suppose x_* minimizes F. Every infinite admissible full-BFGS run has $F(x_k) \to F(x_*)$. If x_* is unique, then $x_k \to x_*$. A representation only near a unique minimizer gives a low-starting local theorem by Lemma 2.1.*

Proof. Translate $x_* = 0$ and let R bound the initial sublevel. At differentiable endpoints, γ is differentiable because F and ψ are. Write $y = y_\psi + y_\gamma$, and set

$$a = y_\psi^T s, \quad b = y_\gamma^T s, \quad \xi = y_\psi^T x', \quad \zeta = y_\gamma^T x'.$$

Then $a, b \geq 0$ and $t = a + b$. Convex local L-smoothness gives $\|y_\psi\|^2 \leq La$, hence $\xi^2 \leq LR^2 a$. Here L is a uniform constant chosen for the compact endpoint set inside the open convex neighborhood. A local proof subtracts a supporting tangent plane and applies the descent estimate at a short step opposite the gradient difference, with the test step kept inside a larger buffered neighborhood. Applying the resulting Bregman estimate in both directions gives the stated inequality. The constant may be enlarged to keep those test steps inside the neighborhood; no assertion that an arbitrarily prescribed Lipschitz constant is sharp is needed.

If $h \in \partial\gamma(x)$, homogeneity and the supporting inequality give $h^T x = \gamma(x)$ and $h^T z \leq \gamma(z)$. Hence $0 \leq \zeta \leq b$. All subgradients of γ are bounded by some G_γ, so $\zeta \leq 2G_\gamma R$. Weighted Cauchy–Schwarz gives

$$\frac{(y^T x')^2}{t} \leq \frac{\xi^2}{a} + \frac{\zeta^2}{b} \leq LR^2 + 2G_\gamma R.$$

Terms with zero denominator are zero since their numerators vanish. Apply the fixed-reference energy bound and the reciprocal-average theorem in Chapter 2. Compactness and uniqueness give the iterate conclusion. □

The original summands need not separately be minimized at the center. Optimality implies $-\nabla\psi(x_*) \in \partial\gamma(0)$. After translating, subtract the smooth tangent plane from ψ and add it to γ; both adjusted summands are then nonnegative and minimized at zero. This is a permissible rewriting of the same objective, not invariance of BFGS under a change of objective.

6.2 Smooth increasing transforms of coercive gauges

Theorem 6.2 (Transformed-gauge convergence)**.** *Let γ be finite convex, degree-one homogeneous and coercive. Let $\Phi : [0, \infty) \to \mathbb{R}$ be continuous, convex, strictly increasing, with $\Phi(0) = 0$, $\Phi \in$*

$C^2((0,\infty))$ *and* $\Phi' > 0$ *there. Then every infinite admissible full-BFGS run on* $F(x) = \Phi(\gamma(x - x_*))$ *converges to* x_*.

Proof. Coercivity of γ and unboundedness of Φ give compact sublevels. Translate $x_* = 0$. Suppose $F_k \downarrow L > 0$. Euler's identity and the chain rule give

$$a_k = g_k^T x_k = \gamma(x_k)\Phi'(\gamma(x_k)) = \Theta(F_k), \quad \Theta(t) = \Phi^{-1}(t)\Phi'(\Phi^{-1}(t)).$$

On $[L, F_0]$, the derivative

$$\Theta'(t) = 1 + \frac{r\Phi''(r)}{\Phi'(r)}, \qquad r = \Phi^{-1}(t),$$

is bounded. Thus $|a_{k+1} - a_k| \le C\Delta_k$. Since

$$y_k^T x_{k+1} = d_k + a_{k+1} - a_k,$$

(2.1) implies

$$R_k \le \frac{(1+C)^2}{(1-c_2)c_1}\Delta_k.$$

The corrections are summable, contradicting a positive gap in (2.9). Hence $F_k \to 0$, and coercivity and strict increase give $x_k \to 0$. □

The same correction estimate can be obtained without introducing an inverse: on a compact positive gauge annulus, bound the derivative of $r\Phi'(r)$ and bound Φ' away from zero, then apply the scalar mean-value theorem twice. The formal convergence argument uses this annulus route; the inverse identities and derivative formula displayed above are separately proved. Neither route claims invariance of BFGS under a nonlinear scalar transform. It includes cases whose scalar second derivative is unbounded at zero. Finite sums of differently powered gauges are instead covered by the radial argument in Chapter 4, including $1 < p < 2$.

6.3 Finite max-affine terms and a certified basin

Let $F = \psi + p$, with ψ convex locally $C^{1,1}$ and $p(x) = \max_i \ell_i(x)$, $\ell_i(x) = a_i^T x + b_i$, a finite real-valued maximum. At a unique minimizer x_*, put $I_* = \{i : \ell_i(x_*) = p(x_*)\}$. Inactive pieces have positive gaps, so on a sufficiently small ball

$$p(x_* + z) = p(x_*) + \max_{i \in I_*} a_i^T z. \tag{6.1}$$

Indeed, with any fixed $i_0 \in I_*$, the difference between its value and an inactive piece j is

$$\delta_j + (a_{i_0} - a_j)^T z, \qquad \delta_j = p(x_*) - \ell_j(x_*) > 0,$$

which stays positive if $\|z\| \le \delta_j/(2\|a_{i_0} - a_j\|)$; a parallel inactive piece stays dominated everywhere. This exact local identity and Theorem 6.1 prove local convergence without strict complementarity or Hessian nondegeneracy.

For an explicit sufficient level, define the closed active region

$$P_j = \{x : \ell_j(x) = p(x)\} = \{x : (a_j - a_i)^T x \ge b_i - b_j \ \forall i\}$$

and

$$\tau_* = \min_{j \notin I_*} \inf_{x \in P_j} F(x), \tag{6.2}$$

with empty infima or minima equal to $+\infty$. Then $\tau_* > F_*$. To prove it, note that unique-minimizer convexity gives compact sublevels; a finite infimum on a nonempty closed P_j is attained. It cannot equal F_* because $x_* \notin P_j$. Finitely many inactive pieces preserve the strict gap.

Every admissible run with $F(x_0) < \tau_*$ converges. On its full compact initial sublevel, only indices in I_* can be active, so (6.1) and the centered correction estimate apply throughout. A tail that first enters this sublevel enjoys the same conclusion for its current arbitrary SPD matrix. The threshold is sufficient, not a characterization of every successful initialization. Each inner infimum is a convex program over the displayed linear inequalities.

6.4 Residual penalties and affine-fiber lifting

For a scalar affine residual nonzero at the solution, its absolute-value or hinge term is locally affine. A zero optimal residual produces a centered sublinear term. The same decomposition applies to an affine-composed Euclidean norm: a nonzero optimal vector residual is locally smooth, and a zero residual becomes a centered norm. Finite sums with a convex locally $C^{1,1}$ smooth part reduce to Theorem 6.1. For a more general gauge, local $C^{1,1}$ smoothness at that particular nonzero residual must be checked; a spectral penalty may fail it.

Theorem 6.3 (Exact linear-image lifting)**.** *Let $A \in \mathbb{R}^{r\times n}$ have full row rank and $F(x) = \bar{F}(Ax)$. Suppose the projected admissible run converges to a unique optimizer z_* of $\bar{F}$, under any of the retained sufficient hypotheses. Then*

$$x_k \to x_0 + H_0A^T(AH_0A^T)^{-1}(z_* - Ax_0).$$

Proof. With $z_k = Ax_k$ and $\bar{H}_k = AH_kA^T$, projection of the inverse-BFGS update gives exactly full BFGS on $\bar{F}$, with the same step lengths and line-search comparisons. Let $S = \operatorname{range} A^T$. Since gradients and secant differences lie in S, the update gives $H_{k+1}S \subset H_kS$; both spaces have dimension r, so equality holds. All displacements lie in H_0S. The restriction of A to that space has right inverse

$$L_0 = H_0A^T(AH_0A^T)^{-1}.$$

Thus $x_k - x_0 = L_0(z_k - z_0)$, proving the result. This is the H_0^{-1}-metric projection onto the affine fiber, not a claim about arbitrary nonunique minimizer sets. □

6.5 Finite-horizon rates and strengthened asymptotic corollaries

Let $S_N = \sum_{k<N} q_k^{-1}$. Retain the explicit lower bound (2.6), which is meaningful at a fixed horizon. All rates below concern fixed dimension, problem, initialization and line-search constants, and count accepted iterations only. Big-O versions of these rates appeared in the source notes. The little-o versions below follow from Theorem 2.3; they do not assert a sharp rate.

Uniformly bounded reference correction. If $R_k \le \bar{R}$, then (2.9) gives

$$F(x_N) - F_* \le \sqrt{\frac{E_0 + \bar{R}N}{S_N}},$$

and along runs with bounded gradients this is $o(N^{-1/2})$, because $S_N/N^2 \to \infty$. The same estimate holds for a linear cumulative correction budget. This is not a rate for arbitrary band-dependent certificates.

Bounded-below degree-one objectives. Chapter 7 proves

$$\sum_{k<N} \frac{F(x_k)^2}{q_k} \le \Phi_0, \quad \Phi_0 = x_0^T H_0^{-1} x_0 + \frac{1-c_1}{c_1} F(x_0).$$

Hence $F(x_N) \le \sqrt{\Phi_0/S_N} = o(N^{-1})$. Coercivity gives $F(x) \ge \mu\|x\|$ and the same order for the iterate norm. The little-o conclusion uses the superquadratic theorem; the finite-horizon bound retains its explicit constants. No rate sharpness is asserted.

Quadratic plus coercive gauge. For $F(u,v) = u^TQu/2 + \gamma(v)$, $Q \succ 0$, let $\mu\|v\| \le \gamma(v) \le \Lambda\|v\|$. Split $t = a + b$ with

$$a = \delta u^TQ\delta u, \quad b = (h' - h)^T\delta v, \quad \xi = (Q\delta u)^Tu', \quad \zeta = (h' - h)^Tv',$$

where $h \in \partial\gamma(v)$ and $h' \in \partial\gamma(v')$. Then

$$\xi^2/a \le u'^T Q u', \qquad 0 \le \zeta \le b, \qquad \zeta \le (1+\Lambda/\mu)\gamma(v').$$

Thus

$$R_k \le C_\gamma F(x_{k+1}), \qquad C_\gamma = \max\{2, 1+\Lambda/\mu\}.$$

Choose $a_0 > 0, N_0$ with $S_N \ge a_0 N^2$ for $N \ge N_0$, and put

$$A_N = E_0 + C_\gamma \sum_{j=1}^{N} F(x_j).$$

Then $F(x_N) \le \sqrt{A_N}/(\sqrt{a_0}N)$, and

$$\sqrt{A_N} - \sqrt{A_{N-1}} = \frac{C_\gamma F(x_N)}{\sqrt{A_N} + \sqrt{A_{N-1}}} \le \frac{C_\gamma}{\sqrt{a_0}N}.$$

Summing first gives the bounds $\sqrt{A_N} = O(\log N)$ and $F(x_N) = O(\log N/N)$. For the strengthened conclusion, keep the exact denominator:

$$\sqrt{A_N} - \sqrt{A_{N-1}} \le \frac{C_\gamma}{\sqrt{S_N}} = o(N^{-1}).$$

A sum of $o(1/N)$ increments is $o(\log N)$: for any fixed $\epsilon > 0$, bound the sufficiently late increments by ϵ/N, then divide their sum by $\log N$ and let $\epsilon \downarrow 0$. Thus $\sqrt{A_N} = o(\log N)$. Dividing by $\sqrt{S_N} = N\sqrt{S_N/N^2}$ yields

$$\boxed{F(x_N) = o(\log N/N).}$$

These strengthened asymptotic corollaries follow from already recorded estimates; they are not uniform complexity bounds over all instances and do not establish linear or superlinear convergence.

6.6 Matrix limits and line-search existence are different questions

The convergence theorems do not assert a universal inverse-Hessian limit. For $u^2 + |v|$, choose $A > 0$, $v_0 > 0$ and $0 < \rho < (1-c_1)/(1+c_1)$. With

$$x_0 = (0, v_0), \quad H_0 = \operatorname{diag}\left(A, \frac{1+\rho}{2\rho}v_0\right), \quad \alpha_k = 2\rho,$$

ordinary BFGS gives

$$x_k = (0, (-\rho)^k v_0), \quad H_k = \operatorname{diag}\left(A, \frac{1+\rho}{2\rho}\rho^k v_0\right).$$

The Armijo ratio is $(1-\rho)/(1+\rho) > c_1$ and the weak-Wolfe ratio is -1. The unexcited tangential entry tends to its arbitrary initial value A, not necessarily the inverse true tangential curvature.

For a finite convex line restriction φ bounded below on the positive ray, suppose $\varphi'(0) = a < 0$. Choose $\sigma = (c_1 + c_2)/2$ and set $\psi(t) = \varphi(t) - \sigma a t$. Its derivative at zero is $(1-\sigma)a < 0$, while the lower bound on φ makes $\psi(t) \to +\infty$ as $t \to +\infty$. Choose a compact interval $[0, M]$ whose interior contains a minimizer $z > 0$ of ψ, with $\psi(z) < \psi(0)$. Then

$$\varphi(z) < \varphi(0) + \sigma a z < \varphi(0) + c_1 a z.$$

Continuity preserves strict Armijo on an interval immediately to the right of z. For each t in a sufficiently small such interval, minimality and convexity give

$$\varphi'(t) \ge \frac{\varphi(t) - \varphi(z)}{t - z} \ge \sigma a > c_2 a$$

whenever the scalar derivative exists. Thus there is a nonempty open interval of strict Armijo points whose actual scalar derivatives satisfy strict weak Wolfe. If full-space nondifferentiability occurs at only finitely many positive ray times, choose a point in that interval outside this finite set; its actual full gradient gives a differentiable acceptable step. This is the proved tilted-minimum construction. It applies to finite affine absolute-value/hinge sums and suitable Euclidean residual sums through their proved full-space bad-ray-set finiteness. It gives existence, not a trial-complexity guarantee for every implementation.

For arbitrary max-functions there is a genuine exception. The coercive function

$$h(x, y, z) = \max\{-x, y + z, -y + z, x/4, -z\}$$

has unique minimizer zero. At $(-2, 0, 1)$ with $H_0 = I$, the direction is $(1, 0, 0)$ and the ray function is $\max\{2 - \alpha, 1, (\alpha - 2)/4\}$. For $(c_1, c_2) = (1/4, 1/2)$, times $0 < \alpha < 1$ fail weak Wolfe; $1 \le \alpha \le 6$ are full-space nondifferentiable; and $\alpha > 6$ fails Armijo, which would require $\alpha \le 5$. Scaling the point and matrix places the same finite obstruction arbitrarily close to the minimizer. It is not an infinite admissible run and does not contradict the positive theorems.

Relation to earlier work

The following comparisons are bibliographic scholarship, separate from the formal theorem assertions.

Guo–Lewis [10, Theorem 4.1 and Corollary 4.4] uses smooth extension and proves a weaker subsequential conclusion for $u^2 + |v|$. The theorem in this chapter gives full convergence for that example within its centered family. Its smooth secant estimate uses established cocoercivity [4, Theorem 2.5]: local regularity is used on an open convex neighborhood, with compact buffers and, where needed, an enlarged constant. It does not require a globally Lipschitz gradient or minimization of each original summand at the center.

Invertible affine invariance is established in Lewis–Overton [2, Section 2] and Xie–Wächter [1, Theorem II.1]. The rectangular reduction in this chapter additionally identifies the point selected on an affine fiber. Schöpfer's smooth noncompact convergence results [11, Theorems 3.4 and 3.7] do not identify that selector. Exact publication priority for this formula remains unresolved; no originality claim is made.

Related quantitative work includes smooth strongly convex Armijo–Wolfe rates [18], smooth exact-search rates [19], and smooth self-concordant analysis [20]. These assumptions and line searches differ from the nonsmooth settings in this chapter. Their matrix-potential techniques are precedents, but their rate theorems do not imply these estimates. The present rates count accepted iterations, with no uniform bound on line-search trials.

Lewis–Overton [2, Theorem 4.5 and Proposition 4.11] supplies scalar Armijo–weak-Wolfe existence theory. Differentiability of the restriction to a ray does not imply full-space differentiability at acceptance. The existence argument and the finite obstruction in this chapter preserve that distinction; neither asserts an infinite-run failure.

CHAPTER 7

Homogeneous objectives without compactness

Coercive homogeneous objectives provide a broad global convergence family. This chapter goes beyond that compact case: a different potential gives noncompact value alternatives, recurrent-gradient obstructions and conditional iterate conclusions, which remain distinct from finite line-search escape and from whole-iterate convergence.

7.1 Scope and main conclusions

This chapter continues the investigation of ordinary, full-memory BFGS under the algorithmic formulation in Xie–Wächter [1]. The accepted steps satisfy fixed Armijo and **weak** Wolfe inequalities. Accepted iterates are differentiable; the inverse matrices are positive definite; no damping, restarts, smoothing, spectral safeguards, or particular line-search implementation is imposed.

The principal family is

$$f(x) = \max_{1\le i\le m} a_i^\top x, \qquad m < \infty.$$

The objective-value theorem below holds more generally for every finite convex, positively homogeneous function on the whole space. "Finite" in this broader statement means real-valued everywhere, not a finite number of pieces.

Proof status. The value and iterate statements below have reviewed Lean proofs under their stated assumptions. Their formal hypotheses and dependencies are indexed with the source claims. Finite numerical tests and publication-priority assertions are not premises of those proofs. The reusable conditional lifting implications retain any projected-run limit explicitly stated in their hypotheses.

7.1.1 Theorem 7.A: an unconditional objective-value dichotomy

Let $f : \mathbb{R}^n \to \mathbb{R}$ be finite, convex, and positively homogeneous of degree one. Every infinite admissible full-BFGS run satisfies

$$\boxed{f(x_k) \longrightarrow 0 \quad \text{or} \quad f(x_k) \longrightarrow -\infty.}$$

There is **no assumption** that the iterates, displacements, accepted step lengths, or matrices are bounded. There is also no assumption of a finite gradient alphabet for this theorem.

Consequently, if f is bounded below, then $f(x_k) \to 0 = \min f$. If f is coercive, then $x_k \to 0$, its unique minimizer. If an infinite run ever attains $f(x_k) \le 0$, its values must tend to $-\infty$; the finite-value branch has $f(x_k) > 0$ at every iteration.

7.1.2 Theorem 7.B: a many-piece extension of the common-tilt geometry

For a finite max-linear objective, let G be the set of gradients occurring infinitely often. If the objective values have a finite limit, then

$$\boxed{0 \in \operatorname{aff} G.}$$

This no longer needs a bounded-displacement hypothesis. In particular,

$$0 \notin \operatorname{aff}\{a_1, \dots, a_m\} \quad \Longrightarrow \quad f(x_k) \longrightarrow -\infty$$

for every infinite admissible run. A useful arbitrary-many-piece family is

$$\boxed{f(u,t) = t + \max_{1 \le i \le m} b_i^\top u.}$$

All its gradients have last component one, so every infinite admissible run drives the values to $-\infty$. This is an infinite-run result, not a proof of finite termination for every many-piece objective.

7.1.3 Theorem 7.C: complete iterate dichotomy in the planar unbounded-below case

For a finite max-linear objective on $\mathbb{R}^2$ with $\inf f = -\infty$, every infinite admissible run satisfies exactly one of

$$\boxed{x_k \longrightarrow 0, \quad f(x_k) \downarrow 0,}$$

or

$$\boxed{f(x_k) \longrightarrow -\infty, \quad \|x_k\| \longrightarrow \infty.}$$

In the first alternative the limit is nonstationary. The previously constructed exact three-piece example attains that alternative. In the second, convergence of the *directions* $x_k/\|x_k\|$ is not asserted.

More generally, for any finite max-linear objective, if the span of the recurrent gradients has dimension at most two, every finite-value-limit run has convergent iterates. Section 7.7 proves this broader statement, including the noncoercive bounded-below case.

7.1.4 What this does not yet establish

The objective-value classification is now unconditional in every dimension. The remaining unresolved iterate question in this investigation is the zero-value branch for noncompact objectives when the recurrent gradient span has dimension at least three. We have neither proved that all such iterate sequences converge nor constructed a nonconvergent zero-value-limit trajectory.

This distinction is essential: the results close the previous unbounded-displacement gap for **function values**, but are not being presented as a full all-dimensional classification of the **iterates**.

7.2 Notation and BFGS identities

Write

$$\begin{aligned} g_k &= \nabla f(x_k), & p_k &= -H_k g_k, & s_k &= \alpha_k p_k, \\ x_{k+1} &= x_k + s_k, & y_k &= g_{k+1} - g_k, & B_k &= H_k^{-1}, \\ q_k &= g_k^\top H_k g_k > 0, & d_k &= -g_k^\top s_k = \alpha_k q_k > 0, & & \\ t_k &= y_k^\top s_k, & \Delta_k &= f(x_k) - f(x_{k+1}). & & \end{aligned}$$

The line-search conditions are

$$f(x_{k+1}) \le f(x_k) + c_1 g_k^\top s_k, \qquad g_{k+1}^\top s_k \ge c_2 g_k^\top s_k, \qquad 0 < c_1 < c_2 < 1.$$

In particular, with $c = 1 - c_2 > 0$,

$$t_k \geq c\, d_k > 0, \qquad \beta_k := \frac{t_k}{d_k} = -\frac{y_k^\top H_k g_k}{q_k} \geq c.$$

The full inverse-BFGS update is

$$H_{k+1} = \left(I - \frac{s_k y_k^\top}{t_k}\right) H_k \left(I - \frac{y_k s_k^\top}{t_k}\right) + \frac{s_k s_k^\top}{t_k}.$$

Its Hessian form, after substituting $s_k = -\alpha_k H_k g_k$, is

$$\boxed{B_{k+1} = B_k - \frac{g_k g_k^\top}{q_k} + \frac{y_k y_k^\top}{t_k}.} \tag{7.2.1}$$

A zero gradient corresponds to a stationary termination rather than an infinite strict-descent run. Throughout, "infinite admissible" therefore includes nonzero accepted gradients and well-defined positive-curvature updates.

7.5 Homogeneous potential and proof of Theorem 7.A

7.5.1 Bounded subgradients and Euler's identity

For finite convex positively homogeneous f, every subgradient $g \in \partial f(x)$ satisfies

$$g^\top x = f(x), \qquad g^\top z \leq f(z) \quad \text{for every } z.$$

The first equality follows by applying the subgradient inequality to 0 and $2x$; the second follows by substituting that equality back into the inequality. In particular, all subgradients belong to $\partial f(0)$. In positive dimension, the unit sphere is nonempty and f is finite and continuous there, so

$$\|g\| \leq \max_{\|z\|=1} f(z) < \infty.$$

In dimension zero the space is a singleton and every gradient is zero; boundedness and the later bounded-sublevel or coercive convergence conclusions hold directly. All uses of a sphere minimum or maximum below concern positive dimension. This leaves the value theorem in arbitrary finite dimension; an infinite admissible run with nonzero gradients cannot occur in dimension zero.

Thus the bounded-gradient assumption in Theorem 2.2 is automatic, even for an infinite-piece sublinear function.

7.5.2 The exact energy identity

Set $f_k = f(x_k)$ and

$$h_k = y_k^\top x_{k+1} = d_k - \Delta_k, \qquad \ell_k = -y_k^\top x_k.$$

Convexity and homogeneity give $h_k, \ell_k \geq 0$ and $t_k = h_k + \ell_k$. Armijo and convexity give

$$c_1 d_k \leq \Delta_k \leq d_k.$$

Expanding (7.2.1) at $x_{k+1} = x_k - \alpha_k H_k g_k$ yields

$$\boxed{x_{k+1}^\top B_{k+1} x_{k+1} = x_k^\top B_k x_k - \frac{f_k^2}{q_k} + \frac{h_k^2}{t_k}.} \tag{7.5.1}$$

The explicit terms involving α_k cancel because $g_k^\top x_k = f_k$:

$$-2\alpha_k f_k + \alpha_k^2 q_k - \frac{(f_k - \alpha_k q_k)^2}{q_k} = -\frac{f_k^2}{q_k}.$$

Also,

$$\frac{h_k^2}{t_k} \le h_k = d_k - \Delta_k \le \frac{1 - c_1}{c_1}\Delta_k.$$

Consequently, with $\gamma = (1 - c_1)/c_1$,

$$\Phi_k = x_k^\top B_k x_k + \gamma f_k, \qquad \boxed{\Phi_{k+1} \le \Phi_k - \frac{f_k^2}{q_k}.} \tag{7.5.2}$$

This potential inequality was established in an earlier source note. The new ingredient is Theorem 2.2, which replaces its former bounded-step reciprocal-sum argument.

7.5.3 Every finite value limit is zero

Armijo gives strictly decreasing values. Either they tend to $-\infty$ or they have a finite limit L.

In the latter case,

$$\sum_k d_k \le \frac{f_0 - L}{c_1} < \infty,$$

so $d_k \to 0$ and Theorem 2.2 gives $\liminf q_k = 0$. On the other hand, $\Phi_k \ge \gamma L$, and telescoping (7.5.2) gives

$$\sum_k \frac{f_k^2}{q_k} < \infty.$$

If $L \neq 0$, this would imply $\sum_k 1/q_k < \infty$ and therefore $q_k \to \infty$, a contradiction.

There is also a direct pointwise argument. The potential bounds

$$x_k^\top B_k x_k \le \Phi_0 - \gamma L =: M < \infty.$$

Metric Cauchy–Schwarz and Euler's identity imply

$$f_k^2 = (g_k^\top x_k)^2 \le (g_k^\top H_k g_k)(x_k^\top B_k x_k) \le M q_k.$$

A finite nonzero limit would thus force q_k to be bounded away from zero, again contradicting Theorem 2.2. Hence $L = 0$, proving Theorem 7.A. □

The formal finite-value theorem uses the same energy identity in a shorter alternative arrangement. If $L \neq 0$, shift to a tail with $|f_k| \ge |L|/2$. The actual homogeneous corrections are nonnegative and bounded by d_k, so they are summable. Chapter 2 gives divergent reciprocal sums, and the energy identity would bound $(|L|/2)^2 \sum q_k^{-1}$ by the initial tail energy plus a finite correction sum, a contradiction. The main formal proof invokes the already available superquadratic theorem to obtain this divergence. The displayed weighted-potential budgets and pointwise metric Cauchy–Schwarz bound are also separately proved; their presence does not assert that the implementation of the principal theorem follows both arguments verbatim.

7.5.4 Bounded-below and coercive consequences

A positively homogeneous function bounded below cannot take a negative value: otherwise $f(tx) = tf(x) \to -\infty$. Thus its minimum value is zero, attained at the origin, and Theorem 7.A gives unconditional objective convergence.

If its sublevels are bounded, then $f(x) > 0$ for $x \neq 0$. Compactness of the unit sphere gives $f(x) \geq \mu\|x\|$ for some $\mu > 0$, so $f_k \to 0$ forces $x_k \to 0$.

These consequences now hold for all finite convex sublinear objectives, not only finite polyhedral maxima. No rate or finite-path-length estimate is asserted.

7.6 Affine-hull obstruction and a many-piece escape theorem

Lemma 7.3. Suppose a full-BFGS run satisfies Armijo–weak-Wolfe, has a finite gradient alphabet, and has objective values bounded below along the run. Let G be the recurrent gradients. Then $0 \in \operatorname{aff} G$.

Proof. Finite objective decrease gives $d_k \to 0$. Bounded gradients and Theorem 2.2 give $\liminf q_k = 0$, hence $\sum_k 1/q_k = \infty$.

Discard a finite prefix so that all remaining gradients belong to G. If $0 \notin \operatorname{aff} G$, a vector v exists with

$$v^\top g = 1 \quad (g \in G).$$

In particular $v^\top y_k = 0$. Equation (7.2.1) then implies

$$v^\top B_{k+1} v = v^\top B_k v - \frac{1}{q_k}.$$

Positive definiteness forces the reciprocal sum to be finite, a contradiction. □

This lemma does not need homogeneity, or even convexity, beyond what may be needed to obtain the specified run. It is a necessary condition for any finite-gradient, finite-value-limit trajectory under these BFGS and line-search assumptions.

For max-linear objectives, if $0 \notin \operatorname{aff}\{a_i\}$, no recurrent subset can have an affine hull containing zero. A finite value limit is impossible; monotonicity therefore forces $f_k \to -\infty$. This proves Theorem 7.B.

7.6.1 Connection to the two-piece paper

The model $|u| + t$ has gradients $(1, 1)$ and $(-1, 1)$, both with last component one. The arbitrary-many-piece model

$$f(u, t) = t + \max_i b_i^\top u$$

has precisely the same common-component property. The new theorem proves that an infinite accepted-iterate sequence cannot stagnate on any finite value for this entire family.

Xie–Wächter [1] proves the stronger finite-termination conclusion for its two-piece model: a search direction exposing unboundedness is reached after finitely many iterations. The argument here does **not** establish finite termination for the general many-piece family. It does show that any infinite admissible continuation must decrease the values without bound.

If $r = \dim \operatorname{span} G$, a finite-value-limit run needs at least $r + 1$ distinct recurrent gradients. Indeed, $0 \in \operatorname{aff} G$ implies that the gradient differences span the same r-dimensional space; $|G| - 1$ differences must span it. The three-gradient, rank-two stagnation example of Chapter 12 attains this lower bound.

7.7 Iterate convergence in effective dimension at most two

7.7.1 Exact invariant subspace and lifting

Suppose that, after some index K, all gradients lie in a subspace S. Let U have orthonormal columns spanning S, and define

$$\widehat{x}_k = U^\top x_k, \qquad \widehat{H}_k = U^\top H_k U, \qquad \widehat{g}_k = U^\top g_k.$$

Projection of the inverse-BFGS formula gives exactly the same inverse-BFGS update for the hatted variables and projected steps. The scalars q_k, d_k, t_k and the line-search inequalities are unchanged.

Moreover, all subsequent steps belong to the **fixed** subspace $H_K S$. To see this, initially $H_K S$ is the image of S; if $H_k S \subset H_K S$, then $s_k \in H_K S$, and the inverse update maps S back into $H_K S$, since $g_k, y_k \in S$. Induction proves the claim.

The restriction of $U^\top$ to $H_K S$ is invertible because $U^\top H_K U \succ 0$. Therefore

$$\boxed{x_k = x_K + L_K(\widehat{x}_k - \widehat{x}_K), \qquad L_K = H_K U (U^\top H_K U)^{-1}.} \tag{7.7.1}$$

In particular, convergence of the projected iterates implies convergence of the original iterates. If $\widehat{x}_k \to 0$, the limit is

$$\boxed{x_* = x_K - H_K U (U^\top H_K U)^{-1} U^\top x_K.} \tag{7.7.2}$$

The projected Hessian is $(U^\top H_k U)^{-1}$, not $U^\top B_k U$. This distinction is necessary for a valid reduction.

7.7.2 A two-gradient endpoint observation

Assume $f_k \to 0$. Then $d_k \to 0$, and every endpoint satisfies

$$g_{k+1}^\top x_{k+1} = f_{k+1} \longrightarrow 0, \qquad g_k^\top x_{k+1} = f_k - d_k \longrightarrow 0. \tag{7.7.3}$$

In two dimensions, if g_k and g_{k+1} are linearly independent, these two equations determine x_{k+1}. There are only finitely many distinct pairs, so their inverse norms are uniformly bounded. Thus the endpoints of all independent-gradient transitions converge to zero, even if the full sequence was not known to be bounded.

An unchanged nonzero gradient cannot satisfy weak Wolfe on a strict descent step. Distinct gradients that are positive multiples of one another also cannot both be active at positive-value iterates: among positive multiples of a slope, only the largest multiple can attain a positive maximum. Consequently, any dependent transition on the positive-value branch is between opposite slopes.

7.7.3 The unbounded-below planar case

If the global objective is unbounded below, its slope set cannot contain an opposite pair $a, -\kappa a$ with $\kappa > 0$: their maximum alone is nonnegative everywhere. Every accepted gradient transition is therefore linearly independent in the plane.

Equation (7.7.3) now gives $x_k \to 0$ directly. Combined with Theorem 7.A, this proves Theorem 7.C. In the other branch, global Lipschitz continuity gives $|f(x)| \le M\|x\|$, so $f_k \to -\infty$ implies $\|x_k\| \to \infty$.

For an unbounded-below max-linear objective there is no stationary point: a stationary point of a convex function would be a global minimizer, and no minimizer exists. Thus the zero limit is genuinely nonstationary.

7.7.4 The general planar finite-value branch

For completeness, opposing-gradient transitions can also be handled, giving convergence for **all** planar finite max-linear objectives, including noncompact bounded-below ones.

Call an endpoint of an independent-gradient transition an *anchor*. By (7.7.3), anchors converge to zero.

Consider consecutive anchors x_a, x_b. If there is no intermediate iterate, there is nothing to prove. Otherwise, all gradients from g_a through g_{b-1} lie on a single gradient line; the intermediate dependent transitions alternate between opposite slopes. By the invariant-subspace argument, all steps from x_a up to and including the step arriving at x_b lie on a fixed primal line. Therefore all points in this block are collinear with its endpoints.

Write its initial gradient as g, and the next opposite gradient as $-\kappa g$, with $\kappa > 0$. The first dependent transition has

$$d_a = f_a + f_{a+1}/\kappa.$$

Armijo implies

$$f_{a+1} \le \frac{1 - c_1}{1 + c_1/\kappa} f_a \le (1 - c_1) f_a,$$

so monotonicity gives $f_b \le (1 - c_1) f_a$. Every accepted point in the block obeys

$$-f_a/\kappa \le g^\top x \le f_a,$$

because g and $-\kappa g$ are both slopes of the objective and $f(x) \le f_a$. The endpoint difference satisfies

$$g^\top (x_b - x_a) \le -c_1 f_a.$$

Writing $x = x_a + \theta(x_b - x_a)$ consequently gives

$$0 \le \theta \le \frac{1 + 1/\kappa}{c_1}.$$

Only finitely many slope pairs occur, so this bound is uniform across blocks. Since both anchor endpoints tend to zero, every intermediate point does too. Arbitrarily long blocks cause no difficulty.

If there are only finitely many anchors, the tail gradients lie on a fixed line, and all tail iterates lie on one fixed primal line. The slope coordinate tends to zero because $f_k \to 0$ and only finitely many nonzero slope multipliers occur. Positive definiteness makes that slope coordinate nonconstant along the primal line, so the entire tail converges to its unique intersection with the zero-slope hyperplane.

This proves planar iterate convergence in every finite-value-limit case. Reducing to the span of the recurrent gradients and using (7.7.1) proves the same conclusion whenever that span has dimension at most two in any ambient dimension. The limit may be nonzero in the original coordinates.

7.8 A noncompact convergence theorem in arbitrary dimension

One source of noncompactness is a subspace of directions on which the function is constant. When this is the only source of noncompactness, the case is fully tractable in all dimensions.

Let

$$S = \operatorname{span}\{a_1, \dots, a_m\}, \qquad P = \operatorname{conv}\{a_1, \dots, a_m\},$$

and suppose $0 \in \operatorname{relint} P$, with relative interior taken in S. Then f depends only on the projection to S and is coercive there:

$$f(x) \geq \mu \|U^\top x\| \quad \text{for some } \mu > 0.$$

Theorem 7.A implies $U^\top x_k \to 0$. The invariant subspace formula with $K = 0$ gives

$$\boxed{x_k \longrightarrow x_0 - H_0 U (U^\top H_0 U)^{-1} U^\top x_0.} \tag{7.8.1}$$

The limit is a minimizer, and in fact the projection of x_0 onto $S^\perp$ in the H_0^{-1} metric. It is independent of the choices of admissible accepted lengths. This includes all finite polyhedral seminorms and noncoercive gauges whose minimizer set is a linear subspace.

The same argument applies after a finite prefix if $0 \in \operatorname{relint} \operatorname{conv} G$ for the recurrent gradients. The projected tail objective is then coercive even if some original, nonrecurrent slopes have disappeared. Formula (7.7.2) supplies the limit.

For $f \equiv 0$, the algorithm is already stationary; there is no nonzero-gradient infinite run. The projection formula above concerns a nontrivial slope span.

Relation to earlier work

The following comparisons are bibliographic scholarship, separate from the formal theorem assertions.

The Euclidean-norm consequence is already Guo–Lewis [10, Corollary 4.2], while Lewis–Overton [2, Section 5.1] analyzes the scalar absolute-value model with its stated line-search conventions. The theorem here treats the larger finite convex homogeneous class without assuming bounded iterates.

The precise prior result is Xie–Wächter's 2017 preprint [1, Theorem IV.1]. Its finite detection of unbounded descent in the two-piece model is stronger there than the infinite-run value conclusion proved here for the many-piece family.

The invariant-subspace argument extends the affine-invariance background [1, Theorem II.1]. Exact prior publication of the resulting H_0^{-1}-metric selector for this noncompact homogeneous class remains unresolved, as for Chapter 6's rectangular lifting formula. This uncertainty concerns attribution, not an additional hypothesis of the theorem.

CHAPTER 8

Affine criticality and stratified geometry

Affine balance concerns equal-rate descent, whereas convex balance concerns stationarity. This distinction gives useful local geometry but does not by itself force an algorithm to visit it.

8.1 Equal-rate descent and affine criticality

This chapter preserves the geometric propositions of the original affine-criticality note. They do not depend on the BFGS principal-minor lemma. The originally proposed dynamical step is now proved only in the completed settings of Chapters 9 and 11; no blanket theorem for all finite smooth maxima is asserted.

For a finite convex F, put

$$V(p) = \operatorname{span}(\partial F(p) - \partial F(p)), \quad U(p) = V(p)^{\perp}, \quad a(p) = \operatorname{proj}_{\operatorname{aff} \partial F(p)} 0.$$

The affine hull equals $a(p) + V(p)$ and $a(p) \perp V(p)$. Hence every subgradient satisfies

$$P_{U(p)} g = a(p), \qquad g^T a(p) = \|a(p)\|^2.$$

The condition

$$0 \in \operatorname{aff} \partial F(p) \Longrightarrow 0 \in \partial F(p) \tag{8.1}$$

is therefore equivalent, at every nonoptimal point under consideration, to the existence of a direction v with $g^T v = -1$ for every $g \in \partial F(p)$. One may take $v = -a(p)/\|a(p)\|^2$. Conversely such an equal-rate direction rules out zero in the affine hull. This is stronger than ordinary strict descent, which requires only negative, not equal, inner products.

On a compact admissible run, all cluster points share the limiting value ℓ. If at least one cluster point is affine critical and (8.1) holds on the basin, that point is optimal, so $\ell = F_*$. Compactness gives convergence to the solution set, and uniqueness gives convergence of the entire sequence. This reduction does not assume an iterate limit. It would be incorrect to replace affine criticality of a cluster point by vanishing gradients at differentiable iterates: for example, the gradients of $|x|$ have unit magnitude away from its minimizer.

8.2 Why pointwise affine separation need not be uniform

Let $\gamma(p) = \operatorname{dist}(0, \operatorname{aff} \partial F(p))$. A useful quantitative condition is a positive minimum of γ on every compact positive objective band. Pointwise nonvanishing plus compactness does not establish it when ranks change.

Consider

$$F(u, v) = u + u^2 + v^2 + |uv| = \max\{u + u^2 + v^2 + uv,\ u + u^2 + v^2 - uv\}.$$

Both branch Hessians have eigenvalues 1 and 3, so F is strongly convex, with unique minimizer $(-1/2, 0)$ and value $-1/4$. At $(0, t)$, $t \neq 0$, the active gradients are $(1 + t, 2t)$ and $(1 - t, 2t)$, giving $\gamma(0, t) = 2|t|$. At zero the gradient is $(1, 0)$, so $\gamma(0, 0) = 1$. On the other switching axis $\gamma(u, 0) = |1 + 2u|$, vanishing only at the optimizer. Smooth points are affine critical only when stationary. Thus (8.1) holds globally, but $\gamma(0, t) \to 0$ while the objective gap tends to $1/4$.

This is a geometric rank-loss example, not a BFGS counterexample. A sufficiently small neighborhood of its actual optimizer excludes the rank-loss point.

8.3 Nondegenerate smooth maxima give local affine-critical correctness

Let $F = \max_{i \le m} f_i$ with convex C^1 branches and a minimizer x_*. Restrict to I_*, the active branches there; inactive branches have positive gaps and disappear in a sufficiently small neighborhood. Suppose the active gradients are affinely independent and admit strictly positive coefficients

$$\lambda_i^* > 0, \quad \sum_{i \in I_*} \lambda_i^* = 1, \quad \sum_{i \in I_*} \lambda_i^* \nabla f_i(x_*) = 0.$$

Then (8.1) holds near x_*.

Proof. Let $A(x)$ have columns $(\nabla f_i(x), 1)$ for $i \in I_*$. Full column rank at x_* persists nearby. At an affine-critical point p, extend the affine coefficients of its active gradients by zero on missing indices. They solve

$$A(p)\lambda = (0, \dots, 0, 1)^T.$$

Whenever this system is solvable, its unique solution equals

$$\lambda(p) = (A(p)^T A(p))^{-1} A(p)^T (0, \dots, 0, 1)^T.$$

This continuous expression equals the strictly positive vector λ^* at x_*. It remains positive nearby. No missing index can therefore have zero coefficient, and the affine combination is a convex combination. Thus $0 \in \partial F(p)$. □

This is the source note's barycentric-stability argument, closely related to the geometric nondegeneracy results it cites. It is not itself a BFGS convergence theorem.

There is also local lower semicontinuity of γ. For each nonempty $J \subset I_*$, the distance to the affine hull of the gradients in J is continuous, because the relevant difference Gram matrix remains nonsingular. For $p_j \to p$, pass to a subsequence of constant active set $J \subset I(p)$. Then

$$\lim \gamma(p_j) = \operatorname{dist}(0, \operatorname{aff}\{\nabla f_i(p) : i \in J\}) \ge \gamma(p).$$

On a compact positive band, local (8.1) therefore gives a strictly positive minimum. This establishes visibility, not the full transport needed in arbitrary dimension.

8.4 Semialgebraic objectives have finitely many affine-critical values

Proposition 8.1 (A low-level affine-critical gap)**.** *If $F : \mathbb{R}^n \to \mathbb{R}$ is finite convex and semialgebraic and attains F_*, then some $\epsilon > 0$ satisfies*

$$F(p) < F_* + \epsilon, \quad 0 \in \operatorname{aff} \partial F(p) \quad \Longrightarrow \quad 0 \in \partial F(p).$$

Proof. The subgradient graph is semialgebraic: its global supporting-plane condition is a quantified polynomial condition involving the graph of F, and real projection and Boolean closure eliminate the quantifiers. An affine representation of zero can be chosen with at most $n + 1$ subgradients, with real coefficients summing to one; these coefficients need not be nonnegative. It follows that both the affine-critical set C_{aff} and its lifted graph

$$\widehat{C}_{\text{aff}} = \{(p, F(p)) : p \in C_{\text{aff}}\}$$

are semialgebraic.

Use a finite semialgebraic cover of $\widehat{C}_{\text{aff}}$ whose pieces are internally connected by continuous paths that are C^∞ outside finitely many interior parameters. For a path $(p(t), q(t))$ inside one piece, $q(t) = F(p(t))$. Coordinate projection therefore gives derivatives of both p and the actual composition $F \circ p$ at every interior time outside one finite exceptional set. No differentiability of F at $p(t)$ is assumed.

At such a time, for every $g \in \partial F(p(t))$, the supporting inequality makes

$$u \longmapsto F(p(u)) - g^T p(u)$$

have a local minimum at t. Differentiation gives

$$(F \circ p)'(t) = g^T p'(t) \qquad \text{for every } g \in \partial F(p(t)).$$

An affine combination of these subgradients equal to zero forces $(F \circ p)'(t) = 0$. The mean-value theorem makes the value constant between consecutive exceptional parameters, and continuity joins those constants at the junctions and endpoints. Thus each piece of the finite lifted cover has only one objective value, proving that there are finitely many affine-critical values.

Since F_* is one of these values, choose $\epsilon > 0$ no larger than the least strictly positive difference between an affine-critical value and F_*, or choose any positive ϵ if there is no larger value. The strict inequality in the proposition then forces $F(p) = F_*$. Convexity turns this equality into $0 \in \partial F(p)$. □

Here and below, a piecewise C^1 path is continuous on $[0, 1]$ and C^1 at every interior parameter outside a finite set. No derivative at 0 or 1, or derivative extension across a junction, is required. The geometric facts used above are established by real-polynomial projection, finite root and sign refinements, and graph-and-band path constructions; the references [7] supply mathematical context rather than unproved premises.

The affine-critical set also has finitely many actual semialgebraic path classes, where two points are related when a continuous path with semialgebraic restricted graph joins them inside the set. Each class is semialgebraic and internally connected by such paths. Every continuous semialgebraic path admits the finite-exception regularity just specified on the same curve; for finite convex semialgebraic F, its actual composition with F is likewise continuous and smooth off finitely many interior parameters. The objective is constant on each class. Ordinary topological connected components are separately finite and semialgebraic, with constant objective value; their identification with path classes is not needed here.

This proposition concerns affine criticality rather than merely Clarke criticality. It does not imply local BFGS convergence without a dynamical theorem forcing an affine-critical cluster point. Chapters 9 and 11 provide such theorems under their respective additional hypotheses.

8.5 What remains a conjecture

The recorded general-dimensional conjecture concerns finite convex $C^{2,1}$ maxima with unique minimizer, compact low sublevels, affinely independent active gradients, positive optimal weights, and positive weighted Hessian on

$$U_* = \{v : (\nabla f_i(x_*) - \nabla f_1(x_*))^T v = 0 \ \forall i \in I_*\}.$$

The proposed conclusion is local full-BFGS convergence for arbitrary SPD initialization. The planar theorem proves a strict-contour version; the fixed-normal theorem proves a different arbitrary-dimensional class. Neither establishes this full conjecture. Straightening an active manifold by a nonlinear coordinate change does not preserve ordinary BFGS, and cannot supply the missing implication.

Relation to earlier work

The following comparisons are bibliographic scholarship, separate from the formal theorem assertions.

The common projected subgradient and affine-hull decomposition belong to established VU and partial-smoothness geometry [26, 8]. The equal-rate direction in this chapter is their elementary projection consequence, included to connect that geometry to the BFGS arguments.

Discontinuity of subgradient affine hulls is established in Liu–Sagastizábal–Solodov, Section 3.1 [22]. The example in this chapter supplies the stated finite, strongly convex realization; the general rank-loss mechanism is not claimed as new.

The argument uses standard stability of full-rank augmented gradient systems and positive multipliers. Compare Lewis–Wylie, Propositions 5.5–5.6 of the 2019 preprint [6], and Gebken's published Lemma 3.5 [5]. This geometric deduction is included with its proof.

This is a corollary of established stratified projection and definable critical-value machinery [7]; see Proposition 4, Lemma 8 and the proof of Corollary 9 in the arXiv version. All subgradients have the same tangential projection on a compatible smooth stratum. Projecting an affine combination equal to zero, whose coefficients sum to one, makes that stratum gradient zero. This additional deduction justifies the affine-critical conclusion beyond the printed Clarke-critical corollary.

CHAPTER 9

Planar switching and affine-critical cluster points

Curved switching interfaces can defeat inward radial accuracy even in a convergent problem. The planar contour construction supplies moving comparison geometry; the explicit three-quadratic basin demonstrates its added scope. The signed determinant and one-sided transport proof make the resulting convergence theorem independent of assumed matrix behavior.

9.1 Statements and assumptions

Let $F : \mathbb{R}^2 \to \mathbb{R}$ be finite and convex, attain F_*, and have a compact sublevel

$$K = \{x : F(x) \le b\}, \qquad b > F_*.$$

A point is *affine critical* when

$$p \in \mathcal{C}_{\mathrm{aff}} := \{p : 0 \in \operatorname{aff} \partial F(p)\}.$$

Ordinary stationarity, $0 \in \partial F(p)$, implies affine criticality, but not conversely.

We use two objective-level hypotheses.

Strictly convex positive sublevels. For every $F_* < t \le b$, the convex body $K_t = \{F \le t\}$ is strictly convex: its boundary contains no nontrivial line segment. This is a condition on the contours, not a uniform lower Hessian bound. Strong convexity of F suffices.

Regular switching away from affine criticality. At every $p \in K \setminus \mathcal{C}_{\mathrm{aff}}$, either F is convex C^2 on a neighborhood of p, or on such a neighborhood

$$F = \max\{f_1, f_2\},$$

where f_1, f_2 are convex C^2, both are active at p, and $\nabla f_1(p), \nabla f_2(p)$ are linearly independent. In particular, two different smooth branches meet transversally both to each other and to the positive level. No assumptions are made on the local structure at an affine-critical point.

Theorem 9.1 (Forced affine-critical cluster point)**.** *Under these hypotheses, every infinite admissible full-BFGS run starting in K has at least one affine-critical cluster point. No iterate limit, bounded condition number, limiting eigenspace, matrix accuracy at initialization, or finite path length is assumed.*

Here admissibility means $H_0 \succ 0$, differentiable accepted iterates with nonzero gradients, $s_k = -\alpha_k H_k g_k$, $\alpha_k > 0$, and fixed Armijo–weak-Wolfe conditions

$$F(x_{k+1}) \le F(x_k) + c_1 g_k^T s_k, \qquad g_{k+1}^T s_k \ge c_2 g_k^T s_k, \qquad 0 < c_1 < c_2 < 1. \tag{9.1}$$

The usual full inverse update is

$$H_{k+1} = \left(I - \frac{s_k y_k^T}{t_k}\right) H_k \left(I - \frac{y_k s_k^T}{t_k}\right) + \frac{s_k s_k^T}{t_k}, \quad y_k = g_{k+1} - g_k, \quad t_k = y_k^T s_k > 0. \tag{9.2}$$

The positivity $t_k > 0$ in the update is derived, not an additional assumption on later matrices: $H_k \succ 0$ and $g_k \neq 0$ give $q_k = g_k^T H_k g_k > 0$, and weak Wolfe gives $t_k \ge (1 - c_2)\alpha_k q_k > 0$. The full update then preserves positive definiteness, starting from $H_0 \succ 0$. Existence of differentiable acceptable lengths on every exceptional search ray is not part of the theorem.

Corollary 9.2 (No false affine criticality gives convergence)**.** *If, in addition, $\mathcal{C}_{\text{aff}} \cap K \subseteq \arg\min F$, then every such run satisfies $F(x_k) \to F_*$ and $\operatorname{dist}(x_k, \arg\min F) \to 0$. A unique minimizer gives convergence of the entire iterate sequence. A local version holds when the hypotheses hold only on a sufficiently low compact sublevel inside a structural neighborhood of a unique minimizer.*

Corollary 9.3 (Nondegenerate planar smooth maxima)**.** *Suppose a finite convex $F : \mathbb{R}^2 \to \mathbb{R}$ has a unique minimizer x_* and, near x_*,*

$$F = \max_{i \le m} f_i,$$

with convex C^2 branches. Suppose sufficiently low positive sublevels are strictly convex. Let I_* be the active set at x_*, and assume its gradients are affinely independent and admit strictly positive optimal weights:

$$\lambda_i^* > 0, \quad \sum_{i \in I_*} \lambda_i^* = 1, \quad \sum_{i \in I_*} \lambda_i^* \nabla f_i(x_*) = 0. \tag{9.3}$$

Then some $\varepsilon > 0$ satisfies

$$F(x_0) < F_* + \varepsilon \quad \Longrightarrow \quad x_k \to x_*$$

for every infinite admissible full-BFGS run, with arbitrary $H_0 \succ 0$ and any fixed valid line-search constants. The basin can be chosen independently of these algorithmic choices. A simple sufficient replacement for the strictly convex-sublevel hypothesis is a common positive lower Hessian bound on the finitely many locally relevant branches.

This proves a planar, strictly-convex-contour version of the smooth-max conjecture recorded in Chapter 8. It does not prove its general-dimensional or merely weighted-tangential-curvature version.

9.2 Why one affine-critical cluster point is enough

Armijo makes the objective values decrease to a limit $\ell \ge F_*$. Every cluster point has value ℓ. If any cluster point is stationary, then $\ell = F_*$. Compactness gives distance to the solution set tending to zero; uniqueness gives whole-sequence convergence.

The geometric condition

$$0 \in \operatorname{aff} \partial F(p) \implies 0 \in \partial F(p) \tag{9.4}$$

therefore reduces the problem to Theorem 9.1. This reduction was proved in Chapter 8; the missing algorithmic assertion is the subject of this chapter.

For completeness, if $a(p)$ is the minimum-norm point of $\operatorname{aff} \partial F(p)$, then

$$g^T a(p) = \|a(p)\|^2 \qquad (g \in \partial F(p)).$$

At a non-affine-critical point, $-a(p)/\|a(p)\|^2$ decreases every supporting model at rate one. Thus (9.4) says that every nonoptimal point retains equal-rate descent across its active models. It does not say that the gradients at differentiable accepted points tend to zero.

9.3 Vanishing steps without an assumed iterate limit

Put

$$q_k = g_k^T H_k g_k, \quad d_k = -g_k^T s_k = \alpha_k q_k, \quad \Delta_k = F(x_k) - F(x_{k+1}), \quad c = 1 - c_2 > 0.$$

Convexity and (9.1) give

$$d_k > 0, \qquad c_1 d_k \le \Delta_k \le d_k, \qquad t_k \ge c d_k. \tag{9.5}$$

In particular $\sum_k d_k < \infty$. A finite convex function has bounded subgradients on a compact set: compare supporting planes at $x \in K$ with values on a fixed enlarged compact neighborhood. Hence the accepted gradients are bounded, and Chapter 2 applies.

Lemma 9.4 (Strict contours force vanishing steps)**.** *If $F(x_k) \downarrow \ell > F_*$ and K_ℓ is strictly convex, then $s_k \to 0$.*

Proof. Otherwise compactness gives a subsequence with $x_k \to x$ and $x_{k+1} \to z \neq x$. For every $0 \le \theta \le 1$, the supporting inequality at the accepted point x_k and convexity give

$$F(x_k) - \theta d_k \le F((1-\theta)x_k + \theta x_{k+1}) \le (1-\theta)F(x_k) + \theta F(x_{k+1}).$$

Since $d_k \to 0$ and both endpoint values tend to ℓ, continuity shows that $F = \ell$ on $[x, z]$. A level point of a continuous convex function lies on the boundary of its sublevel whenever some point has strictly smaller value; here a minimizer has value $F_* < \ell$. Thus $[x, z]$ is a nontrivial boundary segment of K_ℓ, contradicting strict convexity. This argument does not require a convergent subsequence of gradients. □

We suppose, for a contradiction, that the compact cluster set Γ contains no affine-critical point. Then $\ell > F_*$, every point of Γ satisfies the regular-switching hypothesis, and Lemma 9.4 applies. This does not assume that Γ is a singleton, or even use its connectedness.

9.4 Joining comparison directions around a regular positive level

The geometric construction is as follows. Fix a compact subset $C \subset \{F = \ell\}$ containing no affine-critical point and satisfying the regularity hypothesis. Eventually we use $C = \Gamma$.

9.4.1 Finitely many relevant corners and their level velocities

Near a nonsmooth point of C, $F = \max\{f_1, f_2\}$ and the matrix with rows $\nabla f_1^T, \nabla f_2^T$ is nonsingular. The equations

$$f_1(p) = f_2(p) = t$$

therefore define a C^2 point curve $p_j(t)$ near $t = \ell$. On the level ℓ, the switching point is isolated. Smooth points of C have smooth neighborhoods. A compactness argument gives finitely many relevant curves $p_j(t)$ and a neighborhood of C whose nonsmooth points on nearby levels lie on these curves. Explicitly, an infinite sequence of distinct level-ℓ corners in a compact regular neighborhood would accumulate at a smooth chart, or at a chart with only one solution of $f_1 = f_2 = \ell$; either is impossible. Shrinking the neighborhood and the level interval removes all other switching curves.

Define

$$v_j(t) = p_j'(t), \qquad \nabla f_{j,1}(p_j(t))^T v_j(t) = \nabla f_{j,2}(p_j(t))^T v_j(t) = 1. \tag{9.6}$$

This vector is tangent to the switching curve and detects a fixed nonzero increase in the objective. It is C^1 in t.

9.4.2 Why strictly convex contours permit global joining

At $p_j(t)$, the nonnegative combinations of its two active gradients generate a closed arc of unit supporting normals of K_t. This arc has width less than π, and every unit normal n in it satisfies $n^T v_j(t) > 0$.

The normal arcs of distinct corners at $t = \ell$ are disjoint. Indeed, a shared normal would support K_ℓ at two different points, making their connecting segment lie on its boundary. Strict convexity prohibits this. The finitely many compact arcs consequently have positive angular separation. After slightly enlarging them and shrinking the interval of t, choose *fixed* smooth angular cutoffs $0 \le \chi_j \le 1$ with disjoint supports, equal to one on neighborhoods of the entire moving normal arcs. The supports can be chosen where $n^T v_j(t) > 0$ uniformly.

Let

$$n(\theta) = (\cos\theta, \sin\theta), \qquad T(\theta) = (-\sin\theta, \cos\theta),$$

and put

$$h(t, \theta) = 1 + \sum_j \chi_j(\theta)\,[n(\theta)^T v_j(t) - 1]. \tag{9.7}$$

The supports are disjoint, so this is a convex combination of positive numbers at each angle. Hence h is bounded above and bounded away from zero. It is C^1 in t and smooth in θ, with the mixed derivatives used below bounded. No convex-support-function property of h is asserted or needed.

At a smooth point near C, write $n(\theta(x)) = \nabla F(x)/\|\nabla F(x)\|$ and define

$$w(x) = h(F(x), \theta(x))n(\theta(x)) + h_\theta(F(x), \theta(x))T(\theta(x)). \tag{9.8}$$

These expressions are periodic in θ, so no choice of a global angle coordinate is needed.

Where $\chi_j = 1$, $h = n^T v_j$ and $h_\theta = T^T v_j$, so

$$w(x) = v_j(F(x)). \tag{9.9}$$

This holds on a whole neighborhood of the corresponding switching curve, on both sides. Define w by (9.9) on the curve itself. The field thus extends continuously and locally Lipschitzly across the gradient jump. Away from the curves it is C^1. On a smaller compact neighborhood of C, all close-pair Lipschitz and piecewise derivative bounds are uniform.

At smooth points,

$$\nabla F(x)^T w(x) = \|\nabla F(x)\| h(F(x), \theta(x)) > 0. \tag{9.10}$$

At a switching point, every subgradient has inner product one with w. Compactness on a positive level neighborhood gives a uniform positive lower bound in (9.10) and uniform bounds on w.

9.4.3 The planar property that makes transport inexpensive

At fixed t, differentiation gives the exact identity

$$\partial_\theta(hn + h_\theta T) = (h + h_{\theta\theta})T. \tag{9.11}$$

Therefore, at a smooth point, $Dw(x)$ maps the one-dimensional tangent line of the level curve into itself. It may vary arbitrarily in the normal direction. In the orthonormal basis (T, n), it has the form

$$Dw = \begin{pmatrix} a & b \\ 0 & e \end{pmatrix}. \tag{9.12}$$

The coefficients are bounded in the compact neighborhood. The sign of $h + h_{\theta\theta}$ is immaterial.

9.5 Three endpoint estimates

An eligible pair means differentiable x, z satisfying (9.5), with $s = z - x$, $g = \nabla F(x)$, $y = \nabla F(z) - g$, and $t = y^T s$. For each fixed valid pair (c_1, c_2), the following three estimates hold for the same objective-constructed field w. There are an open set $U \supset C$, a compact buffer $Q \supset \overline{U}$, and positive uniform distance and visibility constants such that every sufficiently close eligible pair $x, z \in U$ is covered, with its entire segment in Q. On Q, every subgradient at a given point has the same strictly positive pairing with w, bounded uniformly away from zero. All constants and neighborhoods are chosen before the endpoint pair; none depends on a BFGS matrix or on an assumed trajectory limit.

Lemma 9.5 (Uniform transversality)**.** *There is $\kappa > 0$ such that*

$$|\det(s, w(x + \theta s))| \geq \kappa \|s\| \quad (0 \leq \theta \leq 1)$$

for all sufficiently close eligible pairs.

Proof. Otherwise take pairs coalescing to p in the compact neighborhood, unit steps $s/\|s\| \to e$, and segment points where the normalized determinant tends to zero. Continuity of w implies e is parallel to $w(p)$. Subgradient limits $a, b \in \partial F(p)$ have the same strictly positive inner product

with $w(p)$: this is automatic at a smooth point and follows from (9.6) at a corner. Descent selects $e = -w(p)/\|w(p)\|$, so $d/\|s\| \to -a^T e > 0$. But

$$t/\|s\| \to (b-a)^T e = 0,$$

contradicting $t \geq (1-c_2)d$. Compactness makes the bound uniform, including all segment points. □

Lemma 9.6 (Transport is charged to objective decrease)**.** *For the same pairs, there are scalars $\lambda > 0, \beta$ satisfying*

$$w(z) = \lambda w(x) + \beta s, \qquad |\log \lambda| \leq Cd. \tag{9.13}$$

In particular $\lambda^2 \leq 1 + C'd$ for sufficiently small d.

Proof. Let $q(\theta) = x + \theta s$, $\varphi(\theta) = F(q(\theta))$, and $D(\theta) = \det(s, w(q(\theta)))$. Transversality makes the continuous function D nonzero throughout $[0,1]$, hence of constant sign.

On a smooth part of the segment, use (9.12). The ordered orthonormal basis (T, n) has determinant -1. Consequently, with $s = s_T T + s_n n$, the correctly oriented identity is

$$\det(s, Dw\, s) = s_n[(a-e)s_T + bs_n].$$

Its absolute value, the uniform derivative bound, transversality and the positive lower bound on $\|\nabla F\|$ give the required determinant-derivative estimate. In a corner neighborhood, $w \circ q = v_j \circ \varphi$. The convex function φ has an actual right derivative at every $\theta < 1$; the ordinary one-sided chain rule therefore handles corner points as well as smooth points. We obtain, for one $C \geq 0$,

$$\left|\frac{D'_+(\theta)}{D(\theta)}\right| \leq C|\varphi'_+(\theta)|, \qquad 0 \leq \theta < 1.$$

No finite bound on intersections with a switching curve is used.

Set $h(\theta) = \log|D(\theta)|$. Convexity and $\varphi'(0) = -d$ imply $\varphi'_+(\theta) \geq -d$, and hence

$$|h'_+(\theta)| \leq C|\varphi'_+(\theta)| \leq C(\varphi'_+(\theta) + 2d). \tag{9.14}$$

Apply the one-sided derivative comparison theorem to the continuous observable $h - h(0)$ and the barrier

$$B(\theta) = C[\varphi(\theta) - \varphi(0) + 2d\theta].$$

The initial value is zero and the right-derivative norm is bounded by B'_+. Thus $|h(1) - h(0)| \leq B(1) \leq 2Cd$, since $\varphi(1) \leq \varphi(0)$. Because $D(1)/D(0) > 0$, this is $|\log(D(1)/D(0))| \leq 2Cd$. Finally $s, w(x)$ form a basis, and solving in that basis gives $w(z) = \lambda w(x) + \beta s$ with $\lambda = D(1)/D(0) > 0$. Absorb the factor two into the stated constant. This proves the claim. □

Lemma 9.7 (Bounded secant correction)**.** *There is a uniform $C_0 < \infty$ such that*

$$\frac{(y^T w(z))^2}{t} \leq C_0$$

for all sufficiently close eligible pairs in the same neighborhood.

Proof. Use a finite buffered chart cover. For close pairs in a common smooth chart, or with the same active branch in a two-branch chart, the local convex gradient-increment estimate gives $\|y\|^2 \leq Lt$ for a uniform constant L. One proof subtracts a tangent plane, applies the descent estimate to a short step opposite the gradient difference within the larger convex chart, and combines the resulting one-sided inequalities. Compact buffering supplies a single constant; no positive lower Hessian bound is required. Bounded w proves the claim.

For different active branches i, j in a two-branch chart, the segment has a tie point p. At that point the jump is orthogonal to $w(p)$. The branch gradients and w are Lipschitz, so

$$|y^T w(z)| \le C_1 \|s\|. \tag{9.15}$$

For the denominator put $a = \nabla f_i(x)$, $b = \nabla f_j(x)$, and $r = \nabla f_j(z) - \nabla f_j(x)$, with $\|r\| \le L\|s\|$. Then

$$a^T s = -d, \qquad b^T s = t - d - r^T s.$$

The two gradient rows are uniformly independent on the chart. Thus

$$\|s\| \le A(2d + t + L\|s\|^2) \le A\left(1 + \frac{2}{1 - c_2}\right) t + AL\|s\|^2.$$

For small s, absorb the final term to obtain $t \ge c_0\|s\|$. Combining with (9.15) bounds the correction by a constant times $\|s\|$, hence uniformly. This is a crossing-specific linear curvature estimate, not a uniform positive Hessian assumption. □

9.6 The BFGS contradiction

Proof of Theorem 9.1. Suppose the compact cluster set Γ misses $\mathcal{C}_{\text{aff}}$. Then the limiting value ℓ is strictly above F_*; otherwise all cluster points are minimizers. Lemma 9.4 gives $s_k \to 0$. The distance of x_k to Γ tends to zero, so the entire tail lies in the buffered regular neighborhood constructed above with $C = \Gamma$.

Let $B_k = H_k^{-1}$ and

$$E_k = w(x_k)^T B_k w(x_k), \qquad M_k = B_k - \frac{g_k g_k^T}{q_k} \succeq 0, \qquad M_k s_k = 0.$$

The Hessian form of (9.2), the representation in Lemma 9.6, and $M_k s_k = 0$ give the exact identity

$$E_{k+1} = \lambda_k^2 \left(E_k - \frac{(g_k^T w(x_k))^2}{q_k}\right) + \frac{(y_k^T w(x_{k+1}))^2}{t_k}. \tag{9.16}$$

The parenthesized expression is nonnegative. Uniform visibility and the last two lemmas imply, on a tail,

$$E_{k+1} \le (1 + Cd_k)(E_k - m/q_k) + C_0, \qquad m > 0.$$

Set $P_K = 1$ and $P_{k+1} = P_k(1 + Cd_k)$. Since $\sum d_k < \infty$, $1 \le P_k \le \bar{P} < \infty$. Divide by P_{k+1} and telescope. Positivity gives

$$\frac{m}{\bar{P}} \sum_{k=K}^{N-1} \frac{1}{q_k} \le E_K + C_0(N - K).$$

This contradicts Chapter 2, which forces the average of $1/q_k$ to diverge. Thus $\Gamma \cap \mathcal{C}_{\text{aff}} \ne \varnothing$. □

The argument constructs a field near the *whole hypothetical cluster set*. It does not first select one cluster point and assume all later iterates stay near it. Nor is the field an extra algorithmic input: it is an analytical object derived from the objective, independent of every H_k.

9.7 Local consequences and natural assumptions

Proof of Corollary 9.2. All cluster points have the limiting objective value. An affine-critical cluster point is optimal by the extra hypothesis; hence that value is F_*. Compactness gives distance to the solution set tending to zero. Under uniqueness every convergent subsequence has the same limit, proving whole-sequence convergence. For the local statement, compactness and uniqueness put a sufficiently low sublevel inside any given neighborhood of x_*; Armijo then traps the accepted iterates there. □

Proof of Corollary 9.3. Remove branches inactive at x_* by shrinking the neighborhood. Form the augmented matrix $A(x)$ whose columns are $(\nabla f_i(x), 1)$ for $i \in I_*$. Affine independence makes $A(x_*)$ full column rank; that rank persists. Whenever $A(p)\lambda = (0,0,1)^T$ has a solution, it is unique and equals

$$(A(p)^T A(p))^{-1} A(p)^T (0,0,1)^T.$$

This is continuous and agrees with the positive vector λ^* at x_*. Shrink the neighborhood so every component remains positive. An affine representation of zero using the gradients active at p, extended by zero coefficients on inactive indices, must equal this vector. It therefore has no inactive indices and is a convex representation of zero. This proves local (9.4). This geometric argument is the one recorded in Chapter 8, with related nondegeneracy arguments in [5].

In the plane at most three gradients are affinely independent. At a non-affine-critical nearby point, three cannot all be active, since their affine hull is $\mathbb{R}^2$. A singleton gives a smooth chart. For two active gradients, the difference is nonzero; if the gradients were linearly dependent, their affine line would contain zero, again contradicting non-affine-criticality. The other branches have strict gaps at that point, so the required local two-branch representation follows.

Choose a low sublevel contained in this neighborhood. Its strictly convex positive sublevels and the regular-switching property allow Theorem 9.1, then Corollary 9.2. The localization uses only the objective, not the line-search constants or the initial matrix. Uniformly strongly convex branches on a convex local neighborhood make their maximum strongly convex there, supplying the sufficient strict-contour condition. □

9.7.1 A tame-geometric alternative

Chapter 8 proves that a finite convex semialgebraic function has finitely many affine-critical values. Its proof uses a finite piecewise-smooth cover of the lifted critical graph, so the point path and its actual objective composition have simultaneous derivatives off a finite exceptional set. The two-sided supporting inequality and an affine combination of supporting vectors equal to zero then force the composition derivative to vanish. Continuity joins the resulting constants. No finite-value, path-cover, or composition-differentiability premise is assumed in the final result.

The actual affine-critical set also has finitely many piecewise-smooth path classes, each semialgebraic and internally piecewise-smoothly connected. Once finite critical values are known, continuity makes the value image of each such class a singleton, so F is constant along every piecewise-smooth path in the critical set. This avoids presuming that an arbitrary smooth point path has a semialgebraic graph or that its composition with a nonsmooth F is differentiable in advance. These path classes are distinguished from ordinary topological connected components, whose finiteness, semialgebraicity and value constancy are proved separately.

Combining the low affine-critical gap of Proposition 8.1 with Theorem 9.1 gives the following restricted planar basin conclusion. On a compact structural sublevel above an attained minimum, retain strictly convex positive sublevels and the regular smooth/two-branch hypotheses away from affine criticality. Then a sufficiently low objective basin gives convergence in value and distance to the full minimizer set; uniqueness gives convergence of the whole sequence. Alternatively, with a unique minimizer and the structural hypotheses on a neighborhood, localization supplies a sufficiently low trapped compact sublevel. In both versions the basin is chosen before the admissible run, its initial SPD matrix and its fixed valid line-search constants. No positive-multiplier condition is needed. Semialgebraicity alone does not give the dynamical conclusion: degenerate noncritical ties and flat contours remain outside this theorem.

9.8 An explicit quadratic example and basin

Consider

$$f_0(u,v) = u + u^2 + \tfrac{1}{2}v^2,$$
$$f_1(u,v) = v + \tfrac{1}{2}u^2 + \tfrac{3}{2}v^2,$$
$$f_2(u,v) = -u - v + \tfrac{3}{2}u^2 + v^2, \qquad F = \max_i f_i.$$

All branch Hessians lie between I and $3I$. Also $(f_0 + f_1 + f_2)/3 = u^2 + v^2$, so $F \geq u^2 + v^2$ and zero is its unique minimizer. At zero the active gradients are $(1,0), (0,1), (-1,-1)$, with positive weights $1/3$.

The affine coefficient calculation is exact. With $D = 3(1 + v - uv)$, the solution of $\sum \lambda_i \nabla f_i = 0$, $\sum \lambda_i = 1$ is

$$\lambda_0 = \frac{1 - 4u + 3v - 7uv}{D}, \quad \lambda_1 = \frac{1 + 2u - 3v - uv}{D}, \quad \lambda_2 = \frac{1 + 2u + 3v + 5uv}{D}.$$

On $|u|, |v| \leq 1/10$, $D \geq 267/100$ and the numerators are at least $23/100$, $49/100$, and $45/100$. Hence affine independence and local (9.4) hold on this box. Since $F \geq \|x\|^2$,

$$\boxed{F(x_0) < 1/100 \quad \Longrightarrow \quad x_k \to 0} \tag{9.17}$$

for every infinite admissible full-BFGS run with arbitrary SPD initialization.

9.8.1 This is outside the earlier radial criterion

For small $u > 0$, the active tie $f_0 = f_1$ has

$$v = v(u) = \frac{-1 + \sqrt{1 + 4u + 2u^2}}{2} = u - \tfrac{1}{2}u^2 + O(u^3),$$

with f_2 inactive. The gradient jump at the tie is

$$\nabla f_0 - \nabla f_1 = (1 + u, -1 - 2v), \qquad (\nabla f_0 - \nabla f_1)^T(u,v) = \tfrac{1}{2}u^2 - v^2 \neq 0.$$

Take $x = (u, v + \epsilon)$ on branch 1 and $z = (u, v - \epsilon)$ on branch 0. Then $s = (0, -2\epsilon)$ and

$$d = 2\epsilon(1 + 3v + 3\epsilon), \quad t = 2\epsilon(1 + 2v + 4\epsilon), \quad \Delta = \epsilon(1 + 4v) + \epsilon^2.$$

For each sufficiently small fixed $u > 0$ and $\epsilon \downarrow 0$, these are eligible for $c_1 = 1/4, c_2 = 1/2$. The radial numerator tends to the nonzero quantity $\frac{1}{2}u^2 - v^2$ while $t \to 0$. Thus RS fails on a positive band in every neighborhood of zero, even though (9.17) holds. Curved interfaces between arbitrary quadratic branches are not being replaced by finite polyhedral cells or by a nonlinear change of BFGS coordinates.

9.9 What can already be proved in any dimension

There is a useful partial statement before transport. Let K be a compact set containing no affine-critical point. Suppose that a finite convex F has an exact finite maximum representation by convex C^2 branches on a neighborhood of every point of K. For each $p \in K$, let $J = I(p)$ be the fixed family active at p, and assume its gradient-difference matrix has constant rank on a neighborhood of p. Nearby inactive branches can be removed by their strict gaps. Affine independence of the family at p is a sufficient condition for this local rank hypothesis. The noncritical restriction on K is essential: unit pairing with all subgradients is impossible at an affine-critical point.

The minimum-norm point $a_J(x)$ of the affine hull of these gradients is C^1 in a smaller chart and remains nonzero. To see the regularity, choose an affinely independent subfamily at p that spans this affine hull. Independence persists nearby, and constant rank makes the same subfamily span the full fixed-family hull nearby. The inverse-Gram formula for its minimum-norm point is therefore

C^1, because the branch gradients are C^1. No C^∞ conclusion follows from C^2 branch data alone. Hence

$$v_J(x) = a_J(x)/\|a_J(x)\|^2, \qquad \nabla f_i(x)^T v_J(x) = 1 \quad (i \in J).$$

A finite C^∞ partition of unity on buffered charts yields one C^1 field v, bounded on a compact buffer of K, with $g^T v = 1$ for every actual subgradient at every point of that buffer. The cutoff regularity does not increase the regularity of the local fields. For sufficiently close endpoints, each chart contributing at z also contains x. If i is active at x, both $\nabla F(z)^T v(z)$ and $\nabla f_i(z)^T v(z)$ equal one. Therefore

$$y^T v(z) = [\nabla f_i(z) - \nabla f_i(x)]^T v(z).$$

The local one-sided convex smooth estimate and the finite-max supporting gap give

$$\|\nabla f_i(z) - \nabla f_i(x)\|^2 \le 2LD_{f_i}(z, x) \le 2Ly^T s.$$

To justify the local one-sided estimate, subtract the tangent plane at x and apply the descent lemma at z to the step $-(\nabla f_i(z) - \nabla f_i(x))/L$; for sufficiently close buffered endpoints this step stays in the larger convex chart. Thus visibility and the secant correction can be constructed without a BFGS matrix assumption.

What does not follow: C^1 regularity of this field does not provide the projected transport needed by the BFGS energy. The planar proof succeeds by making its derivative along the level tangent parallel to that tangent. In dimension three and higher, mapping a tangent space into itself is not enough: it may rotate or shear the executed step within the tangent space, while $M_k s_k = 0$ removes only the component parallel to that one step.

A concrete warning uses the entirely smooth quadratic $F(x) = x^T Qx/2$, $Q = \mathrm{diag}(1, 2, 3)$, $x = (1, 1, 1)$, and the canonical common-rate field $v = \nabla F/\|\nabla F\|^2$. Take

$$s = (\epsilon, 0, -(\epsilon + \epsilon^2)/3), \quad y = Qs.$$

Then

$$d = \epsilon^2, \quad t = \frac{\epsilon^2(\epsilon^2 + 2\epsilon + 4)}{3}, \quad \det[g, s, y] = \frac{4}{3}\epsilon^2(1 + \epsilon) \neq 0.$$

For small positive ϵ the pair is Armijo–weak-Wolfe eligible with $c_1 = 1/4, c_2 = 1/2$, but $v(x + s) \notin \mathrm{span}\{v(x), s\}$. Its rank-one projected transport therefore fails for any finite multiplier. Another field, the radial field, works for this quadratic. This is not nonexistence of all certificates or a BFGS counterexample; it demonstrates the gap in an argument based only on a smoothly chosen equal-rate direction.

Relation to earlier work

The following comparisons are bibliographic scholarship, separate from the formal theorem assertions.

Gebken's published Theorem 3.1 assumes an iterate limit and the matrix behavior (B1), including spectral bounds and annihilation of gradient differences [5]. The assertions in this chapter derive their cluster conclusions. Lewis–Wylie's local Newton theory instead concerns a bundle algorithm [6]. These are related results with different hypotheses or algorithms; the literature audit does not establish publication priority for the full statements in this chapter.

The normal-angle expression $hn + h_\theta T$ and its derivative cancellation are classical; see Bogosel, Section 2, equation (4) [23]. Here h is a glued comparison function, without an asserted convex-support-function property.

The multiplier argument in this chapter uses the established nondegeneracy geometry discussed in Chapter 8; compare [6, 5]. Its full-BFGS application additionally requires the preceding dynamical theorem.

Chapter 8's finite-value result is a reproof of a consequence of stratified projection and definable critical-value theory [7]. The planar dynamical hypotheses in this chapter are retained in this application.

The audit found no exact match for these coefficients and the certified basin. The subsequent radial obstruction excludes the stated RS condition, rather than every convergence criterion.

Rank-stable projections and partitions of unity are standard tools. The buffered smooth secant estimate uses Baillon–Haddad cocoercivity; the open-domain formulation is Theorem 2.5 of [4]. Neither this antecedent nor the partial statement supplies the missing transport estimate in arbitrary dimension.

CHAPTER 10

Flat contours and parabolic branches

Flat contour segments are allowed in this finite parabolic family. Its explicit field supplies a convergence theorem that is not contained in the strict-contour planar result.

10.5 An application: arbitrarily many curved quadratic branches

Theorem 10.2 (Finite multi-branch parabolic maxima)**.** *Let $m < \infty$ and*

$$F(u,v) = \max_{i \le m}\{a_i u^2 + b_i v\}, \qquad a_i > 0, \quad \min_i b_i < 0 < \max_i b_i.$$

Every infinite admissible unmodified full-BFGS run, for arbitrary $H_0 \succ 0$ and fixed $0 < c_1 < c_2 < 1$, satisfies

$$(u_k, v_k) \to (0,0), \qquad F(u_k, v_k) \to 0.$$

The number of essential branches is arbitrary. In particular, branches with $b_i = 0$, giving a region flat in the normal coordinate, are allowed.

This is a concrete application of the transported-certificate theorem of Chapter 3, not the definition of its scope. The abstract condition is dimension-independent; this explicitly verified class has two variables.

10.5.1 Compactness and the normalized upper envelope

Set $a_{\min} = \min_i a_i > 0$. A convex combination of one positive and one negative normal slope has slope zero, so

$$F(u,v) \ge a_{\min} u^2.$$

The positive extreme slope bounds v above on a sublevel, and the negative extreme bounds it below. Thus every sublevel is compact. At $u = 0$, $F(0,v) > 0$ for $v \neq 0$, proving uniqueness of the minimum zero.

Discard globally redundant pieces, merge equal normal slopes by retaining the largest a_i, and remove pieces active only on a tie. Order the remaining pieces by strictly increasing b_i. They form the upper envelope

$$\varphi(\xi) = \max_i (a_i + b_i \xi), \qquad \xi = v/u^2 \quad (u \neq 0).$$

The neighboring breakpoints ξ_i increase strictly. Put

$$A_i = a_i + b_i \xi_i = a_{i+1} + b_{i+1}\xi_i > 0.$$

Choose constants k_i recursively:

$$k_{i+1} - k_i = -\frac{a_{i+1} - a_i}{2A_i}. \tag{10.7}$$

At a differentiable point with active branch i, define

$$r(u,v) = (u, 2v)^{\mathsf{T}}, \qquad \theta(u,v) = \exp\left\{\frac{a_i u^2}{2F(u,v)} + k_i\right\}, \qquad W = \theta^2 r r^{\mathsf{T}}. \tag{10.8}$$

The constants make θ agree across every switching interface. At a tie $v = \xi_i u^2$ the exponent difference is

$$\frac{a_{i+1} - a_i}{2A_i} + k_{i+1} - k_i = 0.$$

The field is well-defined on $F > 0$, including by continuous extension across ties. It is only used at differentiable endpoints in the theorem. Since $0 \le u^2/F \le 1/a_{\min}$, there are global constants

$$0 < \theta_{\min} \le \theta \le \theta_{\max} < \infty.$$

There is no coordinate change in the BFGS algorithm. The normalized upper envelope is used solely to construct an analytical certificate.

10.5.2 Visibility and secant correction

Fix a compact positive band $\eta \le F \le B$. For old branch i at $x = (u, v)$ and new branch j at $z = (u', v')$, set

$$h = u' - u, \quad \ell = v' - v, \quad s = (h, \ell),$$
$$e = F(z) - (a_i u'^2 + b_i v') \ge 0, \qquad \bar{e} = F(x) - (a_j u^2 + b_j v) \ge 0.$$

Exact identities are

$$d - \Delta = a_i h^2 + e, \tag{10.9}$$
$$t = (a_i + a_j)h^2 + e + \bar{e}, \tag{10.10}$$
$$g^{\mathsf{T}} r(x) = 2F(x), \tag{10.11}$$
$$y^{\mathsf{T}} r(z) = 2a_i h u' + 2e. \tag{10.12}$$

For eligible pairs, $h^2 \le d/a_{\min}$ and $e \le d$. Weighted Cauchy–Schwarz gives

$$\frac{(y^{\mathsf{T}} r(z))^2}{t} \le \frac{4a_i^2}{a_i + a_j} u'^2 + 4e.$$

The right side is uniformly bounded on the compact band for small d. Multiplication by $\theta(z)^2$ proves (S) of Chapter 3. Equation (10.11) proves (V) with $m = 4\theta_{\min}^2 \eta^2$.

10.5.3 The step quotient cannot degenerate on an eligible small-decrease pair

Define

$$D = \det(s, r(x)) = 2vh - u\ell.$$

We next prove, uniformly on the positive band and for sufficiently small eligible d,

$$d \le C|D|, \qquad |h| \le C|D|. \tag{10.13}$$

This is the essential denominator estimate. It does not assume $s \to 0$ in both coordinates or that successive active pieces are adjacent.

Let $N = y^{\mathsf{T}} r(z)$. Since $r(z) - r(x) = \operatorname{diag}(1, 2)s$,

$$y^{\mathsf{T}} r(x) = -2t + N + y_u h.$$

By (10.12), bounded endpoints and gradients, and $e \le d$,

$$y^{\mathsf{T}} r(x) = -2t + \varepsilon, \qquad |\varepsilon| \le C(|h| + d).$$

The two-dimensional determinant identity yields

$$\det(g, y) D = (g^{\mathsf{T}} s)(y^{\mathsf{T}} r(x)) - (g^{\mathsf{T}} r(x))(y^{\mathsf{T}} s) = -d\, y^{\mathsf{T}} r(x) - 2F(x)t.$$

Consequently

$$(2F(x) - 2d)t \le C|D| + Cd(|h| + d).$$

Use $F(x) \ge \eta$, $t \ge (1 - c_2)d$, and $|h| \le \sqrt{d/a_{\min}}$. Taking d sufficiently small absorbs the final term into the left side, proving $d \le C|D|$. The exact identity

$$b_i D = 2F(x)h + ud \tag{10.14}$$

then proves $|h| \le C|D|$, including when $b_i = 0$. In particular $D \ne 0$.

10.5.4 The rescaling cancels first-order transport

Since $D \neq 0$, define

$$\lambda = 1 + \frac{h\ell}{D}.$$

Then $r(z) - \lambda r(x)$ is parallel to s. A direct identity using $d = -2a_i uh - b_i \ell$ is

$$\lambda - 1 + \frac{a_i u}{F(x)} h = -\frac{vhd}{F(x)D}. \tag{10.15}$$

By (10.13), the right side is $O(d)$ uniformly on the band. Since $h^2 = O(d)$,

$$\lambda = 1 + O(\sqrt{d}) > 0, \qquad \log \lambda = -\frac{a_i u}{F(x)} h + O(d).$$

For the old branch's exponent $L_i(x) = a_i u^2/(2F(x)) + k_i$, smoothness in the two scalar arguments (u, F) on the positive band gives

$$L_i(z) - L_i(x) = \frac{a_i u}{F(x)} h + O(h^2 + \Delta) = \frac{a_i u}{F(x)} h + O(d).$$

Thus $\log \lambda + L_i(z) - L_i(x) = O(d)$.

We still need to account for a change from branch i to branch j. The finite-envelope geometry implies

$$|L_j(z) - L_i(z)| \leq C \frac{e}{F(z)} \leq C_\eta d. \tag{10.16}$$

A complete justification is as follows. Normalize a point by

$$t_0 = u'^2/F(z), \qquad v_0 = v'/F(z).$$

It lies on the compact polygonal arc

$$S = \{(t_0, v_0) : t_0 \geq 0,\ \max_i (a_i t_0 + b_i v_0) = 1\}.$$

Compactness follows from $t_0 \leq 1/a_{\min}$ and the two nonzero extreme normal slopes. On the segment where j is active, both

$$e/F(z) = 1 - a_i t_0 - b_i v_0, \qquad L_j - L_i = (a_j - a_i) t_0/2 + k_j - k_i$$

are affine functions. The first is nonnegative. Whenever it vanishes, the second vanishes by (10.7). On an adjacent segment the two functions are proportional; on a nonadjacent segment the first has a positive minimum. There are finitely many segments and pairs, proving (10.16). Removal of zero-length envelope pieces avoids spurious triple junctions; it does not change the objective or its differentiable gradients.

It follows that the compensated transport multiplier

$$\widetilde{\lambda} = \theta(z)\lambda/\theta(x)$$

satisfies

$$|\log \widetilde{\lambda}| \leq C_\eta d. \tag{10.17}$$

Because

$$\Pi_s(\theta(z) r(z)) = \widetilde{\lambda} \Pi_s(\theta(x) r(x)),$$

we obtain

$$\Pi_s W(z) \Pi_s \preceq e^{2C_\eta d} \Pi_s W(x) \Pi_s \preceq (1 + L_\eta d) \Pi_s W(x) \Pi_s$$

on a sufficiently small-decrease tail. This is (T). The transported-certificate theorem of Chapter 3 proves convergence from every admissible initialization. □

10.6 Explicit three-branch example and strict nonnecessity of RS

Consider

$$F(u,v) = \max\{u^2 + v,\ 3u^2,\ 2u^2 - v\}. \tag{10.18}$$

The switching parabolas are $v = -u^2$ and $v = 2u^2$. Between them the objective is flat in v, but the origin is its unique minimizer. The theorem applies globally.

An especially simple scaling for (10.8) is

$$\theta(u,v) = \begin{cases} \exp\{u^2/F - 1/3\}, & v < -u^2, \\ 1, & -u^2 < v < 2u^2, \\ \exp\{u^2/(2F) - 1/6\}, & v > 2u^2. \end{cases}$$

It is continuous across both interfaces, and $e^{-1/3} \le \theta \le 1$. Take $W = \theta^2 (u, 2v)(u, 2v)^{\mathsf{T}}$.

Fix $u \neq 0$, $A > 0$, and $0 < \epsilon < 6u^2$. Let

$$x_\epsilon = (u, 2u^2 + \epsilon), \qquad z_\epsilon = (u, 2u^2 - \epsilon/2),$$

$$H_\epsilon = \begin{pmatrix} A & -2uA \\ -2uA & 4u^2 A + 3\epsilon/2 \end{pmatrix} \succ 0.$$

The gradients are $g = (2u, 1)^{\mathsf{T}}$ and $g' = (6u, 0)^{\mathsf{T}}$. Directly,

$$-H_\epsilon g = z_\epsilon - x_\epsilon, \quad d = t = 3\epsilon/2, \quad \Delta = \epsilon.$$

Under the fixed ordinary line-search parameters, $\alpha = 1$ is an actual first full-BFGS step exactly when the Armijo condition $c_1 \le 2/3$ holds. Armijo is strict for $c_1 < 2/3$, and weak Wolfe is strict for every allowed $c_2 > 0$. The displayed pair alone is not used for $c_1 > 2/3$.

Here $y = (4u, -1)^{\mathsf{T}}$, so

$$\frac{(y^{\mathsf{T}} z_\epsilon)^2}{t} = \frac{(2u^2 + \epsilon/2)^2}{3\epsilon/2} \longrightarrow \infty.$$

The values tend to the positive level $3u^2$ while $d \to 0$. Such pairs exist in every neighborhood of zero, with $u \neq 0$ fixed before taking $\epsilon \downarrow 0$.

The same separation holds for every fixed ordinary pair $0 < c_1 < c_2 < 1$. Choose

$$\rho = \frac{1 - c_1}{2c_1} > 0, \qquad c_1(1 + \rho) = \frac{1 + c_1}{2} < 1,$$

and replace the new point and metric by

$$z_{\epsilon,\rho} = (u, 2u^2 - \rho\epsilon),$$

$$H_{\epsilon,\rho} = \begin{pmatrix} A & -2uA \\ -2uA & 4u^2 A + (1+\rho)\epsilon \end{pmatrix}, \qquad 0 < \rho\epsilon < 3u^2.$$

For any $A > 0$, the quadratic form is $A(w_1 - 2uw_2)^2 + (1+\rho)\epsilon w_2^2$, so the metric is SPD. The active gradients remain $(2u, 1)^T$ and $(6u, 0)^T$, and direct calculation gives

$$-H_{\epsilon,\rho} g = z_{\epsilon,\rho} - x_\epsilon, \qquad d = t = (1+\rho)\epsilon, \qquad \Delta = \epsilon.$$

Thus these are strictly Armijo–weak-Wolfe eligible unit steps. Their radial quotient is

$$\frac{(y^T z_{\epsilon,\rho})^2}{t} = \frac{(2u^2 + \rho\epsilon)^2}{(1+\rho)\epsilon} \longrightarrow \infty.$$

In each neighborhood first fix a sufficiently small nonzero u and the resulting positive band; then the quotient exceeds any proposed bound with arbitrarily small d. Hence the original (RS) condition

fails even on eligible near-stalling pairs for every fixed allowed line-search pair. These metrics supply individual accepted first steps and vary with ϵ; they are not a single infinite bad BFGS history. Every actual infinite admissible run still converges by the main parabolic theorem.

For the original printed choice $\rho = 1/2$, the following exact transported quantities hold.

In contrast, $\theta(z_\epsilon) = 1$ and

$$y^{\mathsf{T}} r(z_\epsilon) = \epsilon, \qquad \frac{y^{\mathsf{T}} W(z_\epsilon) y}{t} = \frac{2\epsilon}{3} \to 0.$$

The unscaled quotient multiplier is $\lambda = 1$, and the compensated multiplier satisfies exactly

$$\log \widetilde{\lambda} = \frac{\epsilon}{6(3u^2 + \epsilon)} = O(d)$$

on each fixed positive band. This displays both parts of the mechanism: removing the radial mismatch makes the secant correction small, while a mild objective-defined rescaling controls accumulated transport.

Relation to earlier work

The following comparisons are bibliographic scholarship, separate from the formal theorem assertions.

For the special case $u^2 + |v|$, Guo–Lewis proves a subsequence approaching the nonsmooth axis, not full convergence (Corollary 4.4) [10]. The theorem in this chapter treats the complete stated finite family. Gebken's related quasi-Newton conclusions retain behavior assumptions [5]; no historical priority claim follows from this comparison.

CHAPTER 11

Fixed normal geometry in arbitrary dimension

A fixed space of jump normals leaves controlled smooth directions in arbitrary dimension. Residual transport can be paid for in those directions; this is a different hypothesis from planar switching geometry.

11.1 A concrete dimension-independent local theorem

Let $a \in \mathbb{R}^n$ be a unit vector, $b \in \mathbb{R}$, and write

$$P = \{x : a^T x = b\}, \qquad S = a^\perp, \qquad P_S = I - aa^T.$$

The normal direction is fixed; the dimension $n-1$ of the tangent space is unrestricted.

Theorem 11.1 (One exceptional hyperplane). Let $F : \mathbb{R}^n \to \mathbb{R}$ be finite and convex, with unique minimizer $x_* \in P$. Let $K = \{F \le F(x_0)\}$ be compact. On an open convex neighborhood of K, assume:

1. F is μ-strongly convex for some $\mu > 0$.
2. All subgradients have the same tangential projection,

$$P_S g = A(x) \qquad (g \in \partial F(x)),$$

   where A is Lipschitz continuous on that neighborhood.
3. F is locally $C^{1,1}$ away from P.

Then every infinite admissible full-BFGS run starting at x_0 satisfies

$$\boxed{F(x_k) \to F_*, \qquad x_k \to x_*.}$$

The hyperplane itself may include differentiable points; it is allowed to contain failures of gradient-Lipschitz regularity as well as actual kinks. No uniform bound on the full gradient-Lipschitz constants as the hyperplane is approached is required.

If these hypotheses hold only in a neighborhood of a unique minimizer, some $\varepsilon > 0$ gives the same conclusion for every start with $F(x_0) < F_* + \varepsilon$. The same basin works for every SPD initialization and every fixed valid line-search pair. Compact sublevel trapping supplies this local statement: choose a low sublevel contained in the structural neighborhood, then use Armijo monotonicity.

The assumptions concern the objective. In particular, the claim that a part of $B_k = H_k^{-1}$ has a controlled growth rate will be **derived**, not assumed.

11.1.1 A simple family and a weaker-curvature version

An immediate family is

$$\boxed{F(u, v) = \psi(u, v) + h(v), \qquad u \in \mathbb{R}^{n-1}, \quad v \in \mathbb{R},}$$

where ψ is convex with locally Lipschitz gradient, h is finite convex and locally $C^{1,1}$ on $\mathbb{R} \setminus \{0\}$, F is strongly convex on the relevant neighborhood, and the minimizer has $v_* = 0$. Here $A = \nabla_u \psi$, irrespective of the singular normal derivative of h. The objective and the initial matrix may be fully coupled.

Full strong convexity can be weakened for this composite family. It is enough that

$$\psi(u,v) - \frac{\mu}{2}\|u\|^2 \quad \text{is convex}, \qquad \mu > 0,$$

that the relevant positive sublevels of F are strictly convex, and that the minimizer is unique with $v_* = 0$. A compact initial sublevel is still required. The proof of this extension is given in Section 11.6. It needs positive curvature only in the smooth tangential directions, not a positive lower bound in the normal direction.

This is not a claim that all convex nonsmooth functions admit such a splitting. Nor does a common hyperplane through the optimizer describe arbitrary curved switching geometry.

11.2 A forced affine-critical cluster-point theorem

The method also fits the affine-criticality program of Chapter 8 directly.

Theorem 11.2 (Parallel singular interfaces). Let F be finite convex with attained minimum and compact sublevel K. Assume F is μ-strongly convex on an open convex neighborhood of K. Let a be a unit vector and $E \subset \mathbb{R}$ closed. On that neighborhood suppose:

- the singular set is $\Sigma = \{x : a^T x \in E\}$, every point of Σ is genuinely nondifferentiable, and F is locally $C^{1,1}$ off Σ;
- $P_S \partial F(x) = \{A(x)\}$ for a Lipschitz field A, where $S = a^\perp$.

Then the cluster set of every infinite admissible full-BFGS run in K contains a point p with

$$\boxed{0 \in \operatorname{aff} \partial F(p).}$$

There may be several or countably many parallel interfaces, subject to the closed-set and regularity assumptions above. No finite affine-critical set is assumed.

At a point of Σ, the subdifferential is a nontrivial compact segment parallel to a, because every subgradient has the same S projection. Hence

$$\operatorname{aff} \partial F(p) = A(p) + \mathbb{R}a, \qquad 0 \in \operatorname{aff} \partial F(p) \iff A(p) = 0.$$

At a smooth point, affine criticality is ordinary stationarity.

Consequently, adding

$$0 \in \operatorname{aff} \partial F(p) \implies 0 \in \partial F(p)$$

on K, or on a sufficiently low trapped neighborhood, gives convergence to the minimum value and, under uniqueness, convergence of the whole sequence.

Theorem 11.1 is slightly different from a literal specialization of Theorem 11.2: it allows the exceptional hyperplane to contain smooth but not locally $C^{1,1}$ points. Having $x_* \in P$ makes every zero of A on P optimal, so genuine nondifferentiability at every exceptional point is unnecessary there.

11.5 Proof of the parallel-interface and single-interface results

We give a common argument that covers Theorems 11.1 and 11.2.

11.5.1 Preliminary bounds and a contradiction set

Strong convexity and convexity give

$$d_k - (F_k - F_{k+1}) \geq \frac{\mu}{2}\|s_k\|^2, \qquad t_k \geq \mu\|s_k\|^2. \tag{11.10}$$

Thus $\sum \|s_k\|^2 < \infty$ and $s_k \to 0$. Let Γ be the compact cluster set. Every point of Γ has the common limit value ℓ.

For Theorem 11.2, assume Γ has no affine-critical point. Then $\ell > F_*$, so it contains no stationary point. For Theorem 11.1, assume directly $\ell > F_*$.

Write $r(x) = a^T x$ in Theorem 11.2, and $r(x) = a^T x - b$ in Theorem 11.1. Let E be the exceptional scalar set; in Theorem 11.1 use $E = \{0\}$.

The common tangential projection and its Lipschitz constant L_A imply

$$\|P_S y_k\|^2 = \|A(x_{k+1}) - A(x_k)\|^2 \le L_A^2 \|s_k\|^2 \le \frac{L_A^2}{\mu} t_k. \tag{11.11}$$

Hence Theorem 2.4's smooth-subspace estimate holds.

11.5.2 Why tangential descent cannot disappear on the exceptional set

For Theorem 11.2, at an exceptional point the nontrivial subgradient segment has affine hull $A(p) + \mathbb{R}a$. Thus $A(p) = 0$ would make it affine critical, contrary to the assumption on Γ.

For Theorem 11.1, if $p \in P$ and $g \in \partial F(p)$, strong monotonicity with $0 \in \partial F(x_*)$ gives

$$A(p)^T(p - x_*) = g^T(p - x_*) \ge \mu \|p - x_*\|^2.$$

Thus $A(p) = 0$ on P is possible only at x_*. Such a point cannot belong to a positive limiting level.

In either case the compact set

$$Z = \{p \in \Gamma : A(p) = 0\}$$

has scalar image $r(Z)$ disjoint from the closed set E. If Z is nonempty, these sets have positive separation $\delta = \operatorname{dist}(r(Z), E) > 0$. If E is empty there is no singular region and we may take the cutoff below identically one; if Z is empty it may be identically zero.

Choose a scalar Lipschitz function $\tau : \mathbb{R} \to [0, 1]$ which is zero where $\operatorname{dist}(r, E) \le \delta/4$, and one where $\operatorname{dist}(r, E) \ge \delta/2$. Such a cutoff is obtained by composing a piecewise-linear cutoff with the distance function. In particular $\tau = 1$ on $r(Z)$. The cutoff is an analytical object chosen from F and the hypothetical cluster set, not an algorithmic input.

11.5.3 A frame on the entire cluster set

Let the columns of U be an orthonormal basis of S, and set

$$\boxed{V(x) = [\, U,\ \tau(r(x))a \,].} \tag{11.12}$$

Its tangential columns are always present. Its normal column is switched off near singular interfaces and restored away from them.

At differentiable points,

$$\|V(x)^T \nabla F(x)\|^2 = \|A(x)\|^2 + \tau(r(x))^2 |a^T \nabla F(x)|^2.$$

On $\Gamma \setminus Z$ the first term is positive. On Z, the cutoff equals one and the full gradient is nonzero. The displayed expression extends continuously across singular interfaces because the cutoff vanishes on a whole strip around them; gradients are bounded and A is continuous. Compactness therefore provides $m > 0$ and a buffered neighborhood of all of Γ in which visibility holds.

The entire late trajectory lies in this neighborhood because its distance to its compact cluster set tends to zero. There is no selection of one cluster point followed by an assumption that the sequence converges to it.

11.5.4 The new secant term is bounded

The tangential part follows from (11.11). If $\tau(r(z)) = 0$, there is no normal contribution.

If $\tau(r(z)) > 0$, the endpoint is a fixed positive distance from the exceptional scalar set. For sufficiently close endpoints, the entire segment lies in a slightly smaller smooth strip. On a compact

buffered subset of that strip, a finite convex-chart cover gives a common local gradient-Lipschitz constant. Convex cocoercivity on each sufficiently small convex chart yields

$$\|\nabla F(z) - \nabla F(x)\|^2 \le L_\delta\, y^T s.$$

Therefore

$$\|V(z)^T y\|^2 \le C_0 t \tag{11.13}$$

for every sufficiently late accepted pair. This uses local smoothness away from the exceptional set; L_δ need not stay bounded as $\delta \downarrow 0$.

11.5.5 The only transport error is tangential and inexpensive

For an endpoint pair put

$$\beta = \begin{cases} [\tau(r(z)) - \tau(r(x))]/(a^T s), & a^T s \neq 0, \\ 0, & a^T s = 0. \end{cases}$$

Then $|\beta| \le \mathrm{Lip}(\tau)$. With e_n the last frame-coordinate vector,

$$\boxed{V(z) = V(x) + s\,\beta e_n^T - \beta P_S s\, e_n^T.} \tag{11.14}$$

Thus (11.14) is the transport hypothesis of Theorem 2.4, with $T = I$, rank-one term $s(\beta e_n)^T$ and

$$R = -\beta P_S s\, e_n^T \in S, \qquad \|R\|_F^2 \le \mathrm{Lip}(\tau)^2 \|P_S s\|^2 \le \frac{2\,\mathrm{Lip}(\tau)^2}{\mu} d. \tag{11.15}$$

All conditions of Theorem 2.4 hold. The contradiction proves Theorem 11.2, and proves $\ell = F_*$ for Theorem 11.1. Compactness and uniqueness give whole-sequence convergence. □

11.6 Only tangential strong convexity is needed for a composite subclass

Consider $F(u, v) = \psi(u, v) + h(v)$ under the weaker hypotheses in Section 11.1.1. Set $S = \mathbb{R}^{n-1} \times \{0\}$. Since $\psi - \mu\|u\|^2/2$ and h are convex,

$$d_k - (F_k - F_{k+1}) \ge \frac{\mu}{2}\|P_S s_k\|^2. \tag{11.16}$$

This is the bound needed for the transport residual, even without a positive normal curvature bound.

At differentiable accepted points,

$$P_S y = P_S[\nabla\psi(z) - \nabla\psi(x)].$$

Convex L-smoothness of ψ on the compact neighborhood, and monotonicity of the scalar subgradient of h, give

$$\|P_S y\|^2 \le L[\nabla\psi(z) - \nabla\psi(x)]^T s \le Lt. \tag{11.17}$$

This replaces (11.11). On the exceptional plane $v = 0$, the restriction of F is a differentiable strongly convex function of u, uniquely minimized at u_*. Thus the needed nonvanishing of $A = \nabla_u \psi$ on a nonoptimal cluster set still holds.

Finally, strict convexity of the positive sublevels supplies $s_k \to 0$ in place of full strong convexity. The proof is dimension independent: if a subsequence had $x_k \to x$, $x_{k+1} \to z \neq x$, $g_k \to g$, and $F_k \to \ell > F_*$, then $d_k \to 0$ gives $g^T(z - x) = 0$. Convexity and the supporting inequality make F constant at ℓ on $[x, z]$, contrary to the strictly convex boundary of $\{F \le \ell\}$.

Equations (11.16)–(11.17) and the same cutoff frame complete the proof. A quadratic-plus-absolute-value objective is included, but those familiar cases were already proved in the source notes. The extension below illustrates the additional normal irregularity allowed here.

11.7 An explicit all-dimensional example outside RS

Let $a_j = 2^{-j}$ for $j \geq 2$, and let

$$\eta(t) = \begin{cases} (1-t^2)^2, & |t| \leq 1, \\ 0, & |t| > 1. \end{cases}$$

This nonnegative bump is C^1, has $\eta(0) = 1$, and integral $16/15$. Its supports at the following scales are pairwise disjoint. Define

$$k(v) = \sum_{j=2}^{\infty} a_j^{-3} \eta\left(\frac{v - a_j}{a_j^5}\right), \qquad v > 0,$$

$$r(v) = \begin{cases} \int_0^v (v-t)k(t)\,dt, & v > 0, \\ 0, & v \leq 0. \end{cases} \tag{11.18}$$

Each bump has mass $(16/15)a_j^2$. The total mass is finite. The sum is locally finite away from zero, so r is convex, C^1 globally, and C^3 on $(0,\infty)$ and $(-\infty,0)$. In particular it is locally $C^{1,1}$ away from zero. Direct bounds give

$$0 \leq r'(v) \leq 6v^2, \qquad 0 \leq r(v) \leq 6v^3$$

for $v \geq 0$. However,

$$r''(a_j) = a_j^{-3} \to \infty. \tag{11.19}$$

(The second derivative at zero itself exists and equals zero, but is not continuous there.)

For any $n \geq 2$, consider

$$\boxed{F(u,v) = \frac{1}{2}\|u\|^2 + \frac{1}{2}v^2 + |v| + r(v), \qquad u \in \mathbb{R}^{n-1}.} \tag{11.20}$$

It is 1-strongly convex and coercive, with unique minimizer zero. Its only nondifferentiable hyperplane is $v = 0$; its tangential gradient is u, which is globally Lipschitz. Theorem 11.1 therefore proves global convergence of every admissible full-BFGS run, for arbitrary coupled SPD initialization.

Yet **RS fails in every neighborhood of zero**. Fix any sufficiently small nonzero u_0, and take $x_j = (u_0, a_j)$. Their objective values approach the positive level $\|u_0\|^2/2$. At these smooth points,

$$x_j^T \nabla^2 F(x_j) x_j = \|u_0\|^2 + a_j^2 + a_j^{-1} \longrightarrow \infty. \tag{11.21}$$

A uniform inward Taylor estimate on that fixed positive band would, by taking its second-order limit at each x_j, bound the left side of (11.21). This is impossible. The equivalence of Theorem 4.2.1 between inward radial smoothness and RS therefore rules out RS on that band. Choosing u_0 arbitrarily small gives failure in every neighborhood of the optimizer.

This is a **function-class** gain beyond RS, but not a claim that the function lacks every prior transported certificate. It also expands the allowed normal curvature rather than settling the original arbitrary curved finite-C^2-max conjecture of Chapter 8. Where each piece has a $C^{1,1}$ extension through the flat interface, the finite-polyhedral-partition Theorem 4.5.1 already gives local convergence.

11.7.1 The infinite series is evaluated exactly in the verifier

There is no need to truncate (11.18). All bumps below a query can be summed using geometric series, with at most one partially included bump. Define

$$P(t) = t - \frac{2}{3}t^3 + \frac{1}{5}t^5, \qquad Q(t) = \frac{1}{2}t^2 - \frac{1}{2}t^4 + \frac{1}{6}t^6.$$

For a query inside the bump centered at $a = a_j$, with $t = (v - a)/a^5$, the partial mass is $a^2[P(t) + 8/15]$, and its first moment is

$$a^3[P(t) + 8/15] + a^7[Q(t) - 1/6].$$

All smaller bumps have total mass $(16/45)a^2$ and first moment $(16/105)a^3$. This gives exact rational function and derivative values for rational queries. At a peak,

$$r(a) = \frac{64}{315}a^3 + \frac{1}{6}a^7, \qquad r'(a) = \frac{8}{9}a^2, \qquad r''(a) = a^{-3}.$$

These are identities for the complete infinite series, not for a truncation. The exact rational-query formulas and their derivative values are proved for this oracle. For the objective (11.20), the convergence theorem applies in every stated dimension with an arbitrary coupled SPD initial inverse matrix; it does not replace full BFGS by a reduced two-dimensional or block-diagonal algorithm.

11.7.2 The objective need not split as a smooth function plus one scalar penalty

For a fixed vector $c \in \mathbb{R}^{n-1}$ with $\|c\| < 1$, another example is

$$F(u, v) = \frac{1}{2}(\|u\|^2 + v^2) + |(1 + c^T u)v| + r(v).$$

The two quadratic functions before adding r have Hessians

$$\begin{pmatrix} I & c \\ c^T & 1 \end{pmatrix}, \qquad \begin{pmatrix} I & -c \\ -c^T & 1 \end{pmatrix},$$

both bounded below by $(1 - \|c\|)I$. Their maximum plus convex $r(v)$ is strongly convex. Since the absolute-value term and r are nonnegative, zero is the unique minimizer.

On a neighborhood with $|c^T u| < 1/2$, the only exceptional hyperplane is $v = 0$, and the tangential gradient is

$$A(u, v) = u + c|v|,$$

which is Lipschitz across that hyperplane. Theorem 11.1 gives an objective-dependent local full-BFGS convergence basin for every $\|c\| < 1$, including $c = 0$.

If $c \neq 0$, the normal-gradient jump magnitude $2(1 + c^T u)$ varies with u on this neighborhood. More precisely, on no open neighborhood of zero can this objective equal $S(u, v) + h(v)$ with S differentiable and h an arbitrary scalar function. One way to see the obstruction is to subtract values at two nearby tangential points: the scalar term cancels, while the varying jump leaves a nonzero multiple of $|v|$, contradicting differentiability at $v = 0$.

The nonzero condition is needed only for this nonsplitting conclusion. When $c = 0$, the objective has the explicit global decomposition

$$S(u, v) = \tfrac{1}{2}(\|u\|^2 + v^2), \qquad h(v) = |v| + r(v),$$

with S smooth. Thus the subsection's existential claim is realized by nonzero c, not by every allowed parameter. The spike argument still makes (RS) fail in every neighborhood of zero for all $\|c\| < 1$, including the zero case.

Relation to earlier work

The following comparisons are bibliographic scholarship, separate from the formal theorem assertions.

The tangent/normal decomposition has established VU and partial-smoothness antecedents [26, 8]. Quantitative tangential projection also appears in [24]. Its stochastic/proximal applications and the proximal-gradient VU algorithm of [25] concern different iterations. The full-BFGS conclusions in this chapter require the proved restricted matrix control; identification results alone do not supply it.

The smooth-region estimates use classical cocoercivity; see the open-domain Baillon–Haddad theorem [4], Theorem 2.5. The compact-buffer argument retains the local regularity assumptions and does not require a uniform bound on normal curvature near the interface.

The exact spike and coupling examples had no match in the audited sources. Their separation concerns RS. The nonsplitting conclusion of Section 11.7.2 requires $c \neq 0$ and excludes only the stated differentiable-plus-scalar decomposition; it does not exclude general partial-smoothness or nonlinear-composite descriptions [26, 8].

CHAPTER 12

The exact deterministic seed and local smooth embedding

A finite algebraic cycle first determines every query of a specified line search. Scaled copies then produce one fixed convex objective with failing initial points approaching its optimizer.

12.1 Main result and exact implementation

For a differentiable current point x, put $g = \nabla F(x)$ and $p = -Hg$, with $H \succ 0$. We use the ordinary full inverse-BFGS update

$$s = \alpha p, \quad y = \nabla F(x+s) - g, \quad t = y^T s, \qquad H_+ = \left(I - \frac{sy^T}{t}\right) H \left(I - \frac{ys^T}{t}\right) + \frac{ss^T}{t}. \tag{12.1}$$

There is no resetting, damping, rescaling of the stored matrix, memory truncation, or skipped update.

The line search is Algorithm 4.6 of Lewis–Overton [2]: initialize a lower bracket $\ell = 0$, an upper bracket $h = +\infty$, and $\alpha = 1$. At each trial:

1. If Armijo fails, set $h = \alpha$.
2. Otherwise, if weak Wolfe fails, set $\ell = \alpha$.
3. Otherwise, accept α and stop.

After a rejection, try $(\ell + h)/2$ if $h < +\infty$, and 2ℓ otherwise. The tests are

$$F(x + \alpha p) < F(x) + c_1 \alpha g^T p, \qquad \nabla F(x + \alpha p)^T p > c_2 g^T p.$$

All decisions below are strict; using non-strict acceptance inequalities produces the identical orbit. Every trial point, not just every accepted point, is differentiable.

Theorem 12.1. *There exist four convex functions $f_i \in C^2(\mathbb{R}^2)$, with*

$$0 \preceq \nabla^2 f_i(x) \preceq 64 I \quad (i = 0, 1, 2, 3;\ x \in \mathbb{R}^2),$$

such that $F = \max_i f_i$ has compact sublevel sets and the unique global minimizer $x_* = 0$, where F is nonsmooth. For the deterministic line search specified above, with $c_1 = 10^{-4}$ and $c_2 = 1/2$, there are starting points $x_0^{(n)} \to 0$, all initialized with $H_0 = I$, whose full-BFGS iterates satisfy

$$x_k^{(n)} \longrightarrow p_n \neq 0, \qquad F(x_k^{(n)}) \downarrow F(p_n) > 0, \qquad 0 \notin \partial F(p_n).$$

Every line search terminates after at most five trials. In fact, the same function, initial points, matrices, and trial decisions work for every fixed

$$0 < c_1 \leq \tfrac{1}{3}, \qquad c_1 < c_2 < 1.$$

The entire accepted and trial trajectories can be contained in an arbitrarily small neighborhood of 0 by choosing n sufficiently large.

The accepted lengths are not assigned externally. They are generated by the deterministic search:

$$\boxed{\alpha_{3m} = 1, \qquad \alpha_{3m+1} = \tfrac{1}{4}, \qquad \alpha_{3m+2} = \tfrac{1}{16}.}$$

Thus the construction resolves the implementation-selection issue. It does not address strong-Wolfe interpolation searches or a positive-measure basin of failing initial points.

12.2 An exactly specified algebraic seed

Let (c, d) be the real pair specified below. Define

$$U = 127c + 16d, \qquad V = 16c - 127d - 16, \qquad R = c - 8d,$$
$$K = 8c^2 + 8d^2 + d - 8, \qquad T = c^2 - c + d^2, \qquad Y = (c-1)^2 + d^2.$$

The defining equations are

$$\boxed{264192RK - UV(U + 129) = 0, \qquad 16VTK - RUY = 0.} \tag{12.2}$$

Choose the unique root in the closed ℓ_∞ box of radius 10^{-30} centered at the rational pair

$$c_0 = 0.31848142762711863064645131250290703717912205023605,$$
$$d_0 = -1.02523211967574872548135087001300414284511909318444.$$

The decimals here are exact terminating rationals defining the box, not replacements for the algebraic root. Set

$$\rho = \frac{129R}{V(U + 129)} \simeq 0.06019920610754376481537366628 7. \tag{12.3}$$

Lemma 12.1. *The designated box contains exactly one solution of* (12.2)*. All denominators in this chapter are nonzero there, and* $0.060 < \rho < 0.061$*.*

Proof. Let $P = (P_1, P_2)^T$ denote the left sides of (12.2), let $\bar{r} = (c_0, d_0)$, and let $M = DP(\bar{r})^{-1}$, an exactly specified rational matrix. On the box X, exact rational interval arithmetic gives

$$\sup_{z \in X} \|I - MDP(z)\|_\infty < 10^{-26}, \qquad \|MP(\bar{r})\|_\infty < 10^{-50}.$$

Hence $z \mapsto z - MP(z)$ is a strict contraction mapping X into itself, because

$$10^{-50} + 10^{-26}\, 10^{-30} < 10^{-30}.$$

The contraction theorem gives the unique root. The interval calculation also bounds the denominators away from zero and certifies (12.3). These bounds are exact rational certificate inequalities. Appendix A specifies the polynomial residual, coefficient bounds, sound interval operations and finite query inequalities. The contraction argument explains how these bounds imply existence and uniqueness. The formal root and coefficient certificates are separately indexed in the theorem ledger. □

Set

$$g_0 = (1, 0)^T, \qquad g_1 = (-1, -1/8)^T, \qquad g_2 = (c, d)^T, \qquad G(z) = \max_{0 \le i \le 2} g_i^T z.$$

Start with $M_0 = I$, and define

$$M_1 = \begin{pmatrix} 129/256 & -1/16 \\ -1/16 & 1 \end{pmatrix}, \qquad M_2 = \mathcal{B}(M_1, g_1, g_2, 1/4),$$

where $\mathcal{B}$ denotes (12.1) with the stated current and next gradients. Then

$$\mathcal{B}(I, g_0, g_1, 1) = M_1, \qquad \boxed{\mathcal{B}(M_2, g_2, g_0, 1/16) = \rho I.} \tag{12.4}$$

The three displacements are

$$s_0 = (-1, 0)^T, \quad s_1 = (127/1024, 1/64)^T, \quad s_2 = \rho(1 - c, -d)^T. \tag{12.5}$$

Define the starting point by

$$z_0 = \frac{1}{1-\rho}\begin{pmatrix} 897/1024 - \rho(1-c) \\ \rho d - 1/64 \end{pmatrix}, \qquad z_{j+1} = z_j + s_j. \tag{12.6}$$

Then

$$\boxed{z_3 = \rho z_0.}$$

For clarity, approximate points are

j	$(z_j)_1$	$(z_j)_2$
0	0.888432624149	−0.082297397686
1	−0.111567375851	−0.082297397686
2	0.012456061649	−0.066672397686
3	0.053482938654	−0.004954238005

The uniquely active gradient at z_j is g_j for $j = 0, 1, 2$, and g_0 at z_3.

12.2.1 Derivation and independent checks of the matrix closure

The identity is not inferred from near-closure. For arbitrary symbolic (c, d) with the denominators nonzero, put

$$\beta = \frac{66048RK}{UV(U+129)}, \qquad \gamma = \frac{VTK}{RUY}, \qquad \widetilde{\rho} = \frac{2064TK}{U(U+129)Y}.$$

Direct rational identities give

$$\mathcal{B}(\mathcal{B}(M_1, g_1, g_2, \beta), g_2, g_0, \gamma) = \widetilde{\rho} I.$$

One way to derive the formulas is to write the middle update as $C + \beta D$. Requiring $(C + \beta D)g_2$ to be parallel to $g_0 - g_2$ is linear in β. Its solution is the formula above. If $w \perp (g_0 - g_2)$, set

$$\widetilde{\rho} = \frac{w^T(C + \beta D)w}{w^T w}$$

and choose γ so that the last displacement is $\widetilde{\rho}(g_0 - g_2)$. The last update then gives $\widetilde{\rho} I$: its two projection factors are the orthogonal projector onto $(g_0 - g_2)^{\perp}$.

Equations (12.2) are exactly $\beta = 1/4$ and $\gamma = 1/16$. Also

$$\widetilde{\rho} - \rho = \frac{2064R}{V(U+129)}\left(\gamma - \frac{1}{16}\right).$$

This proves (12.4) and (12.5). The Lean formalization checks these exact identities from the actual full-BFGS formulas, together with the secant identity and the equivalence of the inverse and Hessian updates. It proves the required positive curvatures and positive definiteness, including the stated algebraic-root box. No execution of the unavailable historical SymPy program is used as evidence.

12.3 Every deterministic oracle decision is certified

Use $c_1 = 10^{-4}$ and $c_2 = 1/2$ for the displayed table. The line searches below use $p_j = -M_j g_j$. At each rejected trial, the objective is strictly greater than the current value, not merely greater than the Armijo threshold.

Phase	Trial α	Active branch	$G(z_j + \alpha p_j)$	Decision
0	1	1	0.121854550561	Accept
1	1	0	0.384526374149	Armijo rejection
1	1/2	0	0.136479499149	Armijo rejection
1	1/4	2	0.072321707900	Accept
2	1	0	0.668886093720	Armijo rejection
2	1/2	0	0.340671077685	Armijo rejection
2	1/4	0	0.176563569667	Armijo rejection
2	1/8	0	0.094509815658	Armijo rejection
2	1/16	0	0.053482938654	Accept

The starting values in the three phases are approximately

$$(0.888432624149,\ 0.121854550561,\ 0.072321707900).$$

The ratios at the accepted steps are

j	$\dfrac{G(z_j) - G(z_{j+1})}{-g_j^T s_j}$	$\dfrac{g_{j+1}^T s_j}{g_j^T s_j}$
0	0.766578073588	-1
1	0.393190937094	-0.186383158092
2	0.375205963573	-0.817119670485

The displayed decimals are explanatory. The Lean certificates prove the stated rational enclosures and strict comparisons for the actual algebraic root. In particular, each accepted Armijo ratio exceeds $1/3$, and every accepted Wolfe ratio is negative. The rejected trial values exceed their respective current values. Therefore the identical trial tree works for *all* $0 < c_1 \le 1/3$ and $c_1 < c_2 < 1$. These are checked statements about the actual objective and trials, rather than an assertion that an unavailable historical verifier was executed.

All nine trial points have a unique active branch. They satisfy the additional uniform bounds

$$|g_i^T z_{\mathrm{trial}}| < 3 \quad (i = 0, 1, 2), \qquad (z_{\mathrm{trial}})_2 < 2, \qquad G(z_{\mathrm{trial}}) > 0. \tag{12.7}$$

The accepted points satisfy these bounds as well.

Positive homogeneity and (12.4)–(12.6) prove by induction that the deterministic full-BFGS run on G is

$$x_{3m+j} = \rho^m z_j, \qquad H_{3m+j} = \rho^m M_j.$$

Every trial at the corresponding phase is exactly ρ^m times its listed base trial. All comparison quantities scale by ρ^m, preserving every decision, not only acceptance at the selected length. Thus $x_k \to 0$ and $G(x_k) \downarrow 0$.

This limit is nonstationary: $d_* = (-1, 16)^T$ satisfies

$$g_0^T d_* = -1, \quad g_1^T d_* = -1, \quad g_2^T d_* = -c + 16d < -1.$$

In particular $G(td_*) = -t$ for $t \ge 0$ and $0 \notin \mathrm{conv}\{g_0, g_1, g_2\}$.

12.3.1 A simple coercive global failure

The same trajectory is a deterministic counterexample for the finite polyhedral function

$$\overline{G}(u, v) = \max\{u, -u - v/8, cu + dv, v - 17\}.$$

All trial values of G are positive while $v - 17 < 0$, by (12.7) and scaling. The fourth piece never affects the line search. The convex combination of branches $0, 1, 3$ with weights $(8/17, 8/17, 1/17)$

is identically -1. Equality fixes the unique minimizer $(-1, 16)$, and branch 2 is strictly below -1 there. Sublevels are bounded, because $u \le t$, $v \le t + 17$, and $-u - v/8 \le t$ give lower bounds on both coordinates. This supplies a coercive deterministic failure, but does not alone refute local convergence at its true minimizer. The next construction does.

12.4 Two smooth convex scalar functions

Set

$$\vartheta = \frac{1}{16}, \qquad a_n = \vartheta^n \ (n \ge 0), \qquad \epsilon_0 = \frac{1}{128}, \qquad b_* = 2c - 32d \in (33, 34).$$

We define scalar functions only for $b \in \{2, b_*\}$. Let

$$c_b = \frac{1 - (1 + \vartheta + \vartheta^2)/b}{1 - \vartheta^2}.$$

Define a monotone C^1 ramp P_b on $[0, 1]$ by

$$P_b(\xi) = \begin{cases} 0, & \xi \le c_b - \epsilon_0, \\ 3t^2 - 2t^3, & c_b - \epsilon_0 < \xi < c_b + \epsilon_0, \quad t = (\xi - c_b + \epsilon_0)/(2\epsilon_0), \\ 1, & \xi \ge c_b + \epsilon_0. \end{cases}$$

Then $0 \le P_b' \le 96$ and $\int_0^1 P_b(\xi)\, d\xi = 1 - c_b$. Its primitive $J_b(\xi) = \int_0^\xi P_b(t)\, dt$ is

$$J_b(\xi) = \begin{cases} 0, & \xi \le c_b - \epsilon_0, \\ 2\epsilon_0(t^3 - t^4/2), & c_b - \epsilon_0 < \xi < c_b + \epsilon_0, \\ \xi - c_b, & \xi \ge c_b + \epsilon_0. \end{cases}$$

For $x \in [b\vartheta a_n, b a_n]$, put

$$\xi = \frac{x/(b a_n) - \vartheta}{1 - \vartheta},$$

and define

$$\varphi_b(x) = (\vartheta a_n)^3 + b a_n^3 (1 - \vartheta) \left[\vartheta^2 \xi + (1 - \vartheta^2) J_b(\xi)\right]. \tag{12.8}$$

Complete its definition by

$$\varphi_b(x) = \begin{cases} 1 - 3(x + 1) + 3(x + 1)^2, & x < -1, \\ -x^3, & -1 \le x \le 0, \\ 1 + x - b, & x \ge b. \end{cases}$$

These intervals cover the real line, with agreeing formulas at shared endpoints.

Lemma 12.2. *Each φ_b is globally convex, C^2, nonnegative, coercive, and uniquely minimized at 0. Moreover,*

$$\varphi_b(b a_n) = a_n^3, \quad \varphi_b'(b a_n) = a_n^2, \quad \varphi_b''(b a_n) = 0, \tag{12.9}$$

$$0 \le \varphi_b''(x) \le \max\{6, 102/b\} \quad (x \in \mathbb{R}), \tag{12.10}$$

$$\boxed{\varphi_b(b a_n + h) = a_n^3 + a_n^2 h \quad \text{for } |h| \le a_n/2.} \tag{12.11}$$

Proof. Differentiate (12.8) on its positive intervals:

$$\varphi_b'(x) = a_n^2[\vartheta^2 + (1 - \vartheta^2) P_b(\xi)], \qquad \varphi_b''(x) = \frac{a_n(1 + \vartheta)}{b} P_b'(\xi).$$

The latter is nonnegative and at most $102 a_n/b$. The identity

$$\vartheta^2 + (1 - \vartheta^2)(1 - c_b) = \frac{1 + \vartheta + \vartheta^2}{b}$$

ensures the required integral and value matching at every node. The derivative is constant near each interval endpoint; thus the first two derivatives match adjoining intervals.

For $b\vartheta a_n \le x \le ba_n$, $a_n \le x/(b\vartheta)$. Consequently as $x \downarrow 0$,

$$\varphi_b(x) = O(x^3), \qquad \varphi_b'(x) = O(x^2), \qquad \varphi_b''(x) = O(x).$$

They match the negative formula at zero, including two derivatives. The joins at -1 and b are also C^2. The derivative is negative on $x < 0$ and positive on $x > 0$; both outer extensions diverge to infinity. This proves convexity, the minimum assertions, coercivity, and (12.10).

The constant-derivative lengths below and above ba_n, divided by a_n, are respectively

$$\ell_b = b(1-\vartheta)(1-c_b-\epsilon_0), \qquad u_b = \frac{b(1-\vartheta)(c_b-\epsilon_0)}{\vartheta}.$$

At $n = 0$ the affine outer extension supplies the upper length. Exact rational interval bounds give

b	c_b	ℓ_b	u_b
2	0.468627450980	0.981675091912	13.824448529412
b_*	0.971910573260	0.635765118557	483.655089309488

Both transition endpoints are strictly inside $(0, 1)$ and both lengths exceed $1/2$. This proves (12.11). □

Additional regularity. The construction even gives globally Lipschitz second derivatives. Within a ramp,

$$|\varphi_b'''(x)| \le \frac{1+\vartheta}{b^2(1-\vartheta)}\frac{3}{2\epsilon_0^2} = \frac{139264}{5b^2}.$$

The second derivative is continuous across every join and at the accumulation point 0. The uniform bound above, together with the bound 6 on the negative cubic part, proves global Lipschitz continuity of φ_b'' by summing over interval lengths. Thus the branches below are in $C^{2,1}$ as well. This does not make the first three branches analytic, nor any branch uniformly strongly convex.

12.5 The single objective with bad starts arbitrarily near its minimizer

Define

$$\begin{aligned} f_0(u,v) &= \varphi_2(u), \\ f_1(u,v) &= \varphi_2(-u - v/8), \\ f_2(u,v) &= \varphi_{b_*}(cu + dv), \\ f_3(u,v) &= v, \qquad F(u,v) = \max_{0\le i\le 3} f_i(u,v). \end{aligned} \tag{12.12}$$

Each branch is convex and C^2. For $i \le 2$, its Hessian is the scalar second derivative times $g_i g_i^T$. Thus

$$\|\nabla^2 f_0\| \le 51, \qquad \|\nabla^2 f_1\| \le 51\frac{65}{64} < 52,$$

while $b_* > 33$ gives $\varphi_{b_*}'' \le 6$ and $c^2 + d^2 < 2$, so

$$\|\nabla^2 f_2\| < 12, \qquad \nabla^2 f_3 = 0.$$

A common global upper bound $64I$ therefore holds. The preceding derivative estimate and $\|g_i\| < 2$ also give a common finite Lipschitz bound for the branch Hessians (for example 60000 suffices).

The first three branches are nonnegative and vanish at zero. If $F(x) = 0$, its first two scalar arguments must vanish; g_0 and g_1 are independent, so $x = 0$. The same two scalar arguments lie in

bounded scalar sublevels whenever $F(x)$ is bounded above, proving compactness of every sublevel of F.

At zero, the first three branches have zero gradient. Therefore

$$F'(0;h) = \max\{0, h_2\},$$

which proves that the unique minimizer is nonsmooth.

Put

$$p_n = a_n(2,-32)^T.$$

Then $g_0^T p_n = g_1^T p_n = 2a_n$ and $g_2^T p_n = b_* a_n$. Lemma 12.2 yields the exact local representation

$$\boxed{F(p_n + h) = a_n^3 + a_n^2 G(h)} \tag{12.13}$$

whenever $|g_i^T h| \le a_n/2$ for $i = 0, 1, 2$. Indeed, all three transformed branches are in their affine intervals and have value at least $a_n^3/2$. The first two slab inequalities imply $|h_1| \le a_n/2$ and $|h_2| \le 8a_n$, so $f_3(p_n + h) < 0$ and is inactive.

At p_n, the direction $d_* = (-1, 16)^T$ strictly decreases all three active branches. Hence

$$F(p_n) = a_n^3 > 0, \qquad F'(p_n; d_*) = -a_n^2 < 0, \qquad 0 \notin \partial F(p_n).$$

12.6 All rejected trials also stay in the local model

For each $n \ge 1$, initialize

$$\boxed{x_0^{(n)} = p_n + a_n^2 z_0, \qquad H_0 = I.}$$

The claimed deterministic trajectory is

$$x_{3m+j}^{(n)} = p_n + a_n^2 \rho^m z_j, \qquad H_{3m+j}^{(n)} = \rho^m M_j, \qquad j = 0, 1, 2. \tag{12.14}$$

At the corresponding phase, *every* trial generated by the line search has the form

$$p_n + a_n^2 \rho^m z_{\text{trial}},$$

where z_{trial} is one of the nine base trials already certified. By (12.7),

$$|g_i^T (a_n^2 \rho^m z_{\text{trial}})| < 3a_n^2 \le \frac{3a_n}{16} < \frac{a_n}{2}.$$

Thus (12.13) holds at every accepted and rejected trial. Their values and gradients are

$$F(p_n + a_n^2 \rho^m z_{\text{trial}}) = a_n^3 + a_n^4 \rho^m G(z_{\text{trial}}), \qquad \nabla F = a_n^2 g_{\text{active}}.$$

Every function difference and every predicted directional decrease is the corresponding seed quantity multiplied by $a_n^4 \rho^m$. The additive a_n^3 cancels from Armijo. Therefore all rejection and acceptance decisions are the same as in the seed table. In particular the accepted lengths are actual outputs of the deterministic search.

Substituting gradients scaled by a_n^2 and steps scaled by $a_n^2 \rho^m$ into (12.1) leaves the projection factors unchanged and scales the rank-one term by ρ^m. This proves the matrix part of (12.14). The point and matrix cycle closures complete the induction. All finite matrices are SPD, and all oracle evaluations are at differentiable points.

Consequently,

$$x_k^{(n)} \to p_n \ne 0, \qquad F(x_k^{(n)}) \downarrow a_n^3 > 0.$$

Yet $x_0^{(n)} \to 0$ and $F(x_0^{(n)}) \to 0$. Also the supremum of the norms of *all accepted and trial points* in the nth trajectory is bounded by

$$a_n \|(2,-32)\| + a_n^2 \max_{\text{base trials}} \|z_{\text{trial}}\| \to 0.$$

This proves Theorem 12.1. The failure is not caused by leaving a local neighborhood.

Relation to earlier work

The following comparisons are bibliographic scholarship, separate from the formal theorem assertions.

The unit-first search in this chapter is Lewis–Overton's Algorithm 4.6 [2]. Earlier negative results include their convex subgradient/exact-search construction with null steps [28], smooth globally nonconvex examples [31, 32, 33, 34, 35], Clarke-stationary limits with descent directions [29], and scaled memoryless BFGS failure [30]. Those settings differ from the convex, differentiable-query, full-update statement here. The recent planar counterexample of Diao uses a smooth nonconvex objective and an escaping exact-search trajectory [36]. No exact match for the complete theorem was identified in the documented search; this is not a priority certification.

Prescribing an algebraic orbit and then constructing an objective that realizes its update equations has substantial precedents in Mascarenhas's divergence constructions [31, 32, 33]. The convex interpolation and all-query containment in this chapter are specific additional properties. Preconditioned residual maps and rigorous root enclosures are standard validated-numerics tools; compare Rump [37, Theorem 13.3]. Here the required rational inequalities have their own Lean proofs, with no external numerical computation imported as a proof.

CHAPTER 13

Failure with positive lower curvature

Positive lower curvature requires a separate construction. The local finite-branch functions have Lipschitz gradients, and three of the four lose second differentiability at the minimizer; no claim combines uniform strong convexity with the preceding local $C^{2,1}$ branch regularity.

13.1 Statements and retained implementation

All runs use ordinary full BFGS, $H_0 \succ 0$, differentiable oracle queries, and a fixed unit-first Armijo–weak-Wolfe search. The search starts with $\alpha = 1$, lower bracket zero and upper bracket infinity; an Armijo failure updates the upper bracket, a Wolfe failure the lower bracket; use the midpoint after bracketing and doubling otherwise. Fix

$$0 < c_1 \le \tfrac{1}{3}, \qquad c_1 < c_2 < 1.$$

The orbits, objectives and initial points below are chosen before this pair. The update is unmodified:

$$s = -\alpha H g, \qquad y = g^+ - g, \qquad H^+ = \left(I - \frac{sy^\mathsf{T}}{y^\mathsf{T}s}\right) H \left(I - \frac{ys^\mathsf{T}}{y^\mathsf{T}s}\right) + \frac{ss^\mathsf{T}}{y^\mathsf{T}s}.$$

The seed inequalities have strict margins, so there are no equality-convention issues.

Theorem 13.1 (Global failure with constant positive branch Hessians)**.** *There are three fixed slopes $g_0, g_1, g_2 \in \mathbb{R}^2$ such that, for every $\mu > 0$,*

$$Q_\mu(x) = \tfrac{\mu}{2}\|x\|^2 + \max_{i=0,1,2} g_i^\mathsf{T} x$$

is μ-strongly convex, coercive, and has a unique nonsmooth minimizer $x_* \neq 0$, yet admits initial states (x_0, H_0) that produce, for every such pair (c_1, c_2), the same deterministic full-BFGS iterates converging to the nonstationary origin. Initial states and every future trial may be chosen arbitrarily close to the origin in the sense made precise in Section 13.3. Every branch has Hessian exactly μI globally. The accepted lengths repeat $1, 1/4, 1/16$.

This is *global* failure: approaching the wrong vertex is not the same as approaching the actual minimizer. It does not contradict our local PLQ theorem.

Theorem 13.2 (Local failure under the pointwise Hessian bounds)**.** *There is a finite strongly convex objective $F : \mathbb{R}^2 \to \mathbb{R}$ with compact sublevels and a unique nonsmooth minimizer at zero, and constants $0 < \underline{\mu} \le \overline{L} < \infty$, such that*

$$\underline{\mu} I \preceq \nabla^2 F(x) \preceq \overline{L} I \quad \text{at every twice-differentiable point,}$$

yet the fixed deterministic full-BFGS implementation with $H_0 = I$ has bad initial points $x_0^{(n)} \to 0$ whose runs, the same for every such pair (c_1, c_2), converge to nonstationary points $p_n \to 0$. All trial points of the nth run remain uniformly near zero as $n \to \infty$.

The function can be chosen as a maximum of four globally $C^{1,1}$, uniformly strongly convex, globally gradient-Lipschitz branches. Alternatively, it can be chosen so that its Hessian is one fixed positive-definite matrix wherever it is twice differentiable; the latter version has a countable polyhedral nonsmooth part.

The local theorem satisfies the literal classical-Hessian assumption, and even actual strong convexity. It is *not* a local counterexample with four globally C^2 branches having uniform lower curvature. That stronger combination remains unproved here.

13.2 The inherited algebraic seed

The seed is reproduced sufficiently to specify the new calculations. Its exact symbolic closure and nine strict query decisions were certified in Chapter 12.

$$g_0 = (1,0)^{\mathsf{T}}, \qquad g_1 = (-1,-1/8)^{\mathsf{T}}, \qquad g_2 = (c,d)^{\mathsf{T}}.$$

Put

$$\begin{aligned} &U = 127c + 16d, && V = 16c - 127d - 16, && R = c - 8d,\\ &K = 8c^2 + 8d^2 + d - 8, && T = c^2 - c + d^2, && Y = (c-1)^2 + d^2. \end{aligned}$$

The algebraic constants solve

$$264192RK - UV(U + 129) = 0, \qquad 16VTK - RUY = 0$$

in the rational ℓ_∞ box of radius 10^{-30} about

$$\begin{aligned} c_0 &= 0.318481427627118630646451312502907037179122050236 05,\\ d_0 &= -1.025232119675748725481350870013004142845119093184 44. \end{aligned}$$

The Lean formalization proves existence and uniqueness of this root by a contracting self-map with exact rational bounds checked in Lean. Define

$$\rho = \frac{129R}{V(U+129)} \in (0.060, 0.061), \qquad z_0 = \frac{1}{1-\rho}\begin{pmatrix} 897/1024 - \rho(1-c) \\ \rho d - 1/64 \end{pmatrix}.$$

Let $M_0 = I$ and use consecutive gradients g_0, g_1, g_2, g_0 with lengths $1, 1/4, 1/16$. Exact full-BFGS algebra gives

$$M_1 = \begin{pmatrix} 129/256 & -1/16 \\ -1/16 & 1 \end{pmatrix}, \qquad M_3 = \rho I, \qquad z_3 = \rho z_0.$$

Here $z_{j+1} = z_j - \alpha_j M_j g_j$. Every displayed accepted gradient is uniquely active for $G(z) = \max_i g_i^{\mathsf{T}} z$. The finite line-search query lists are

$$[1], \qquad [1, 1/2, 1/4], \qquad [1, 1/2, 1/4, 1/8, 1/16].$$

Every rejected trial has a value larger than the current value. The accepted decrease ratios exceed $1/3$, and the accepted new-to-old directional-derivative ratios are negative. All query points have a unique active branch. Thus the query decisions and branch itinerary persist in a sufficiently small open neighborhood of the normalized seed state. This strictness is essential below.

The vector $(-1, 16)^{\mathsf{T}}$ has negative inner product with every g_i, so $0 \notin \operatorname{conv}\{g_i\}$. On the other hand,

$$\bar{g} = (1,-16)^{\mathsf{T}}/257 = \frac{129}{257} g_0 + \frac{128}{257} g_1$$

is the least-norm point of this convex hull. Indeed $g_0^{\mathsf{T}}\bar{g} = g_1^{\mathsf{T}}\bar{g} = 1/257 = \|\bar{g}\|^2$ and $g_2^{\mathsf{T}}\bar{g} > 1/257$. Consequently

$$x_* = -\bar{g}/\mu, \qquad \min Q_\mu = -\frac{1}{514\mu}. \tag{13.1}$$

At x_* precisely the first two branches are active; their gradients differ, so the unique minimizer is nonsmooth. At the origin $Q_\mu(0) = 0$ and the subdifferential is the same nonstationary triangle as for G.

13.3 Blowing up the shrinking cycle

Adding a quadratic changes the BFGS iterates. We do not claim the old exact orbit survives unchanged. Instead we prove existence of nearby perturbed orbits.

Use the five normalized state coordinates

$$x = \tau z, \qquad H = \tau M(b,e), \qquad M(b,e) = \begin{pmatrix} 1 & b \\ b & e \end{pmatrix}, \qquad \tau > 0.$$

The limiting seed state is

$$P_* = (z_0, 0, 1, 0),$$

where the last three entries denote (b, e, τ). To define a three-iteration return map near this boundary state, keep the input scale τ fixed during the three substeps. Start with $(Z, M) = (z, M(b,e))$ and, for $j = 0, 1, 2$, set

$$\gamma_j = g_j + \mu\tau Z, \qquad S = -\alpha_j M\gamma_j, \qquad Z^+ = Z + S, \tag{13.2}$$

$$Y_j = g_{j+1} - g_j + \mu\tau S, \tag{13.3}$$

$$M^+ = \left(I - \frac{SY_j^\mathsf{T}}{Y_j^\mathsf{T}S}\right) M \left(I - \frac{Y_jS^\mathsf{T}}{Y_j^\mathsf{T}S}\right) + \frac{SS^\mathsf{T}}{Y_j^\mathsf{T}S}. \tag{13.4}$$

Gradient indices are cyclic and $\alpha = (1, 1/4, 1/16)$. Let the final quantities be (Z_3, M_3) and let $r = (M_3)_{11}$. The normalized return map is

$$\mathcal{T}_\mu(z, b, e, \tau) = \left(Z_3/r,\ (M_3)_{12}/r,\ (M_3)_{22}/r,\ \tau r\right). \tag{13.5}$$

All denominators are nonzero at P_*, and $r = \rho > 0$ there. This is a real-analytic map on an open neighborhood of P_*, allowing an artificial negative τ in that analytic extension. Only its positive-τ part represents actual BFGS.

Lemma 13.1 (Agreement with the actual implementation)**.** *There is an open neighborhood $\mathcal{N}$ of P_* such that for every state in $\mathcal{N}$ with $\tau > 0$ and every such pair (c_1, c_2), the actual three consecutive full-BFGS iterations on Q_μ have exactly the certified trial lists and return map (13.5).*

Proof. At a query τZ, the common quadratic does not affect which slope is active. Its function value divided by τ is

$$G(Z) + \tfrac{\mu\tau}{2}\|Z\|^2,$$

and its gradient is $g_i + \mu\tau Z$. All Armijo and weak-Wolfe comparisons, divided by their common positive scale when needed, are continuous in the normalized state. The seed has strict branch gaps and strict decisions at its finite list of nine queries: accepted queries pass Armijo with $c_1 = 1/3$ and have positive new directional derivatives, and rejected queries raise the value, so the same finitely many strict inequalities decide every such pair (c_1, c_2). Intersecting the corresponding open neighborhoods preserves the entire decision tree. The normalized secant and update formulas are exactly (13.2)–(13.3); no reset of the actual matrix is made. □

13.4 A certified hyperbolic fixed point

At $\tau = 0$ the matrix return depends only on (b, e), not on z. Its normalized shape map is

$$(b, e) \longmapsto ((M_3)_{12}/(M_3)_{11}, (M_3)_{22}/(M_3)_{11}).$$

Let J be its derivative at $(0, 1)$. The supplied rational-interval automatic-differentiation program computes rigorous enclosures of all entries; their decimal displays are

$$J \simeq \begin{pmatrix} 1.112398625073290 & 0.645970102731598 \\ 0.751501570427498 & 0.490513688583175 \end{pmatrix}.$$

More importantly, using exact rational interval signs of $P_J(\lambda) = \lambda^2 - \operatorname{tr}(J)\lambda + \det J$, it proves

$$0.038 < \lambda_s < 0.039, \qquad 1.564 < \lambda_u < 1.565. \tag{13.6}$$

Specifically $P_J(0.038) > 0$, $P_J(0.039) < 0$, $P_J(1.564) < 0$, and $P_J(1.565) > 0$ throughout the algebraic root box. The two disjoint sign-change intervals contain the two distinct real roots. No floating-point eigensolver is used for this assertion.

The full derivative has the analytically determined block-triangular form

$$D\mathcal{T}_\mu(P_*) = \begin{pmatrix} \rho^{-1} I_2 & C & \mu a \\ 0 & J & \mu b \\ 0 & 0 & \rho \end{pmatrix}. \tag{13.7}$$

The unneeded blocks C, a, b are finite. The upper left block follows because, at $\tau = 0$, $Z_3 = z + v(b, e)$ and r is independent of z. The last row follows from $\tau^+ = \tau r$.

The eigenvalues are therefore

$$\rho^{-1},\ \rho^{-1},\ \lambda_s,\ \lambda_u,\ \rho.$$

None lies on the unit circle and none is zero. There are exactly two stable eigenvalues. The derivative is invertible, so the inverse function theorem makes $\mathcal{T}_\mu$ a local diffeomorphism.

Crucially ρ is different from every eigenvalue of the upper left 4×4 block. Thus its eigenvector can be chosen with last coordinate one: solve the top four linear equations after assigning $v_\tau = 1$. The stable subspace is not contained in the artificial boundary $\{\tau = 0\}$.

Proof of Theorem 13.1. Apply the local stable-manifold theorem for a hyperbolic fixed point of a smooth local diffeomorphism; see [3, Sections 3–4]. It gives a smooth two-dimensional local stable manifold tangent to the stable eigenspace at P_*. Since this tangent space has a vector with nonzero τ component, the stable manifold contains points with arbitrarily small positive τ.

Choose such a point sufficiently close that its entire forward return-map orbit remains in the neighborhood of Lemma 13.1 and converges to P_*. The positive half-space is preserved because $\tau^+ = \tau r$ and $r > 0$ in this neighborhood. At every return, $M(b, e)$ is positive definite and close to I; all actual substep matrices are positive definite as well.

The return-map orbit represents the actual deterministic full-BFGS orbit, with accepted lengths $1, 1/4, 1/16$. Its physical states satisfy

$$x_{3m} = \tau_m z_m \to 0, \qquad H_{3m} = \tau_m M(b_m, e_m) \to 0.$$

The intermediate accepted states and all rejected queries are τ_m times bounded continuous functions of the normalized state, so they tend to zero too. They retain strict branch gaps and strict decisions. Their objective values are positive near the seed and decrease to zero. Equation (13.1) and $0 \notin \partial Q_\mu(0)$ prove actual nonstationary convergence.

The same argument shows more: given any neighborhood of physical zero, the initial stable-manifold point can be chosen so that *the initial point and all queries of the orbit* lie inside it. This small-orbit property will be used for the embedding. The initial matrix is supplied by the stable manifold, not assumed to be I in these particular coordinates. □

This is an existence proof of an actual trajectory, not a claim that a numerically truncated stable eigenvector produces it forever. Expanding eigenvalues make ordinary floating-point forward shooting an inappropriate certificate of membership in the stable manifold.

13.5 A local accumulation embedding with positive lower curvature

We now construct a single objective satisfying the literal Hessian bounds and having no guaranteed local basin. Fix

$$\mu = 1/4096, \quad c_* = (2, -32)^{\mathsf{T}}, \quad q_i = g_i - \mu c_*, \quad b_i = q_i^{\mathsf{T}} c_*, \quad r = 1/128, \quad a_n = r^n.$$

The seed interval certificate proves

$$b_0 = b_1 = 1791/1024, \qquad 1.7 < b_i < 34.$$

Only two distinct values of b_i occur. For either value write

$$\vartheta_b = \frac{1 - (1+r)/b}{1-r}, \qquad \epsilon = 1/1024.$$

Define a monotone C^1 ramp P_b on $[0,1]$: it is zero for $\xi \le \vartheta_b - \epsilon$, one for $\xi \ge \vartheta_b + \epsilon$, and on the intervening interval equals

$$P_b(\xi) = 3t^2 - 2t^3, \qquad t = \frac{\xi - \vartheta_b + \epsilon}{2\epsilon}.$$

Let $J_b(\xi) = \int_0^\xi P_b(v)\,dv$. The elementary formulas give

$$\int_0^1 P_b = 1 - \vartheta_b, \qquad 0 \le P_b' \le 768.$$

On $[ba_{n+1}, ba_n]$ put

$$\xi = \frac{x/(ba_n) - r}{1-r}, \tag{13.8}$$

$$\varphi_b(x) = (ra_n)^2 + ba_n^2(1-r)\{r\xi + (1-r)J_b(\xi)\}. \tag{13.9}$$

Complete the definition by $\varphi_b(0) = 0$, $\varphi_b(x) = x^2/2$ for $x < 0$, and $\varphi_b(x) = 1 + x - b$ for $x \ge b$.

Lemma 13.2 (Scalar regularity and exact affine neighborhoods)**.** *Each φ_b is convex, nonnegative, coercive, uniquely minimized at zero, and globally $C^{1,1}$. It is C^2 on $(0,\infty)$ and on $(-\infty,0)$, but is not twice differentiable at zero. At every node,*

$$\varphi_b(ba_n) = a_n^2, \quad \varphi_b'(ba_n) = a_n,$$

and

$$\varphi_b(ba_n + h) = a_n^2 + a_n h \qquad (|h| \le a_n/2). \tag{13.10}$$

Where its second derivative exists, $0 \le \varphi_b'' \le \max\{1, 768/b\}$.

Proof. On the positive intervals,

$$\varphi_b' = a_n\{r + (1-r)P_b(\xi)\}, \qquad \varphi_b'' = P_b'(\xi)/b.$$

The defining value of ϑ_b gives $r + (1-r)(1-\vartheta_b) = (1+r)/b$, which proves value matching. The derivative rises monotonically from ra_n to a_n and is constant near the endpoints. Thus values and derivatives match at every node. At zero $\varphi_b' = O(x)$ and $\varphi_b = O(x^2)$, giving differentiability with derivative zero. Uniform bounded derivative slopes on all intervals imply a global Lipschitz bound, including across the accumulation point by taking limits. The negative and outer extensions match in value and first derivative.

The constant-derivative lengths on the two sides of ba_n, divided by a_n, are

$$b(1-r)(1-\vartheta_b-\epsilon), \qquad b(1-r)(\vartheta_b-\epsilon)/r.$$

Both exceed $1/2$ for the certified values of b. This proves (13.10). At zero, along $x = ba_n$ the ratio $\varphi_b'(x)/x$ equals $1/b$, whereas along $x = (b+1/4)a_n$ it equals $1/(b+1/4)$. Both sequences tend to zero, so $\varphi_b''(0)$ does not exist. This is the deliberate regularity distinction from the previous $C^{2,1}$ embedding. □

Define four functions and their maximum:

$$f_i(x) = \tfrac{\mu}{2}\|x\|^2 + \varphi_{b_i}(q_i^\mathsf{T} x) \quad (i = 0, 1, 2), \qquad f_3(x) = \tfrac{\mu}{2}\|x\|^2 + x_2, \qquad F = \max_i f_i. \tag{13.11}$$

Each branch is globally μ-strongly convex and has a globally Lipschitz gradient. The Lean formalization proves the common gradient-Lipschitz constant 512 from the actual scalar series and exact rational bounds: the first three Hessian upper bounds are less than $439, 446, 27$, respectively, wherever those Hessians exist, and the fourth has Hessian μI. The global gradient bounds are proved directly across the scalar nonsmooth-second-derivative points. Thus the maximum is μ-strongly convex.

Since the first three scalar terms are nonnegative,

$$F(x) \geq \tfrac{\mu}{2}\|x\|^2, \qquad F(0) = 0.$$

This proves compact sublevels and the unique minimum at zero. The first three gradients vanish there and the fourth is $(0, 1)^\mathsf{T}$; consequently $F'(0; h) = \max\{0, h_2\}$ and the minimizer is nonsmooth.

At any differentiable point of a finite maximum of L-smooth functions, all active gradients agree. Inactive pieces remain inactive locally, and the upper smooth Taylor bounds for the active pieces therefore give a local upper Taylor bound $L\|h\|^2/2$ for the maximum. Together with strong convexity, this proves

$$\mu I \preceq \nabla^2 F(x) \preceq 512 I \quad \text{whenever } F \text{ is twice differentiable.} \tag{13.12}$$

This conclusion does not assume the maximum has a globally Lipschitz gradient.

13.6 Copying one perturbed bad orbit at every scale

Set $p_n = a_n c_*$ and $C_* = 1 + \mu\|c_*\|^2/2$. For any y with $|q_i^\mathsf{T} y| \leq 1/2$,

$$\begin{aligned} f_i(p_n + a_n y) &= a_n^2 \left\{ \tfrac{\mu}{2}\|c_* + y\|^2 + 1 + q_i^\mathsf{T} y \right\} \\ &= a_n^2 \{ C_* + g_i^\mathsf{T} y + \tfrac{\mu}{2}\|y\|^2 \}. \end{aligned}$$

The fourth branch is negative relative to its quadratic part there if y is sufficiently small and $n \geq 1$. Hence, on a fixed small neighborhood of $y = 0$,

$$F(p_n + a_n y) = a_n^2 \{ C_* + Q_\mu(y) \}. \tag{13.13}$$

Use Theorem 13.1, whose orbit does not depend on (c_1, c_2), to choose one entire bad orbit $\{\bar{x}_k, \bar{H}_k\}$ of Q_μ whose initial point and accepted and rejected queries are all in that neighborhood. The same orbit can be made as small as needed, by the last part of its proof. Set

$$x_0^{(n)} = p_n + a_n \bar{x}_0, \qquad H_0^{(n)} = \bar{H}_0.$$

Every query in the nth run is exactly $p_n + a_n$ times the corresponding seed query, and

$$x_k^{(n)} = p_n + a_n \bar{x}_k, \qquad H_k^{(n)} = \bar{H}_k.$$

Indeed, function differences and directional-derivative products scale by a_n^2, while both gradients and steps scale by a_n. In the inverse-BFGS formula both scale factors cancel. The line search therefore makes the same decisions, including all rejections. Equation (13.13) applies at the initial point and to every query by the choice of the whole seed orbit. Induction proves the claim without numerical approximation.

Since $\bar{x}_k \to 0$,

$$x_k^{(n)} \to p_n, \qquad F(x_k^{(n)}) \downarrow a_n^2 C_* > 0.$$

At p_n the three active gradients are $a_n g_i$, so $0 \notin \partial F(p_n)$. Meanwhile $x_0^{(n)} \to 0$ and the supremum of the norms of all queries is at most

$$a_n \left(\|c_*\| + \sup_{\text{seed queries}} \|\bar{x}\| \right) \to 0.$$

The initial matrix is one fixed SPD matrix for every n.

Fixing identity initialization. Let $A = \bar{H}_0^{1/2}$ and define $\widetilde{F}(z) = F(Az)$. Full-BFGS affine invariance preserves the complete line-search decisions and gives initial matrix I in the z coordinates. The branch strong-convexity and gradient-Lipschitz constants become, for example,

$$\underline{\mu} = \mu \lambda_{\min}(\bar{H}_0) > 0, \qquad \overline{L} = 512 \lambda_{\max}(\bar{H}_0) < \infty.$$

These constants are fixed for the one transformed objective, not changed between starts. Its bad initial and limit points approach zero, and all queries remain arbitrarily close. This proves the four-$C^{1,1}$-branch part of Theorem 13.2.

13.7 The literal Hessian condition permits an even simpler variant

Replace the smooth ramp by the step $P_b(\xi) = \mathbf{1}_{\{\xi > \vartheta_b\}}$ and its primitive by $J_b(\xi) = (\xi - \vartheta_b)_+$. Use (13.9) on the positive intervals, the negative extension $\varphi_b(x) = -x$ for $x < 0$, and the same affine positive outer extension. This scalar function is convex and piecewise affine with countably many pieces; it has the same node values, node slopes, and affine neighborhoods (13.10). Thus the entire local-copy construction and the same fixed initial matrix work unchanged.

For this variant $F = \mu \|x\|^2 / 2 + p(x)$, with convex countably piecewise-affine p. At a nonzero point, any branch with $q_i^{\mathsf{T}} x = 0$ is strictly below the maximum: its scalar value is zero, whereas at least one of $q_0^{\mathsf{T}} x, q_1^{\mathsf{T}} x$ is nonzero and has positive scalar value. Here q_0, q_1 are independent. All potentially active scalar arguments are therefore nonzero and encounter only finitely many breakpoints in a small neighborhood. Hence p is locally finite-polyhedral near every nonzero point. At any differentiable such point it is locally affine. The origin is nonsmooth. Consequently

$$\nabla^2 F(x) = \mu I \quad \text{at every twice-differentiable point.}$$

After the fixed affine transformation making $H_0 = I$, that Hessian is the one fixed SPD matrix $\mu \bar{H}_0$. This establishes the remaining version of Theorem 13.2. No claim of four smooth branches is made for this alternative.

Relation to earlier work

The following comparisons are bibliographic scholarship, separate from the formal theorem assertions.

The negative-results comparisons in Chapter 12 apply here with the stated stronger convexity and different branch regularity. Like Chapter 12, the chapter covers every pair $0 < c_1 \le 1/3$, $c_1 < c_2 < 1$. The first three branches of the local finite-branch construction fail to be twice differentiable at the minimizer. The simpler global quadratic-plus-maximum example does not establish failure arbitrarily near its true minimizer. These distinctions are essential when comparing the theorem with smooth nonconvex counterexamples [32, 33, 36].

The stable-manifold principle and graph-transform method are classical; this chapter already uses Dyatlov's exposition [3, Sections 3–4]. The report-specific work identifies the derivative of the actual BFGS return map, certifies its spectral bounds, and applies the principle to that map. Its exact formal orbit construction is recorded with the proof correspondence.

The final change of coordinates fixing the common SPD initialization to identity uses the established affine invariance of full BFGS; see Lewis–Overton [2, Section 2] and Xie–Wächter [1, Theorem II.1]. It is a single fixed transformation for the constructed family, not a per-iteration algorithm change.

CHAPTER 14

Formal correspondence and the evidence boundary

Mathematical proof, formal statement correspondence, computation and historical execution are different kinds of evidence. The dated authoring audit is stated first; the preserved release record that follows is historical evidence, not a new run count.

The October 2026 Lean review

An independent review rebuilt the main library from an empty build directory with Lean 4.19.0 and the pinned mathlib cache: all 736 modules compiled, with no errors and no `sorry`. The new package `bfgs_lean_article/` also compiles the article supplements and the lower-curvature failure theorems for the non-strict tests (Armijo $\leq$, weak Wolfe $\geq$). Both failures hold for every $0 < c_1 \leq 1/3$ and $c_1 < c_2 < 1$, with one orbit, objective and set of starts for all such pairs. The axiom audit of 15 October 2026 checked 5420 Lean names, namely every name then cited by the claim ledger or by the Lean coverage registers of both documents (`editorial/lean_coverage/register/`) and the principal declarations of the package; they depend only on `propext`, `Classical.choice` and `Quot.sound` (`bfgs_lean_article/AXIOM_AUDIT.json`). A kernel replay with lean4checker re-checked every project module except the two root modules, which contain only imports (783 modules; `bfgs_lean_article/KERNEL_REPLAY.json`).

The 12 September 2026 narrative-revision replay

The recorded 12 September replay used Lean 4.19.0 and mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The command `lake build BfgsVerification` completed successfully in the existing pinned build environment. Fresh main and supplemental declaration audits also passed. This was an incremental build and a fresh audit, not a claim of recompiling every dependency from an empty cache. All nine dependency Git revisions and clean states agreed before and after the run; no audited source input changed.

The main project contributed 9,672 safe declarations; the union with the supplemental correction modules had 9,781, including 7,954 theorems. The only transitive axiom dependencies were `propext`, `Classical.choice` and `Quot.sound`. The checks reject `sorryAx`, custom mathematical axioms, and reachable unsafe or partial implementation artifacts, examining types, proof values and structural dependencies. This restriction identifies the foundations actually used; it does not establish their consistency or replace trust in the kernel.

The unmodified audit runner and its source provenance are under `editorial/formal_audit/`. Actual per-command output, input/output hashes, elapsed times, package states and the overall PASS are in `editorial/formal_audit/runs/20260912_narrative/SUMMARY.json` and its `logs/` directory. The preceding sealed master-report validation also passed, with output retained in `editorial/logs/baseline_release_gate.json`. Reproduction from the repository root is:

```
python3 editorial/formal_audit/run.py --run
  --run-dir editorial/formal_audit/runs/local_replay
```

The command is one shell line; the displayed break is typographic.

The narrative revision also adds two separately compiled article supplements. The literal mixed-benchmark theorem quantifies over every actual admissible run for $u^2 + |v|$ and proves both value and whole-iterate convergence. Its three module-owned theorems have 31,960 checked transitive

dependencies. The matched-scaling supplement proves eleven declarations for the update, pairings and Armijo–weak-Wolfe comparisons, with 10,697 checked transitive dependencies. It establishes covariance; the existing construction contracts supply orbit existence and all-query containment. Both audits passed with only the three permitted axioms, agreement between kernel and collected type/value dependencies, and agreement with the ordinary axiom collector. The runs, exact types and hashed compiler outputs are registered in `editorial/narrative_rebuild/FORMAL_SUPPLEMENTS.json`. These totals overlap the main library and are not additive. A further skeptical review freshly compiled the unchanged fourteen article-geometry declarations and checked their types, proof values and complete structural dependencies, including mutual inductives and recursor rules. The 26,496-declaration closure agrees with the earlier audit, uses only the three permitted axioms, and agrees independently with Lean's ordinary axiom collector. Exact theorem types are unchanged. The runner and actual compiler/audit evidence are under `editorial/satisfaction_review/formal/`; the earlier evidence remains under `editorial/article_formal/`.

The same review also formalized the restriction and fixed-coordinate-change argument for the first three lower-counterexample branches: each actual whitened branch fails differentiability of its Fréchet derivative at zero. The combined export includes this clause together with the original full local-failure target. Its three declarations and 37,525 transitive dependencies passed the same structural, kernel and axiom checks; see `editorial/satisfaction_review/branch_formal/`.

The following inherited verification material describes the earlier corrected release and its limits. Its dates identify those earlier executions.

Operative verification record. The evidence for this edition consists of actual pinned Lean builds, checked theorem statements and module-wide transitive axiom audits. The original report's old job counts, timings, precision claims and unprovided archive assertions are not adopted. They remain visible only in the preserved original and edit ledger.

14.1 What the checks mean

Lean checks the formal proof term against its precise statement. Exact rational inequalities, algebraic roots, actual matrix updates, objective derivatives, infinite sequences and geometric existence arguments have different mathematical roles, but each retained assertion here has a reviewed formal correspondence. An external numerical result is not imported as a proof. Compiler logs are evidence of actual executions; they are not axioms and do not prove that an older execution happened.

The principal results concern ordinary full BFGS. The negative seeds and their embeddings are actual objective/run constructions. The positive framework derives its scalar hierarchy from actual principal determinants. The stable-manifold conclusion in Chapter 13 follows from the actual return map and a proved analytic argument, not a numerical starting vector assumed to lie on that manifold. Exact formal hypotheses and dependencies appear in the complete claim index.

14.2 Actual build and audit register

The pinned project uses Lean 4.19.0 and mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b`. All nine package revisions and clean dependency trees are checked. The preserved main source set contains 737 Lean files. The current module-wide audit supplements the older namespace-based theorem audit; the two scopes must not be conflated.

Record	Meaning
Main project	9,672 safe declarations, including 7,874 theorems.
Main plus exact supplemental corrections	9,781 distinct safe declarations, including 7,954 theorems.

Record	Meaning
Transitive foundations	Only `propext`, `Classical.choice`, `Quot.sound`. No safe proof may reach an unsafe or partial artifact.
Preserved theorem audit	All 7,813 formerly enumerated theorem types and axiom sets are reconciled with the wider audit.
Claim review 20	Six actual supplemental compiler jobs; 182 ordinary declaration reports, representing 141 distinct names. These occurrences are not added to the unique-declaration totals.
Module-wide audit	Main-only and combined environments are checked separately, including private/generated declarations and actual type/value dependencies.

The actual command records, exit codes, compiler output and source hashes are delivered under `master_report/audit/`. The earlier closed records under `audit/claim_review20/` and the reconstruction audit remain byte-preserved. New runs use new output directories. The safe declaration set is counted by names: lazy equation lemmas can receive different importing-module metadata without changing their names, types or proof dependencies.

14.3 Mathematical scope of the reconstructed suites

The labels below are the original report's suite names, whose letters are not chapter letters; they preserve correspondence with that report. They describe present formal results, not a claim that the unavailable old scripts ran with their printed sample counts. Each subsection's exact exports are listed in the claim index.

14.3.1 D-root

Actual algebraic seed existence and isolation, denominator signs, accepted and rejected trial decisions, and the global embedding geometry are proved. The root and containment statements apply to the exact specified objects, rather than a rounded seed.

14.3.2 D-symbolic

The inverse and Hessian updates, their secant identities, determinant and positivity properties, and the literal seed-cycle closure are Lean equalities or inequalities. No symbolic-program output is accepted as a mathematical premise.

14.3.3 D-full-oracle

The exact deterministic failure and embedded local failure theorems establish the actual run and every stated line-search trial. They imply the specified infinite behavior; a finite list of sampled trials is not their proof.

14.3.4 L-hyperbolicity

The normalized Jacobian is the derivative of the actual return map. The spectral contract and rational precision statements are proved for that derivative, and the actual stable-manifold construction is linked to it. The embedding's Hessian claims retain their stated local scope.

14.3.5 L-embedding

Scalar node, peak, ramp and regularity formulas prove the complete local model. The smooth and step constructions then give their respective actual failure copies. The two regularity constructions retain their distinct hypotheses and conclusions.

14.3.6 A-minors

Compression identities and the Gram numerator are proved in arbitrary finite dimension. The principal Gram-sum identity and full-BFGS principal-minor hierarchy use actual determinants. Selected finite-dimensional cases are not substituted for this theorem.

14.3.7 H-lifting

Actual projected full updates and the invariant matrix range give affine-fiber lifting. Reusable conditional limit implications retain their projected-run hypotheses. The final low-rank applications derive the limits they use.

14.3.8 M-mixed

Along runs with bounded gradients, the actual full-matrix hierarchy yields the normalized hierarchy, weighted potential, qualitative reciprocal growth and superquadratic reciprocal growth under summable predicted decreases. A separately specified unexcited-metric example proves its literal matrix-limit statement; no such limit is assumed in the principal convergence framework.

14.3.9 P-polyhedral

The smooth-plus-polyhedral and smooth PLQ basin theorems, linear-image lifting and the stated scalar regularity example are formal results. Their hypotheses distinguish local smoothness from the stronger twice-differentiable conditions that some examples fail.

14.3.10 R-secant

Finite maxima, mixed powers, inward inequalities and homogeneous gauge composition lead to the stated radial basins and actual-run convergence. The exponents, domains and differentiability assumptions are present in the exact theorem types.

14.3.11 R-interpretation

Supporting-plane, radial and inward formulations are related by proved equivalences on their stated compact domains. The corrected homogeneous reference formula and compact optimizer deductions are included. The explicit four-piece objective in Chapter 4 is a new specified reconstruction with an actual full-BFGS failure run; it is not asserted to be an unidentified historical objective.

14.3.12 G-obstruction

Pointwise PSD feasibility, eligible-pair realization, incoming-jump obstructions and accumulating-kink certificate nonexistence are proved. Accessible and all-active jump spans remain distinct. Failure of a certificate is not represented as failure of BFGS.

14.3.13 V-ridges

The moving-frame identity and the scalar-ridge implication into the planar convergence theorem are operative formal results. No unavailable independent ridge archive is needed for that implication.

14.3.14 T-certificate

Projected PSD energy, objective-value rescaling, certificate normalization, transport and actual BFGS convergence are proved. The scalar nonnecessity result establishes a separate logical limitation of the sufficient condition.

14.3.15 C-planar

The correctly oriented determinant identity, actual angular derivative, all-pair endpoint estimates, quadratic basin and higher-dimensional warnings are formal statements. The higher-dimensional example does not invalidate the restricted planar theorem.

14.3.16 U-multidimensional

Residual-frame contradiction, cutoff transport, finite-horizon bounds and the fixed-normal convergence theorem are proved. The spike oracle is the complete infinite-series objective with exact rational-query identities and proved peak derivatives. It is not a finite truncation used as a proxy for the function.

14.3.17 C-planar-numerical

The operative statement is analytic: the stated planar example converges from its basin (9.17), and hence every positive tolerance on its objective value is eventually attained. This does not establish the original execution count, trial count, decimal precision or a particular iteration at which a tolerance was attained.

14.3.18 U-multidimensional-numerical

The actual fixed-normal spike objective has a complete oracle and the stated convergence basin in each allowed dimension. Its convergence implies eventual attainment of every positive tolerance. Historical run lengths, starts and precision remain unsupported rather than retrospectively certified.

14.3.19 G-oscillatory-numerical

The explicitly defined accumulating-kink objective has proved eligible crossings, secant curvature and a divergent obstruction quotient. It disproves local certificate existence while scalar admissible runs converge. This construction replaces an unavailable auxiliary; it is not identified with that auxiliary or used as an infinite bad BFGS trajectory.

14.4 Reviewed analytic arguments and remaining boundaries

The actual matrix hierarchy, ordered tail limits, objective certificate equivalences, planar joining, fixed-normal residual budgets, exact failure embeddings, stable-manifold existence and semialgebraic finite critical values all have formal proof chains in the delivered project. The claim index identifies their exact final contracts and internal dependencies. Assumptions such as regular switching, fixed normals, compact structural sublevels or low effective rank remain substantive.

The broader questions explicitly marked open in Chapters 8 and 7 are not proved. Nor does the report claim convergence for every finite convex nonsmooth objective, a positive-measure basin for the deterministic seed, robustness to floating-point perturbation, or a rounded initial point lying exactly on the stable manifold. No publication-priority claim is made. Those boundaries are part of the corrected statement, not gaps concealed by the compiler pass.

14.5 Reproduce the audit

From `bfgs_lean_stage1/`, the original pinned bootstrap remains the entry point for obtaining the toolchain and building the project:

```
bash scripts/bootstrap.sh
```

The closed milestone evidence should be preserved when rerunning tools that write their historical output paths. The current edition's public runner instead creates fresh output directories:

```
python3 master_report/audit/run.py --run
```

The audit records the actual commands and compiler output, checks the main source hashes, verifies the pinned clean dependencies before and after, enumerates the compiled module-owned declarations, and reconciles the main and supplemental environments. Its README describes the read-only evidence check; the runner prints the new timestamped output directory. A nonzero exit or mismatch fails the gate.

The earlier source/correspondence records also have read-only validators:

```
python3 scripts/validate_report_reconstruction_19.py
```

```
python3 scripts/verify_report_claim_review_20.py
```

The axiom audit verifies proof dependencies, not the consistency of the permitted standard axioms. No external symbolic, interval or numerical result is a trusted mathematical input.

14.6 Rebuild the corrected report

From `bfgs_lean_stage1/`:

```
python3 scripts/build_rigorous_master_report.py
python3 scripts/build_rigorous_master_report.py --check
python3 scripts/render_rigorous_master_report.py
```

The first command applies uniquely anchored, source-hash-bound edits to the preserved chapters and generates the complete claim/proof index. The second checks reproducible textual assembly. The third runs the LaTeX build and records PDF checks. The resulting PDF is `output/pdf/Rigorous_BFGS_Master_Report.pdf`. These are document tools; they perform no unchecked computation for a Lean proof. The edition status and final review record bind the generated report, source correspondence and actual compiler evidence.

CHAPTER A

The finite rational seed certificate

This appendix gives the objects and finite inequalities needed for the seed and query assertions in Chapter 12. There are two logically separate tasks. The algebraic equations close the three-step matrix and displacement cycle; the strict inequalities force the specified search to choose that cycle. Solving the equations alone would leave the algorithmic claim unproved. All decimals defining a box are terminating rationals. No rounded point is substituted for a root or an infinite orbit.

A.1 Existence and uniqueness of the defining root

The two unknown coordinates of the third gradient $g_2 = (c, d)^T$ are used to enforce the two prescribed dyadic lengths. The matrix calculation in the next section first expresses the lengths needed for closure as rational functions $\beta(c, d)$ and $\gamma(c, d)$. Clearing their positive denominators in $\beta = 1/4$ and $\gamma = 1/16$ gives the following two equations. A contraction certificate then selects one exact solution near a rational center.

Put

$$U = 127c + 16d, \qquad V = 16c - 127d - 16, \qquad R = c - 8d,$$
$$K = 8c^2 + 8d^2 + d - 8, \qquad T = c^2 - c + d^2, \qquad Y = (c-1)^2 + d^2,$$

and let

$$P(c, d) = \begin{pmatrix} 264192RK - UV(U + 129) \\ 16VTK - RUY \end{pmatrix}. \tag{A.1}$$

The root is specified by $P(c, d) = 0$ and $\|(c, d) - \bar{r}\|_\infty \le 10^{-30}$, where

$$\bar{r}_1 = 0.318481427627118630646451312502907037179122050236 05,$$
$$\bar{r}_2 = -1.025232119675748725481350870013004142845119093184 44.$$

Lemma A.1 (Rational root certificate)**.** *This closed box contains exactly one root. On the box all of $U, V, R, K, T, Y, U + 129$ are positive, and (12.3) holds.*

Proof. Let $D = DP(\bar{r})$ and define its inverse by the ordinary two-by-two adjugate formula $A = \operatorname{adj}(D)/\det D$. The following finite rational certificate uses only the coefficients of (A.1) and the two displayed rationals:

$$\det D \neq 0, \qquad \max_i |(AP(\bar{r}))_i| < 10^{-50},$$
$$\max_i \sum_{2 \le |\nu| \le 5} |a_{i,\nu}|\, |\nu| \le 7000, \tag{A.2}$$

where, for $\nu = (\nu_1, \nu_2)$,

$$a_{i,\nu} = -\frac{1}{\nu_1!\nu_2!} \sum_{j=1}^{2} A_{ij}\, \partial^\nu P_j(\bar{r}).$$

These formulas specify every rational in the certificate: the index set contains the eighteen monomials of degrees two through five. The bounds in (A.2) follow by substituting the rational center, differentiating the displayed polynomials, and clearing positive denominators in the three comparisons. Thus they are finite integer inequalities, rather than tolerances for a numerically computed root.

For completeness, their analytic consequence follows directly. The centered residual map $N(h) = h - AP(\bar{r} + h)$ has the exact expansion

$$N_i(h) = -(AP(\bar{r}))_i + \sum_{2 \le |\nu| \le 5} a_{i,\nu} h^\nu;$$

its linear part vanishes because $AD = I$. On $\|h\|_\infty, \|k\|_\infty \le r = 10^{-30}$, telescoping the factors of each monomial gives $|h^\nu - k^\nu| \le |\nu| r \|h - k\|_\infty$, since $r \le 1$ and $|\nu| \ge 2$. Hence N has Lipschitz constant at most $7000r$ and

$$\|N(h)\|_\infty \le 10^{-50} + 7000 r^2 < r.$$

The closed box is complete, so contraction gives a unique fixed point. Since A is invertible, this is exactly a root of P. This is the standard preconditioned-residual principle; compare Rump [37, Theorem 13.3].

The root box lies in $318/1000 < c < 319/1000$ and $-1026/1000 < d < -1025/1000$. Substitution gives positivity of the stated factors. For example,

$$\frac{2397}{100} \le U \le \frac{24113}{1000}, \quad \frac{119263}{1000} \le V \le \frac{59703}{500}, \quad \frac{4259}{500} \le R \le \frac{8527}{1000}.$$

Multiplying positive endpoints in $129R/[V(U + 129)]$ gives $60/1000 < \rho < 61/1000$. In particular all divisions used below are valid. The same interval rules on the root box give the thirty digits in (12.3). □

A.2 The matrix identity

Proof of the algebraic part of the seed and query assertions in Chapter 12. The first update of the cycle, $\mathcal{B}(I, g_0, g_1, 1) = M_1$, is immediate from $s_0 = (-1, 0)^T$ and $g_1 - g_0 = (-2, -1/8)^T$. For an arbitrary point (c, d), not only the root, with the positive factors above, put

$$\beta = \frac{66048RK}{UV(U + 129)}, \quad \gamma = \frac{VTK}{RUY}, \quad \widetilde{\rho} = \frac{2064TK}{U(U + 129)Y}.$$

The rational identities obtained from (12.1) are

$$\begin{aligned} \mathcal{B}(\mathcal{B}(M_1, g_1, g_2, \beta), g_2, g_0, \gamma) &= \widetilde{\rho} I, \\ -\gamma \mathcal{B}(M_1, g_1, g_2, \beta) g_2 &= \widetilde{\rho}(g_0 - g_2). \end{aligned} \tag{A.3}$$

Here is a way to check the identities without guessing matrix entries. The middle update is $C + \beta E$, with projected term C and rank-one term βE, because its two projection factors are independent of the step length and its rank-one term is linear in that length. Requiring $(C + \beta E)g_2$ to be parallel to $g_0 - g_2$ gives a linear equation for β; substitution of M_1, g_1, g_2 gives the displayed value. For a unit vector $w \perp (g_0 - g_2)$, substitution gives $w^T(C + \beta E)w = \widetilde{\rho}$. With the displayed γ, the last step is $\widetilde{\rho}(g_0 - g_2)$. Its projection factors are the orthogonal projector $\Pi = ww^T$ onto $(g_0 - g_2)^\perp$, so its projected term is $\Pi(C + \beta E)\Pi = \widetilde{\rho}\Pi$, and its rank-one term is $\widetilde{\rho}(I - \Pi)$. Their sum is $\widetilde{\rho} I$, proving (A.3). Each substitution is a rational identity and can equivalently be checked by clearing its nonzero denominators and expanding.

Equations (A.1) give exactly $\beta = 1/4$ and $\gamma = 1/16$. Moreover,

$$\widetilde{\rho} - \rho = \frac{2064R}{V(U + 129)} \left(\gamma - \frac{1}{16} \right) = 0.$$

This gives the matrix and displacement closures. The steps (12.5) also satisfy $s_j = -\alpha_j M_j g_j$ with $\alpha_0 = 1$, $\alpha_1 = 1/4$ and $\alpha_2 = 1/16$: for $j = 0, 1$ by substitution, and for $j = 2$ by the second identity in (A.3). The formula for z_0 gives $s_0 + s_1 + s_2 = -(1 - \rho)z_0$. The strict inequalities below imply $(g_{j+1} - g_j)^T s_j > 0$ at each phase. Positive definiteness follows inductively from the full update: its two nonnegative quadratic terms cannot both vanish on a nonzero vector when $y^T s > 0$. □

A.3 Sound enclosure of every query

The remaining task is finite because one cycle has only nine queries. For each query we must certify its unique active branch, the descent direction at the current iterate, and the comparison that accepts or rejects it. A positive derivative at each accepted query supplies weak Wolfe for every permitted curvature constant. Strict value increase at each rejected query makes Armijo fail, so the routine reaches exactly the next displayed bisection. This explains why the certificate below contains more than values at the three accepted points.

An interval $[a, b]$ encloses the corresponding real quantity. Addition, negation, multiplication and reciprocal are evaluated by

$$\begin{aligned}[a,b]+[c,d]&=[a+c,b+d], \qquad -[a,b]=[-b,-a],\\ [a,b][c,d]&=[\min(ac,ad,bc,bd),\max(ac,ad,bc,bd)],\end{aligned}$$

and $[a, b]^{-1} = [1/b, 1/a]$ when $0 \notin [a, b]$. These inclusion rules follow from order on the reals; induction on a rational expression proves the soundness of every resulting bound. All endpoints used here are rational, so no directed floating-point rounding is needed.

For the query calculation it suffices to use the larger box

$$\begin{gathered}c \in \frac{[318481, 318482]}{10^6}, \quad d \in \frac{[-1025233, -1025232]}{10^6},\\ \rho \in \frac{[6019, 6021]}{10^5}.\end{gathered} \tag{A.4}$$

The first two inclusions follow from the root box; substitution in (12.3), with positive denominator, gives the third. Evaluate z_j from (12.6) and use

$$p_0 = (-1, 0)^T, \quad p_1 = (127/256, 1/16)^T, \quad p_2 = 16\rho(1 - c, -d)^T.$$

For a row with active index i, set $q = z_j + \alpha p_j$ and evaluate the following rational expressions by the inclusion rules:

$$(g_i - g_k)^T q \quad (k \neq i), \quad -\alpha g_j^T p_j, \quad \begin{cases} g_j^T z_j + \alpha g_j^T p_j/3 - g_i^T q, & \text{acceptance},\\ g_i^T q - g_j^T z_j, & \text{rejection}.\end{cases}$$

For an acceptance also evaluate $\alpha g_i^T p_j$. Every expression in its row is strictly greater than the following positive rational lower bound, in the order of the table in Section 12.3:

$$\frac{1}{10^4}(729, 2417, 145, 75, 5964, 2682, 1041, 221, 20). \tag{A.5}$$

For each trial value $G(z_j + \alpha p_j)$ in the table of Section 12.3 and each current value $G(z_j)$ in the display after it, the same interval operations give a rational enclosure of width at most $4 \cdot 10^{-4}$; the twelve-digit decimals there are explanatory. For the three base points, the designated active gaps are positive as well. Evaluating $3 \pm g_i^T q$ for every i, $2 - (q)_2$, and the designated active value $g_i^T q$ gives positive lower bounds for all base points and all nine queries. This proves (12.7). All inputs, expressions, interval rules and output bounds of this finite certificate are now specified. The strict active gaps also give differentiability on a neighborhood of each query. Together with the matrix identities, these calculations complete the proof of the seed and query assertions in Chapter 12.

CHAPTER B

Complete correspondence and reproduction

The machine-readable correspondence is part of this report's source release. It is designed to expose whole theorem contracts and all dependencies, rather than offer a sample export as a substitute for a proof.

The frozen corrected mathematical source is Git commit `00362c0249262d182866f14bac756a7c4edac95c`, with paths and hashes in `editorial/BASELINE.json`. The narrative revision starts from the initial author-review release at `7f53b4c`; its source manifest, original contract reviews and unchanged earlier PDFs are preserved under `editorial/narrative_rebuild/` and `release/history/20260912_initial/`. All 172 corrected source spans remain present with their complete exports. The new opening changes their scientific presentation and reading order, not their mathematical assertions. The exceptions of October 2026 are noted at the end of this appendix.

The revised article independently proves global centered and coercive homogeneous convergence, local finite-partition and planar results, the actual-matrix framework and both finite-branch local counterexamples. The noncompact homogeneous alternatives, fixed-normal geometry, flat-contour parabolic family, lifting, rates and scalar baseline remain separate companion results. The infinite constant-Hessian family in Chapter 13 also remains here after its removal from the article's main theorem selection. None of these results is treated as superseded merely because the article selects a smaller statement set.

Record	Purpose
`companion/SOURCE_CORRESPONDENCE.json`	Every corrected source span, classification, new chapter and line location, and its full formal-export list.
`editorial/CLAIM_LEDGER.json`	Article statements, equations and substantive deductions, their proof locations and precise source/contract correspondences.
`editorial/SUPERSESSION_LEDGER.json`	Disposition of every source span; incomparable positive and negative results are retained.
`bfgs_lean_stage1/master_report/DECLARATION_INDEX.json`	Exact Lean signatures, proof excerpts and type/value/structural dependencies of the source declarations.
`bfgs_lean_stage1/master_report/PROOF_INDEX.html`	Human-readable original claim index, full exports, exact types and direct proof dependencies.
`editorial/formal_audit/`	Preserved runner, actual fresh compiler and transitive audit evidence.
`editorial/LITERATURE_LEDGER.json`	Source versions, access levels, point-of-use comparisons and unresolved attribution.

The formal source project remains in `bfgs_lean_stage1/`. Its pinned `lean-toolchain`, `lake file.toml` and `lake-manifest.json` are authoritative. On a new supported system, run `bash scripts/bootstrap.sh` there, then the fresh audit command in Chapter 14. Network access is needed to obtain the pinned toolchain and dependencies when they are absent. The release archive omits those large caches and preserves the source tree's relative paths.

To rebuild the documents from the repository root, run `bash editorial/build.sh`. The normal journal class used for the article is supplied under `editorial/vendor/svjour3/` with the original

notices and download provenance. PDF figures are generated from stated mathematical data; their code is in `paper/figures/`. The figure computations are renderings, not additional formal proof premises. Build logs, page-level rendering evidence and editorial review findings are kept under `editorial/logs/` and `editorial/reviews/`.

Source transformations preserve the corrected mathematical assertions, except the logged October 2026 generalization of the lower-curvature failures to every admissible pair of line-search constants and the October 2026 corrections recorded in `TODO_MANUSCRIPT_REVIEW.md`. Exact Lean contracts can contain definitions that abbreviate substantial hypotheses; the declaration index exposes their definitions and dependency closure. Reading one exported name without expanding those contracts would not validate the prose. Correspondence review is a separate logical bridge from kernel checking, and author approval remains necessary before submission.

Bibliography

Numbered source locators below identify the inspected version. Journal publication metadata and preprint proof text are distinguished; a citation does not assert an independent audit of the entire cited proof. Per-source access records are in `editorial/LITERATURE_LEDGER.json`.

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